

Zolotarev problem case 1a

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Abstract

Zolotarev

Contents

1	Description of the report and overview of the results	3
1.1	Methodology	3
1.2	Overview of the results	3
1.3	Preliminary conclusions	3
2	Approximation objective $r = 3$	4
2.1	Numerator and denominator coefficients	4
2.2	Poles	5
2.3	Zeros	5
2.4	Poles and zeros visualisation	6
2.5	Approximation visualization	6
3	Approximation objective $r = 4$	7
3.1	Numerator and denominator coefficients	7
3.2	Poles	8
3.3	Zeros	9
3.4	Poles and zeros visualisation	10
3.5	Approximation visualization	10
4	Approximation objective $r = 5$	11
4.1	Numerator and denominator coefficients	11
4.2	Poles	12
4.3	Zeros	13
4.4	Poles and zeros visualisation	14
4.5	Approximation visualization	14

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5	Approximation objective $r = 6$	15
5.1	Numerator and denominator coefficients	15
5.2	Poles	16
5.3	Zeros	17
5.4	Poles and zeros visualisation	18
5.5	Approximation visualization	18
6	Approximation objective $r = 7$	19
6.1	Numerator and denominator coefficients	19
6.2	Poles	20
6.3	Zeros	21
6.4	Poles and zeros visualisation	22
6.5	Approximation visualization	23
7	Approximation objective $r = 8$	24
7.1	Numerator and denominator coefficients	24
7.2	Poles	25
7.3	Zeros	26
7.4	Poles and zeros visualisation	27
7.5	Approximation visualization	28
8	Approximation objective $r = 9$	29
8.1	Numerator and denominator coefficients	29
8.2	Poles	30
8.3	Zeros	31
8.4	Poles and zeros visualisation	32
8.5	Approximation visualization	33
9	Approximation objective $r = 10$	34
9.1	Numerator and denominator coefficients	34
9.2	Poles	35
9.3	Zeros	36
9.4	Poles and zeros visualisation	38
9.5	Approximation visualization	38

1 Description of the report and overview of the results

1.1 Methodology

1.2 Overview of the results

1.3 Preliminary conclusions

2 Approximation objective $r = 3$

2.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00044$		AAA-sign-L1000, $\sigma_3 = 0.0005$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	-0.011	$6.5 \cdot 10^{-4}$	-70.0	12.0
z^2	2.6	0	3.7	$1.8 \cdot 10^{-16}$	2.6	$-2.2 \cdot 10^{-3}$	0.95	0.064
z	0	0	$-6.8 \cdot 10^{-15}$	$-1.2 \cdot 10^{-15}$	$9.9 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	-150.0	27.0
1.0	0.65	0	1.5	$1.9 \cdot 10^{-16}$	0.65	$-6.1 \cdot 10^{-4}$	-0.36	0.082
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-4.2 \cdot 10^{-15}$	$-1.3 \cdot 10^{-15}$	$-6.6 \cdot 10^{-3}$	$2.1 \cdot 10^{-3}$	-180.0	31.0
z	2.2	0	4.2	$4.4 \cdot 10^{-16}$	2.3	$-1.9 \cdot 10^{-3}$	-0.38	0.084
1.0	0	0	$-2.8 \cdot 10^{-15}$	$-2.0 \cdot 10^{-16}$	$7.3 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	-44.0	7.6

Table 1: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00044$		AAA-sign-L1000, $\sigma_3 = 0.0005$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	-0.011	$6.5 \cdot 10^{-4}$	-70.0	12.0
z^2	2.6	0	3.7	$6.2 \cdot 10^{-16}$	2.6	$-2.2 \cdot 10^{-3}$	0.95	0.064
z	0	0	$-7.6 \cdot 10^{-15}$	$1.4 \cdot 10^{-16}$	$9.9 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	-150.0	27.0
1.0	0.65	0	1.5	$3.3 \cdot 10^{-17}$	0.65	$-6.1 \cdot 10^{-4}$	-0.36	0.082
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-3.8 \cdot 10^{-15}$	$2.7 \cdot 10^{-16}$	$-6.6 \cdot 10^{-3}$	$2.1 \cdot 10^{-3}$	-180.0	31.0
z	2.2	0	4.2	$6.2 \cdot 10^{-16}$	2.3	$-1.9 \cdot 10^{-3}$	-0.38	0.084
1.0	0	0	$-3.4 \cdot 10^{-15}$	$-9.1 \cdot 10^{-17}$	$7.3 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	-44.0	7.6

Table 2: Case 1a (cpv MACI64 MR2022b), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00037$		AAA-sign-L1000, $\sigma_3 = 0.00037$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	$-2.4 \cdot 10^{-11}$	$2.7 \cdot 10^{-13}$	$-9.5 \cdot 10^{-15}$	$5.6 \cdot 10^{-13}$
z^2	2.6	0	3.7	$-1.8 \cdot 10^{-16}$	2.6	$-5.8 \cdot 10^{-15}$	2.6	$-4.2 \cdot 10^{-15}$
z	0	0	$-5.1 \cdot 10^{-15}$	$6.8 \cdot 10^{-16}$	$-5.4 \cdot 10^{-11}$	$6.3 \cdot 10^{-13}$	$-2.1 \cdot 10^{-14}$	$1.2 \cdot 10^{-12}$
1.0	0.65	0	1.5	$-2.4 \cdot 10^{-16}$	0.65	$-9.7 \cdot 10^{-15}$	0.65	$-6.4 \cdot 10^{-15}$
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-3.2 \cdot 10^{-15}$	$3.0 \cdot 10^{-18}$	$-6.2 \cdot 10^{-11}$	$7.1 \cdot 10^{-13}$	$-2.3 \cdot 10^{-14}$	$1.4 \cdot 10^{-12}$
z	2.2	0	4.2	$-4.0 \cdot 10^{-16}$	2.2	$-9.9 \cdot 10^{-15}$	2.2	$-9.8 \cdot 10^{-15}$
1.0	0	0	$-1.8 \cdot 10^{-15}$	$4.3 \cdot 10^{-16}$	$-1.6 \cdot 10^{-11}$	$1.9 \cdot 10^{-13}$	$-7.1 \cdot 10^{-15}$	$3.6 \cdot 10^{-13}$

Table 3: Case 1a (cpv PCWIN64 MR2025b), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00037$		AAA-sign-L1000, $\sigma_3 = 0.00037$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	$-2.4 \cdot 10^{-11}$	$2.7 \cdot 10^{-13}$	$-9.5 \cdot 10^{-15}$	$5.6 \cdot 10^{-13}$
z^2	2.6	0	3.7	$-1.8 \cdot 10^{-16}$	2.6	$-5.8 \cdot 10^{-15}$	2.6	$-4.2 \cdot 10^{-15}$
z	0	0	$-5.1 \cdot 10^{-15}$	$6.8 \cdot 10^{-16}$	$-5.4 \cdot 10^{-11}$	$6.3 \cdot 10^{-13}$	$-2.1 \cdot 10^{-14}$	$1.2 \cdot 10^{-12}$
1.0	0.65	0	1.5	$-2.4 \cdot 10^{-16}$	0.65	$-9.7 \cdot 10^{-15}$	0.65	$-6.4 \cdot 10^{-15}$
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-3.2 \cdot 10^{-15}$	$3.0 \cdot 10^{-18}$	$-6.2 \cdot 10^{-11}$	$7.1 \cdot 10^{-13}$	$-2.3 \cdot 10^{-14}$	$1.4 \cdot 10^{-12}$
z	2.2	0	4.2	$-4.0 \cdot 10^{-16}$	2.2	$-9.9 \cdot 10^{-15}$	2.2	$-9.8 \cdot 10^{-15}$
1.0	0	0	$-1.8 \cdot 10^{-15}$	$4.3 \cdot 10^{-16}$	$-1.6 \cdot 10^{-11}$	$1.9 \cdot 10^{-13}$	$-7.1 \cdot 10^{-15}$	$3.6 \cdot 10^{-13}$

Table 4: Case 1a (AlexReis PCWIN64 MR2025b), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

2.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	0	$6.58 \cdot 10^{-16} + 4.81 \cdot 10^{-17}i$	$-0.00323 - 9.0 \cdot 10^{-5}i$	$-0.00181 + 0.493i$
2.0	$1.5i$	$2.14 \cdot 10^{-15} - 2.06i$	$0.00557 + 1.5i$	$-0.00165 - 0.494i$
3.0	$-1.5i$	$1.49 \cdot 10^{-15} + 2.06i$	$0.00428 - 1.5i$	$180.0 - 31.3i$

Table 5: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	0	$8.09 \cdot 10^{-16} + 2.14 \cdot 10^{-17}i$	$-0.00323 - 9.0 \cdot 10^{-5}i$	$-0.00181 + 0.493i$
2.0	$1.5i$	$1.75 \cdot 10^{-15} - 2.06i$	$0.00557 + 1.5i$	$-0.00165 - 0.494i$
3.0	$-1.5i$	$1.29 \cdot 10^{-15} + 2.06i$	$0.00428 - 1.5i$	$180.0 - 31.3i$

Table 6: Case 1a (cpv MACI64 MR2022b), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	0	$4.35 \cdot 10^{-16} - 1.02 \cdot 10^{-16}i$	$6.9 \cdot 10^{-12} - 8.47 \cdot 10^{-14}i$	$3.42 \cdot 10^{-15} - 1.59 \cdot 10^{-13}i$
2.0	$1.5i$	$1.25 \cdot 10^{-15} - 2.06i$	$2.76 \cdot 10^{-11} + 1.5i$	$1.22 \cdot 10^{-14} + 1.5i$
3.0	$-1.5i$	$1.47 \cdot 10^{-15} + 2.06i$	$2.76 \cdot 10^{-11} - 1.5i$	$7.73 \cdot 10^{-15} - 1.5i$

Table 7: Case 1a (cpv PCWIN64 MR2025b), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	0	$4.35 \cdot 10^{-16} - 1.02 \cdot 10^{-16}i$	$6.9 \cdot 10^{-12} - 8.47 \cdot 10^{-14}i$	$3.42 \cdot 10^{-15} - 1.59 \cdot 10^{-13}i$
2.0	$1.5i$	$1.25 \cdot 10^{-15} - 2.06i$	$2.76 \cdot 10^{-11} + 1.5i$	$1.22 \cdot 10^{-14} + 1.5i$
3.0	$-1.5i$	$1.47 \cdot 10^{-15} + 2.06i$	$2.76 \cdot 10^{-11} - 1.5i$	$7.73 \cdot 10^{-15} - 1.5i$

Table 8: Case 1a (AlexReis PCWIN64 MR2025b), $r = 3$, Z4: poles.

2.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	NaN	NaN	$-0.00241 + 0.5i$	$-0.00236 + 1.2 \cdot 10^{-4}i$
2.0	$0.5i$	$9.48 \cdot 10^{-16} - 0.637i$	$-0.00245 - 0.501i$	$0.00667 - 1.48i$
3.0	$-0.5i$	$8.89 \cdot 10^{-16} + 0.637i$	$237.0 + 13.9i$	$0.00871 + 1.48i$

Table 9: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4: zeros.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	NaN	NaN	$-0.00241 + 0.5i$	$-0.00236 + 1.2 \cdot 10^{-4}i$
2.0	$0.5i$	$9.48 \cdot 10^{-16} - 0.637i$	$-0.00245 - 0.501i$	$0.00667 - 1.48i$
3.0	$-0.5i$	$1.1 \cdot 10^{-15} + 0.637i$	$237.0 + 13.9i$	$0.00871 + 1.48i$

Table 10: Case 1a (cpv MACI64 MR2022b), $r = 3$, Z4: zeros.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	NaN	NaN	$9.21 \cdot 10^{-12} + 0.5i$	$4.98 \cdot 10^{-15} + 0.5i$
2.0	$0.5i$	$7.16 \cdot 10^{-16} + 0.637i$	$9.2 \cdot 10^{-12} - 0.5i$	$2.0 \cdot 10^{-15} - 0.5i$
3.0	$-0.5i$	$6.45 \cdot 10^{-16} - 0.637i$	$1.09 \cdot 10^{11} + 1.22 \cdot 10^9i$	$7.82 \cdot 10^{10} + 4.66 \cdot 10^{12}i$

Table 11: Case 1a (cpv PCWIN64 MR2025b), $r = 3$, Z4: zeros.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	NaN	NaN	$9.21 \cdot 10^{-12} + 0.5i$	$4.98 \cdot 10^{-15} + 0.5i$
2.0	$0.5i$	$7.16 \cdot 10^{-16} + 0.637i$	$9.2 \cdot 10^{-12} - 0.5i$	$2.0 \cdot 10^{-15} - 0.5i$
3.0	$-0.5i$	$6.45 \cdot 10^{-16} - 0.637i$	$1.09 \cdot 10^{11} + 1.22 \cdot 10^9i$	$7.82 \cdot 10^{10} + 4.66 \cdot 10^{12}i$

Table 12: Case 1a (AlexReis PCWIN64 MR2025b), $r = 3$, Z4: zeros.

2.4 Poles and zeros visualisation

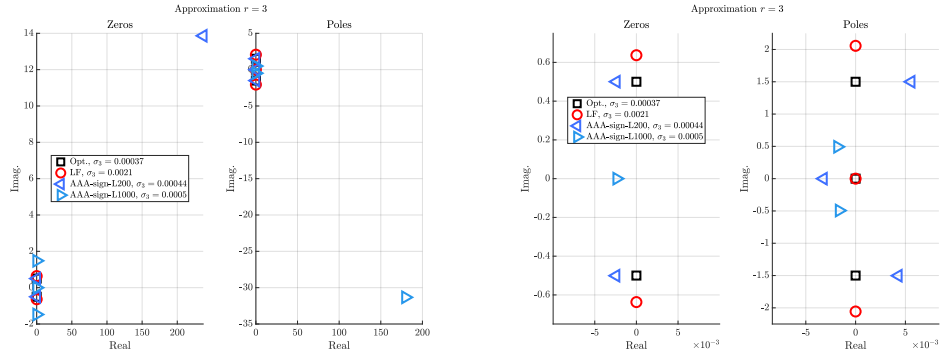


Figure 1: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

2.5 Approximation visualization

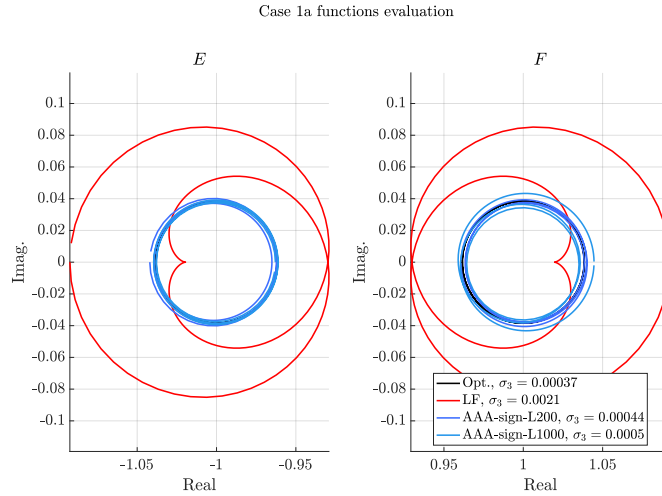


Figure 2: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4. E and F evaluations.

3 Approximation objective $r = 4$

3.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	-140.0	16.0	-150.0	7.3
z^3	3.5	0	3.7	$4.1 \cdot 10^{-16}$	4.7	-0.29	4.8	-0.22
z^2	0	0	$-4.6 \cdot 10^{-15}$	$3.8 \cdot 10^{-16}$	-650.0	70.0	-660.0	33.0
z	2.6	0	3.2	$2.0 \cdot 10^{-16}$	5.5	-0.52	5.6	-0.34
1.0	0	0	$-1.3 \cdot 10^{-15}$	$-1.6 \cdot 10^{-15}$	-80.0	8.8	-82.0	4.1
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-2.7 \cdot 10^{-15}$	$-8.4 \cdot 10^{-16}$	-500.0	54.0	-510.0	25.0
z^2	4.5	0	5.2	$8.2 \cdot 10^{-16}$	7.9	-0.7	7.9	-0.49
z	0	0	$-3.5 \cdot 10^{-15}$	$-1.1 \cdot 10^{-15}$	-370.0	41.0	-380.0	19.0
1.0	0.56	0	0.75	$1.0 \cdot 10^{-16}$	1.4	-0.13	1.4	-0.078

Table 13: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	-140.0	16.0	-150.0	7.3
z^3	3.5	0	3.7	$6.4 \cdot 10^{-16}$	4.7	-0.29	4.8	-0.22
z^2	0	0	$-4.8 \cdot 10^{-15}$	$3.7 \cdot 10^{-16}$	-650.0	70.0	-660.0	33.0
z	2.6	0	3.2	$1.8 \cdot 10^{-15}$	5.5	-0.52	5.6	-0.34
1.0	0	0	$-1.0 \cdot 10^{-15}$	$2.6 \cdot 10^{-15}$	-80.0	8.8	-82.0	4.1
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-9.8 \cdot 10^{-16}$	$8.7 \cdot 10^{-16}$	-500.0	54.0	-510.0	25.0
z^2	4.5	0	5.2	$1.9 \cdot 10^{-15}$	7.9	-0.7	7.9	-0.49
z	0	0	$-4.0 \cdot 10^{-15}$	$2.8 \cdot 10^{-15}$	-370.0	41.0	-380.0	19.0
1.0	0.56	0	0.75	$5.9 \cdot 10^{-16}$	1.4	-0.13	1.4	-0.078

Table 14: Case 1a (cpv MACI64 MR2022b), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	$-5.4 \cdot 10^{-3}$	$-3.1 \cdot 10^{-3}$	$-4.8 \cdot 10^{-3}$	$-2.8 \cdot 10^{-3}$
z^3	3.5	0	3.7	$1.8 \cdot 10^{-16}$	3.5	$-3.7 \cdot 10^{-3}$	3.5	$-3.8 \cdot 10^{-3}$
z^2	0	0	$-2.8 \cdot 10^{-15}$	$-9.9 \cdot 10^{-16}$	-0.07	$-2.6 \cdot 10^{-3}$	-0.078	$-2.5 \cdot 10^{-3}$
z	2.6	0	3.2	$-5.5 \cdot 10^{-16}$	2.6	-0.011	2.6	-0.011
1.0	0	0	$-1.8 \cdot 10^{-15}$	$-7.7 \cdot 10^{-16}$	-0.015	$1.3 \cdot 10^{-3}$	-0.017	$1.4 \cdot 10^{-3}$
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-4.6 \cdot 10^{-16}$	$-1.2 \cdot 10^{-15}$	-0.036	$-6.9 \cdot 10^{-3}$	-0.038	$-6.5 \cdot 10^{-3}$
z^2	4.5	0	5.2	$2.7 \cdot 10^{-16}$	4.5	-0.011	4.5	-0.011
z	0	0	$-4.2 \cdot 10^{-15}$	$-4.2 \cdot 10^{-16}$	-0.054	$2.2 \cdot 10^{-3}$	-0.062	$2.5 \cdot 10^{-3}$
1.0	0.56	0	0.75	$-5.2 \cdot 10^{-16}$	0.57	$-3.3 \cdot 10^{-3}$	0.57	$-3.3 \cdot 10^{-3}$

Table 15: Case 1a (cpv PCWIN64 MR2025b), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	$-5.4 \cdot 10^{-3}$	$-3.1 \cdot 10^{-3}$	$-4.8 \cdot 10^{-3}$	$-2.8 \cdot 10^{-3}$
z^3	3.5	0	3.7	$1.8 \cdot 10^{-16}$	3.5	$-3.7 \cdot 10^{-3}$	3.5	$-3.8 \cdot 10^{-3}$
z^2	0	0	$-2.8 \cdot 10^{-15}$	$-9.9 \cdot 10^{-16}$	-0.07	$-2.6 \cdot 10^{-3}$	-0.078	$-2.5 \cdot 10^{-3}$
z	2.6	0	3.2	$-5.5 \cdot 10^{-16}$	2.6	-0.011	2.6	-0.011
1.0	0	0	$-1.8 \cdot 10^{-15}$	$-7.7 \cdot 10^{-16}$	-0.015	$1.3 \cdot 10^{-3}$	-0.017	$1.4 \cdot 10^{-3}$
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-4.6 \cdot 10^{-16}$	$-1.2 \cdot 10^{-15}$	-0.036	$-6.9 \cdot 10^{-3}$	-0.038	$-6.5 \cdot 10^{-3}$
z^2	4.5	0	5.2	$2.7 \cdot 10^{-16}$	4.5	-0.011	4.5	-0.011
z	0	0	$-4.2 \cdot 10^{-15}$	$-4.2 \cdot 10^{-16}$	-0.054	$2.2 \cdot 10^{-3}$	-0.062	$2.5 \cdot 10^{-3}$
1.0	0.56	0	0.75	$-5.2 \cdot 10^{-16}$	0.57	$-3.3 \cdot 10^{-3}$	0.57	$-3.3 \cdot 10^{-3}$

Table 16: Case 1a (AlexReis PCWIN64 MR2025b), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

3.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$3.21 \cdot 10^{-16} - 0.384i$	$0.00377 + 6.65 \cdot 10^{-5}i$	$0.00375 - 2.07 \cdot 10^{-5}i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$3.09 \cdot 10^{-16} + 0.384i$	$0.00535 - 0.864i$	$0.00513 + 0.864i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$1.12 \cdot 10^{-15} - 2.25i$	$0.00516 + 0.864i$	$0.00532 - 0.864i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$9.09 \cdot 10^{-16} + 2.25i$	$499.0 - 54.4i$	$507.0 - 25.2i$

Table 17: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$3.19 \cdot 10^{-16} + 0.384i$	$0.00377 + 6.65 \cdot 10^{-5}i$	$0.00375 - 2.07 \cdot 10^{-5}i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$4.66 \cdot 10^{-16} - 0.384i$	$0.00535 - 0.864i$	$0.00513 + 0.864i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$4.91 \cdot 10^{-16} - 2.25i$	$0.00516 + 0.864i$	$0.00532 - 0.864i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$-3.02 \cdot 10^{-16} + 2.25i$	$499.0 - 54.4i$	$507.0 - 25.2i$

Table 18: Case 1a (cpv MACI64 MR2022b), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$5.75 \cdot 10^{-16} + 0.384i$	$0.00636 + 0.36i$	$0.00731 + 0.36i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$2.75 \cdot 10^{-16} - 0.384i$	$0.0052 - 0.361i$	$0.00612 - 0.361i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$-1.15 \cdot 10^{-16} - 2.25i$	$0.00935 - 2.1i$	$0.00961 - 2.09i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$-2.71 \cdot 10^{-16} + 2.25i$	$0.0147 + 2.1i$	$0.0149 + 2.1i$

Table 19: Case 1a (cpv PCWIN64 MR2025b), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$5.75 \cdot 10^{-16} + 0.384i$	$0.00636 + 0.36i$	$0.00731 + 0.36i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$2.75 \cdot 10^{-16} - 0.384i$	$0.0052 - 0.361i$	$0.00612 - 0.361i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$-1.15 \cdot 10^{-16} - 2.25i$	$0.00935 - 2.1i$	$0.00961 - 2.09i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$-2.71 \cdot 10^{-16} + 2.25i$	$0.0147 + 2.1i$	$0.0149 + 2.1i$

Table 20: Case 1a (AlexReis PCWIN64 MR2025b), $r = 4$, Z4: poles.

3.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00403 - 0.358i$	$0.00404 + 0.358i$
2.0	0	$4.1 \cdot 10^{-16} + 5.92 \cdot 10^{-16}i$	$0.00406 + 0.358i$	$0.00402 - 0.358i$
3.0	$0.866i$	$3.69 \cdot 10^{-16} + 0.929i$	$0.0128 - 2.08i$	$0.0127 - 2.09i$
4.0	$-0.866i$	$4.46 \cdot 10^{-16} - 0.929i$	$0.0117 + 2.09i$	$0.0116 + 2.09i$

Table 21: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4: zeros.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00403 - 0.358i$	$0.00404 + 0.358i$
2.0	0	$3.12 \cdot 10^{-16} - 7.29 \cdot 10^{-16}i$	$0.00406 + 0.358i$	$0.00402 - 0.358i$
3.0	$0.866i$	$6.57 \cdot 10^{-16} - 0.929i$	$0.0128 - 2.08i$	$0.0127 - 2.09i$
4.0	$-0.866i$	$3.16 \cdot 10^{-16} + 0.929i$	$0.0117 + 2.09i$	$0.0116 + 2.09i$

Table 22: Case 1a (cpv MACI64 MR2022b), $r = 4$, Z4: zeros.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00562 - 4.73 \cdot 10^{-4}i$	$0.00656 - 5.2 \cdot 10^{-4}i$
2.0	0	$5.68 \cdot 10^{-16} + 2.27 \cdot 10^{-16}i$	$0.00528 - 0.87i$	$0.00611 - 0.869i$
3.0	$0.866i$	$3.38 \cdot 10^{-18} - 0.929i$	$0.00792 + 0.871i$	$0.00875 + 0.87i$
4.0	$-0.866i$	$1.73 \cdot 10^{-16} + 0.929i$	$486.0 - 282.0i$	$539.0 - 319.0i$

Table 23: Case 1a (cpv PCWIN64 MR2025b), $r = 4$, Z4: zeros.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00562 - 4.73 \cdot 10^{-4}i$	$0.00656 - 5.2 \cdot 10^{-4}i$
2.0	0	$5.68 \cdot 10^{-16} + 2.27 \cdot 10^{-16}i$	$0.00528 - 0.87i$	$0.00611 - 0.869i$
3.0	$0.866i$	$3.38 \cdot 10^{-18} - 0.929i$	$0.00792 + 0.871i$	$0.00875 + 0.87i$
4.0	$-0.866i$	$1.73 \cdot 10^{-16} + 0.929i$	$486.0 - 282.0i$	$539.0 - 319.0i$

Table 24: Case 1a (AlexReis PCWIN64 MR2025b), $r = 4$, Z4: zeros.

3.4 Poles and zeros visualisation

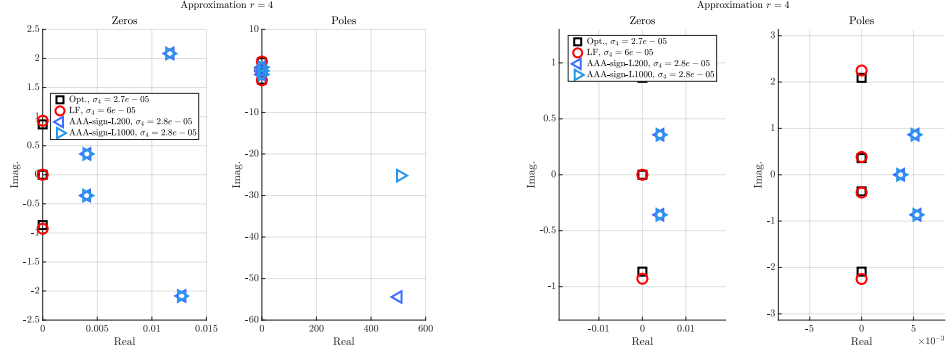


Figure 3: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

3.5 Approximation visualization

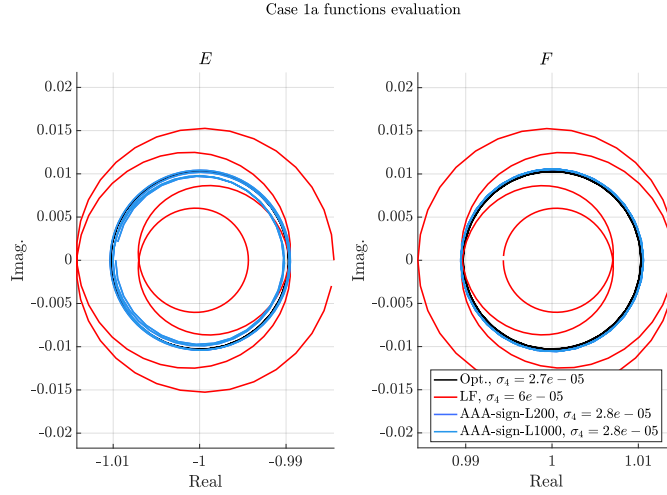


Figure 4: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4. E and F evaluations.

4 Approximation objective $r = 5$

4.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$-1.1 \cdot 10^{-11}$	$-4.8 \cdot 10^{-11}$	$2.1 \cdot 10^{-13}$	$4.5 \cdot 10^{-11}$
z^4	4.3	0	5.5	$-5.9 \cdot 10^{-15}$	4.3	$-6.4 \cdot 10^{-13}$	4.3	$1.2 \cdot 10^{-12}$
z^3	0	0	$-4.1 \cdot 10^{-15}$	$-3.4 \cdot 10^{-15}$	$-8.3 \cdot 10^{-11}$	$-3.6 \cdot 10^{-10}$	$1.5 \cdot 10^{-12}$	$3.4 \cdot 10^{-10}$
z^2	6.5	0	12.0	$-1.7 \cdot 10^{-14}$	6.5	$-2.9 \cdot 10^{-12}$	6.5	$5.2 \cdot 10^{-12}$
z	0	0	$-4.0 \cdot 10^{-15}$	$-9.0 \cdot 10^{-15}$	$-3.1 \cdot 10^{-11}$	$-1.4 \cdot 10^{-10}$	$5.5 \cdot 10^{-13}$	$1.3 \cdot 10^{-10}$
1.0	0.49	0	1.1	$-1.3 \cdot 10^{-15}$	0.49	$-3.6 \cdot 10^{-13}$	0.49	$6.3 \cdot 10^{-13}$
z^5	1.0	0	1.0	0	1.0	0	1.0	0
z^4	0	0	$-3.6 \cdot 10^{-15}$	$-3.0 \cdot 10^{-15}$	$-4.8 \cdot 10^{-11}$	$-2.1 \cdot 10^{-10}$	$8.8 \cdot 10^{-13}$	$2.0 \cdot 10^{-10}$
z^3	7.5	0	11.0	$-1.6 \cdot 10^{-14}$	7.5	$-2.2 \cdot 10^{-12}$	7.5	$4.0 \cdot 10^{-12}$
z^2	0	0	$-4.4 \cdot 10^{-15}$	$-1.0 \cdot 10^{-14}$	$-7.2 \cdot 10^{-11}$	$-3.1 \cdot 10^{-10}$	$1.3 \cdot 10^{-12}$	$3.0 \cdot 10^{-10}$
z	2.8	0	5.8	$-7.1 \cdot 10^{-15}$	2.8	$-1.7 \cdot 10^{-12}$	2.8	$3.0 \cdot 10^{-12}$
1.0	0	0	$-9.8 \cdot 10^{-16}$	$-1.1 \cdot 10^{-15}$	$-5.4 \cdot 10^{-12}$	$-2.4 \cdot 10^{-11}$	$9.5 \cdot 10^{-14}$	$2.3 \cdot 10^{-11}$

Table 25: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$-4.1 \cdot 10^{-11}$	$3.9 \cdot 10^{-11}$	$-2.9 \cdot 10^{-14}$	$1.2 \cdot 10^{-11}$
z^4	4.3	0	5.5	$-6.5 \cdot 10^{-16}$	4.3	$-5.4 \cdot 10^{-13}$	4.3	$1.3 \cdot 10^{-13}$
z^3	0	0	$-2.1 \cdot 10^{-14}$	$4.3 \cdot 10^{-15}$	$-3.1 \cdot 10^{-10}$	$2.9 \cdot 10^{-10}$	$-2.9 \cdot 10^{-13}$	$9.0 \cdot 10^{-11}$
z^2	6.5	0	12.0	$7.5 \cdot 10^{-15}$	6.5	$-2.4 \cdot 10^{-12}$	6.5	$5.9 \cdot 10^{-13}$
z	0	0	$-1.3 \cdot 10^{-14}$	$1.6 \cdot 10^{-14}$	$-1.2 \cdot 10^{-10}$	$1.1 \cdot 10^{-10}$	$-6.8 \cdot 10^{-14}$	$3.4 \cdot 10^{-11}$
1.0	0.49	0	1.1	$2.0 \cdot 10^{-16}$	0.49	$-3.0 \cdot 10^{-13}$	0.49	$8.0 \cdot 10^{-14}$
z^5	1.0	0	1.0	0	1.0	0	1.0	0
z^4	0	0	$-5.8 \cdot 10^{-15}$	$1.8 \cdot 10^{-15}$	$-1.8 \cdot 10^{-10}$	$1.7 \cdot 10^{-10}$	$-1.7 \cdot 10^{-13}$	$5.2 \cdot 10^{-11}$
z^3	7.5	0	11.0	$3.2 \cdot 10^{-15}$	7.5	$-1.8 \cdot 10^{-12}$	7.5	$4.5 \cdot 10^{-13}$
z^2	0	0	$-2.3 \cdot 10^{-14}$	$1.8 \cdot 10^{-14}$	$-2.7 \cdot 10^{-10}$	$2.5 \cdot 10^{-10}$	$-2.2 \cdot 10^{-13}$	$7.8 \cdot 10^{-11}$
z	2.8	0	5.8	$4.1 \cdot 10^{-15}$	2.8	$-1.4 \cdot 10^{-12}$	2.8	$3.5 \cdot 10^{-13}$
1.0	0	0	$-3.4 \cdot 10^{-15}$	$2.4 \cdot 10^{-15}$	$-2.0 \cdot 10^{-11}$	$1.9 \cdot 10^{-11}$	$-2.3 \cdot 10^{-15}$	$5.9 \cdot 10^{-12}$

Table 26: Case 1a (cpv MACI64 MR2022b), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$1.1 \cdot 10^3$	-31.0	$1.1 \cdot 10^3$	-31.0
z^4	4.3	0	5.5	$-3.7 \cdot 10^{-15}$	3.8	0.021	3.8	0.021
z^3	0	0	$-6.7 \cdot 10^{-15}$	$2.5 \cdot 10^{-15}$	$8.1 \cdot 10^3$	-240.0	$8.1 \cdot 10^3$	-230.0
z^2	6.5	0	12.0	$-1.4 \cdot 10^{-14}$	4.1	0.086	4.1	0.085
z	0	0	$-1.4 \cdot 10^{-14}$	$6.9 \cdot 10^{-17}$	$3.1 \cdot 10^3$	-89.0	$3.1 \cdot 10^3$	-87.0
1.0	0.49	0	1.1	$-1.9 \cdot 10^{-15}$	0.19	$9.6 \cdot 10^{-3}$	0.19	$9.5 \cdot 10^{-3}$
z^5	1.0	0	1.0	0	1.0	0	1.0	$-5.8 \cdot 10^{-17}$
z^4	0	0	$1.9 \cdot 10^{-16}$	$-1.4 \cdot 10^{-15}$	$4.7 \cdot 10^3$	-140.0	$4.7 \cdot 10^3$	-130.0
z^3	7.5	0	11.0	$-1.1 \cdot 10^{-14}$	5.7	0.071	5.7	0.07
z^2	0	0	$-1.7 \cdot 10^{-14}$	$3.9 \cdot 10^{-15}$	$7.1 \cdot 10^3$	-200.0	$7.1 \cdot 10^3$	-200.0
z	2.8	0	5.8	$-8.6 \cdot 10^{-15}$	1.4	0.047	1.4	0.046
1.0	0	0	$-3.3 \cdot 10^{-15}$	$5.1 \cdot 10^{-16}$	530.0	-15.0	530.0	-15.0

Table 27: Case 1a (cpv PCWIN64 MR2025b), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$1.1 \cdot 10^3$	-31.0	$1.1 \cdot 10^3$	-31.0
z^4	4.3	0	5.5	$-3.7 \cdot 10^{-15}$	3.8	0.021	3.8	0.021
z^3	0	0	$-6.7 \cdot 10^{-15}$	$2.5 \cdot 10^{-15}$	$8.1 \cdot 10^3$	-240.0	$8.1 \cdot 10^3$	-230.0
z^2	6.5	0	12.0	$-1.4 \cdot 10^{-14}$	4.1	0.086	4.1	0.085
z	0	0	$-1.4 \cdot 10^{-14}$	$6.9 \cdot 10^{-17}$	$3.1 \cdot 10^3$	-89.0	$3.1 \cdot 10^3$	-87.0
1.0	0.49	0	1.1	$-1.9 \cdot 10^{-15}$	0.19	$9.6 \cdot 10^{-3}$	0.19	$9.5 \cdot 10^{-3}$
z^5	1.0	0	1.0	0	1.0	0	1.0	$-5.8 \cdot 10^{-17}$
z^4	0	0	$1.9 \cdot 10^{-16}$	$-1.4 \cdot 10^{-15}$	$4.7 \cdot 10^3$	-140.0	$4.7 \cdot 10^3$	-130.0
z^3	7.5	0	11.0	$-1.1 \cdot 10^{-14}$	5.7	0.071	5.7	0.07
z^2	0	0	$-1.7 \cdot 10^{-14}$	$3.9 \cdot 10^{-15}$	$7.1 \cdot 10^3$	-200.0	$7.1 \cdot 10^3$	-200.0
z	2.8	0	5.8	$-8.6 \cdot 10^{-15}$	1.4	0.047	1.4	0.046
1.0	0	0	$-3.3 \cdot 10^{-15}$	$5.1 \cdot 10^{-16}$	530.0	-15.0	530.0	-15.0

Table 28: Case 1a (AlexReis PCWIN64 MR2025b), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

4.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$1.69 \cdot 10^{-16} + 1.84 \cdot 10^{-16}i$	$1.92 \cdot 10^{-12} + 8.44 \cdot 10^{-12}i$	$-3.41 \cdot 10^{-14} - 8.04 \cdot 10^{-12}i$
2.0	$3.89 \cdot 10^{-16} + 2.67i$	$2.45 \cdot 10^{-16} - 0.729i$	$2.84 \cdot 10^{-12} - 0.629i$	$-2.12 \cdot 10^{-13} + 0.629i$
3.0	$3.89 \cdot 10^{-16} - 2.67i$	$-1.57 \cdot 10^{-16} + 0.729i$	$3.03 \cdot 10^{-12} + 0.629i$	$1.08 \cdot 10^{-13} - 0.629i$
4.0	$-5.55 \cdot 10^{-17} + 0.629i$	$-1.18 \cdot 10^{-15} - 3.3i$	$1.98 \cdot 10^{-11} - 2.67i$	$-1.09 \cdot 10^{-12} + 2.67i$
5.0	$-5.55 \cdot 10^{-17} - 0.629i$	$4.72 \cdot 10^{-15} + 3.3i$	$2.05 \cdot 10^{-11} + 2.67i$	$3.47 \cdot 10^{-13} - 2.67i$

Table 29: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4: poles.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$5.89 \cdot 10^{-16} - 4.22 \cdot 10^{-16}i$	$7.1 \cdot 10^{-12} - 6.72 \cdot 10^{-12}i$	$-8.34 \cdot 10^{-16} - 2.1 \cdot 10^{-12}i$
2.0	$3.89 \cdot 10^{-16} + 2.67i$	$4.48 \cdot 10^{-16} + 0.729i$	$1.09 \cdot 10^{-11} + 0.629i$	$-8.47 \cdot 10^{-15} + 0.629i$
3.0	$3.89 \cdot 10^{-16} - 2.67i$	$8.9 \cdot 10^{-16} - 0.729i$	$1.08 \cdot 10^{-11} - 0.629i$	$3.14 \cdot 10^{-14} - 0.629i$
4.0	$-5.55 \cdot 10^{-17} + 0.629i$	$1.51 \cdot 10^{-15} + 3.3i$	$7.46 \cdot 10^{-11} + 2.67i$	$-7.54 \cdot 10^{-15} + 2.67i$
5.0	$-5.55 \cdot 10^{-17} - 0.629i$	$2.43 \cdot 10^{-15} - 3.3i$	$7.4 \cdot 10^{-11} - 2.67i$	$1.54 \cdot 10^{-13} - 2.67i$

Table 30: Case 1a (cpv MACI64 MR2022b), $r = 5$, Z4: poles.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$5.62 \cdot 10^{-16} - 8.81 \cdot 10^{-17}i$	$-7.91 \cdot 10^{-5} + 0.281i$	$-7.91 \cdot 10^{-5} + 0.281i$
2.0	$3.89 \cdot 10^{-16} + 2.67i$	$7.72 \cdot 10^{-16} + 0.729i$	$-7.96 \cdot 10^{-5} - 0.281i$	$-7.95 \cdot 10^{-5} - 0.281i$
3.0	$3.89 \cdot 10^{-16} - 2.67i$	$2.77 \cdot 10^{-16} - 0.729i$	$-3.61 \cdot 10^{-4} + 1.19i$	$-3.61 \cdot 10^{-4} + 1.19i$
4.0	$-5.55 \cdot 10^{-17} + 0.629i$	$-2.67 \cdot 10^{-15} - 3.3i$	$-3.65 \cdot 10^{-4} - 1.19i$	$-3.65 \cdot 10^{-4} - 1.19i$
5.0	$-5.55 \cdot 10^{-17} - 0.629i$	$1.2 \cdot 10^{-15} + 3.3i$	$-4700.0 + 136.0i$	$-4700.0 + 134.0i$

Table 31: Case 1a (cpv PCWIN64 MR2025b), $r = 5$, Z4: poles.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$5.62 \cdot 10^{-16} - 8.81 \cdot 10^{-17}i$	$-7.91 \cdot 10^{-5} + 0.281i$	$-7.91 \cdot 10^{-5} + 0.281i$
2.0	$3.89 \cdot 10^{-16} + 2.67i$	$7.72 \cdot 10^{-16} + 0.729i$	$-7.96 \cdot 10^{-5} - 0.281i$	$-7.95 \cdot 10^{-5} - 0.281i$
3.0	$3.89 \cdot 10^{-16} - 2.67i$	$2.77 \cdot 10^{-16} - 0.729i$	$-3.61 \cdot 10^{-4} + 1.19i$	$-3.61 \cdot 10^{-4} + 1.19i$
4.0	$-5.55 \cdot 10^{-17} + 0.629i$	$-2.67 \cdot 10^{-15} - 3.3i$	$-3.65 \cdot 10^{-4} - 1.19i$	$-3.65 \cdot 10^{-4} - 1.19i$
5.0	$-5.55 \cdot 10^{-17} - 0.629i$	$1.2 \cdot 10^{-15} + 3.3i$	$-4700.0 + 136.0i$	$-4700.0 + 134.0i$

Table 32: Case 1a (AlexReis PCWIN64 MR2025b), $r = 5$, Z4: poles.

4.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$2.08 \cdot 10^{-12} - 0.281z$	$-1.06 \cdot 10^{-13} + 0.281z$
2.0	$5.55 \cdot 10^{-17} + 1.19z$	$2.26 \cdot 10^{-16} - 0.32z$	$2.16 \cdot 10^{-12} + 0.281z$	$3.17 \cdot 10^{-14} - 0.281z$
3.0	$5.55 \cdot 10^{-17} - 1.19z$	$1.36 \cdot 10^{-16} + 0.32z$	$5.38 \cdot 10^{-12} - 1.19z$	$-4.15 \cdot 10^{-13} + 1.19z$
4.0	$6.94 \cdot 10^{-17} + 0.281z$	$5.92 \cdot 10^{-16} + 1.42z$	$5.74 \cdot 10^{-12} + 1.19z$	$2.15 \cdot 10^{-13} - 1.19z$
5.0	$6.94 \cdot 10^{-17} - 0.281z$	$-1.68 \cdot 10^{-16} - 1.42z$	$1.99 \cdot 10^{10} - 8.58 \cdot 10^{10}z$	$-4.5 \cdot 10^8 + 9.58 \cdot 10^{10}z$

Table 33: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4: zeros.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$7.88 \cdot 10^{-12} + 0.281z$	$-7.88 \cdot 10^{-15} + 0.281z$
2.0	$5.55 \cdot 10^{-17} + 1.19z$	$5.94 \cdot 10^{-16} + 0.32z$	$7.81 \cdot 10^{-12} - 0.281z$	$1.46 \cdot 10^{-14} - 0.281z$
3.0	$5.55 \cdot 10^{-17} - 1.19z$	$4.39 \cdot 10^{-16} - 0.32z$	$2.07 \cdot 10^{-11} + 1.19z$	$-9.47 \cdot 10^{-15} + 1.19z$
4.0	$6.94 \cdot 10^{-17} + 0.281z$	$1.88 \cdot 10^{-15} - 1.42z$	$2.04 \cdot 10^{-11} - 1.19z$	$6.07 \cdot 10^{-14} - 1.19z$
5.0	$6.94 \cdot 10^{-17} - 0.281z$	$9.89 \cdot 10^{-16} + 1.42z$	$5.49 \cdot 10^{10} + 5.28 \cdot 10^{10}z$	$8.85 \cdot 10^8 + 3.6 \cdot 10^{11}z$

Table 34: Case 1a (cpv MACI64 MR2022b), $r = 5$, Z4: zeros.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$-6.3 \cdot 10^{-5} - 4.98 \cdot 10^{-6}z$	$-6.3 \cdot 10^{-5} - 4.9 \cdot 10^{-6}z$
2.0	$5.55 \cdot 10^{-17} + 1.19z$	$5.2 \cdot 10^{-16} - 0.32z$	$-1.45 \cdot 10^{-4} + 0.629z$	$-1.45 \cdot 10^{-4} + 0.629z$
3.0	$5.55 \cdot 10^{-17} - 1.19z$	$7.03 \cdot 10^{-16} + 0.32z$	$-1.46 \cdot 10^{-4} - 0.629z$	$-1.46 \cdot 10^{-4} - 0.629z$
4.0	$6.94 \cdot 10^{-17} + 0.281z$	$4.37 \cdot 10^{-16} + 1.42z$	$-0.00156 + 2.66z$	$-0.00156 + 2.66z$
5.0	$6.94 \cdot 10^{-17} - 0.281z$	$-4.16 \cdot 10^{-16} - 1.42z$	$-0.00158 - 2.67z$	$-0.00158 - 2.67z$

Table 35: Case 1a (cpv PCWIN64 MR2025b), $r = 5$, Z4: zeros.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$-6.3 \cdot 10^{-5} - 4.98 \cdot 10^{-6}z$	$-6.3 \cdot 10^{-5} - 4.9 \cdot 10^{-6}z$
2.0	$5.55 \cdot 10^{-17} + 1.19z$	$5.2 \cdot 10^{-16} - 0.32z$	$-1.45 \cdot 10^{-4} + 0.629z$	$-1.45 \cdot 10^{-4} + 0.629z$
3.0	$5.55 \cdot 10^{-17} - 1.19z$	$7.03 \cdot 10^{-16} + 0.32z$	$-1.46 \cdot 10^{-4} - 0.629z$	$-1.46 \cdot 10^{-4} - 0.629z$
4.0	$6.94 \cdot 10^{-17} + 0.281z$	$4.37 \cdot 10^{-16} + 1.42z$	$-0.00156 + 2.66z$	$-0.00156 + 2.66z$
5.0	$6.94 \cdot 10^{-17} - 0.281z$	$-4.16 \cdot 10^{-16} - 1.42z$	$-0.00158 - 2.67z$	$-0.00158 - 2.67z$

Table 36: Case 1a (AlexReis PCWIN64 MR2025b), $r = 5$, Z4: zeros.

4.4 Poles and zeros visualisation

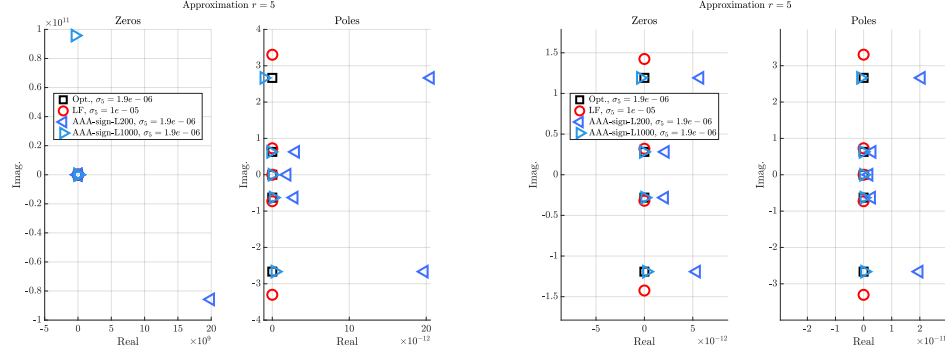


Figure 5: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

4.5 Approximation visualization

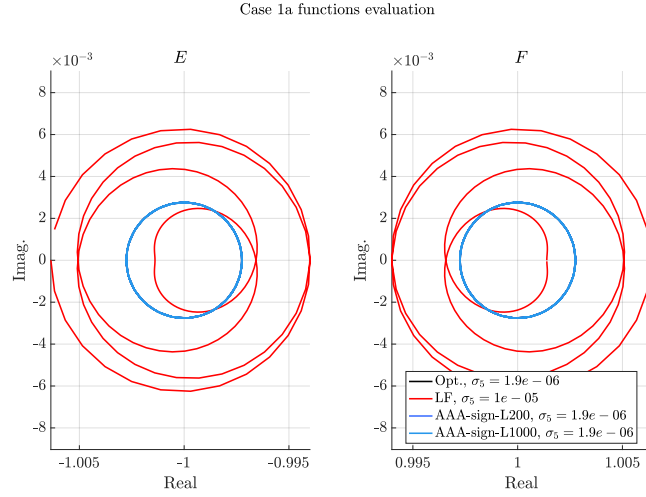


Figure 6: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4. E and F evaluations.

5 Approximation objective $r = 6$

5.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-1.2 \cdot 10^{-10}$	$-2.0 \cdot 10^{-10}$	$1.9 \cdot 10^{-12}$	$-8.0 \cdot 10^{-11}$
z^5	5.2	0	5.5	$-5.7 \cdot 10^{-15}$	5.2	$-6.3 \cdot 10^{-12}$	5.2	$-4.6 \cdot 10^{-12}$
z^4	0	0	$-2.3 \cdot 10^{-14}$	$-9.3 \cdot 10^{-15}$	$-1.4 \cdot 10^{-9}$	$-2.3 \cdot 10^{-9}$	$2.1 \cdot 10^{-11}$	$-9.2 \cdot 10^{-10}$
z^3	13.0	0	15.0	$-1.1 \cdot 10^{-14}$	13.0	$-4.7 \cdot 10^{-11}$	13.0	$-3.5 \cdot 10^{-11}$
z^2	0	0	$-4.2 \cdot 10^{-14}$	$-2.7 \cdot 10^{-14}$	$-1.1 \cdot 10^{-9}$	$-1.7 \cdot 10^{-9}$	$1.5 \cdot 10^{-11}$	$-7.0 \cdot 10^{-10}$
z	2.9	0	3.7	0	2.9	$-1.7 \cdot 10^{-11}$	2.9	$-1.3 \cdot 10^{-11}$
1.0	0	0	$-4.1 \cdot 10^{-15}$	$-3.2 \cdot 10^{-15}$	$-5.3 \cdot 10^{-11}$	$-8.7 \cdot 10^{-11}$	$7.8 \cdot 10^{-13}$	$-3.6 \cdot 10^{-11}$
z^6	1.0	0	1.0	0	1.0	0	1.0	0
z^5	0	0	$-6.9 \cdot 10^{-15}$	$-2.9 \cdot 10^{-15}$	$-6.5 \cdot 10^{-10}$	$-1.0 \cdot 10^{-9}$	$9.8 \cdot 10^{-12}$	$-4.2 \cdot 10^{-10}$
z^4	11.0	0	12.0	$-1.5 \cdot 10^{-14}$	11.0	$-2.7 \cdot 10^{-11}$	11.0	$-2.0 \cdot 10^{-11}$
z^3	0	0	$-4.3 \cdot 10^{-14}$	$-2.8 \cdot 10^{-14}$	$-1.6 \cdot 10^{-9}$	$-2.6 \cdot 10^{-9}$	$2.4 \cdot 10^{-11}$	$-1.1 \cdot 10^{-9}$
z^2	8.4	0	10.0	$-1.5 \cdot 10^{-15}$	8.4	$-4.0 \cdot 10^{-11}$	8.4	$-3.0 \cdot 10^{-11}$
z	0	0	$-2.0 \cdot 10^{-14}$	$-1.1 \cdot 10^{-14}$	$-3.6 \cdot 10^{-10}$	$-6.0 \cdot 10^{-10}$	$5.3 \cdot 10^{-12}$	$-2.5 \cdot 10^{-10}$
1.0	0.42	0	0.56	$-5.3 \cdot 10^{-16}$	0.42	$-3.0 \cdot 10^{-12}$	0.42	$-2.3 \cdot 10^{-12}$

Table 37: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-1.2 \cdot 10^{-10}$	$-6.5 \cdot 10^{-11}$	$-1.8 \cdot 10^{-12}$	$3.7 \cdot 10^{-10}$
z^5	5.2	0	5.5	$-8.5 \cdot 10^{-16}$	5.2	$4.5 \cdot 10^{-12}$	5.2	$5.7 \cdot 10^{-12}$
z^4	0	0	$-2.2 \cdot 10^{-14}$	$-7.0 \cdot 10^{-15}$	$-1.4 \cdot 10^{-9}$	$-7.1 \cdot 10^{-10}$	$-2.0 \cdot 10^{-11}$	$4.2 \cdot 10^{-9}$
z^3	13.0	0	15.0	$-2.3 \cdot 10^{-15}$	13.0	$3.3 \cdot 10^{-11}$	13.0	$4.2 \cdot 10^{-11}$
z^2	0	0	$-3.0 \cdot 10^{-14}$	$-6.1 \cdot 10^{-15}$	$-1.0 \cdot 10^{-9}$	$-5.2 \cdot 10^{-10}$	$-1.5 \cdot 10^{-11}$	$3.2 \cdot 10^{-9}$
z	2.9	0	3.7	$-4.8 \cdot 10^{-15}$	2.9	$1.2 \cdot 10^{-11}$	2.9	$1.6 \cdot 10^{-11}$
1.0	0	0	$-8.2 \cdot 10^{-16}$	$-2.6 \cdot 10^{-16}$	$-5.2 \cdot 10^{-11}$	$-2.5 \cdot 10^{-11}$	$-7.2 \cdot 10^{-13}$	$1.6 \cdot 10^{-10}$
z^6	1.0	0	1.0	$3.5 \cdot 10^{-17}$	1.0	$-3.6 \cdot 10^{-18}$	1.0	0
z^5	0	0	$-4.6 \cdot 10^{-15}$	$-1.1 \cdot 10^{-15}$	$-6.4 \cdot 10^{-10}$	$-3.3 \cdot 10^{-10}$	$-9.4 \cdot 10^{-12}$	$1.9 \cdot 10^{-9}$
z^4	11.0	0	12.0	$5.7 \cdot 10^{-16}$	11.0	$1.9 \cdot 10^{-11}$	11.0	$2.4 \cdot 10^{-11}$
z^3	0	0	$-3.3 \cdot 10^{-14}$	$-3.1 \cdot 10^{-15}$	$-1.6 \cdot 10^{-9}$	$-8.1 \cdot 10^{-10}$	$-2.3 \cdot 10^{-11}$	$4.9 \cdot 10^{-9}$
z^2	8.4	0	10.0	$-9.1 \cdot 10^{-15}$	8.4	$2.9 \cdot 10^{-11}$	8.4	$3.6 \cdot 10^{-11}$
z	0	0	$-1.2 \cdot 10^{-14}$	$-6.9 \cdot 10^{-15}$	$-3.6 \cdot 10^{-10}$	$-1.8 \cdot 10^{-10}$	$-5.0 \cdot 10^{-12}$	$1.1 \cdot 10^{-9}$
1.0	0.42	0	0.56	$3.5 \cdot 10^{-17}$	0.42	$2.1 \cdot 10^{-12}$	0.42	$2.7 \cdot 10^{-12}$

Table 38: Case 1a (cpv MACI64 MR2022b), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-4.4 \cdot 10^8$	$-2.2 \cdot 10^9$	$-6.7 \cdot 10^7$	$-6.4 \cdot 10^9$
z^5	5.2	0	5.5	$-4.0 \cdot 10^{-15}$	5.2	$-4.6 \cdot 10^{-4}$	5.2	$5.3 \cdot 10^{-4}$
z^4	0	0	$-2.4 \cdot 10^{-16}$	$1.1 \cdot 10^{-14}$	$-5.0 \cdot 10^9$	$-2.5 \cdot 10^{10}$	$-7.5 \cdot 10^8$	$-7.2 \cdot 10^{10}$
z^3	13.0	0	15.0	$-1.9 \cdot 10^{-14}$	13.0	$-3.2 \cdot 10^{-3}$	13.0	$3.8 \cdot 10^{-3}$
z^2	0	0	$-1.8 \cdot 10^{-15}$	$8.2 \cdot 10^{-15}$	$-3.7 \cdot 10^9$	$-1.9 \cdot 10^{10}$	$-5.6 \cdot 10^8$	$-5.4 \cdot 10^{10}$
z	2.9	0	3.7	$-5.4 \cdot 10^{-15}$	2.9	$-1.3 \cdot 10^{-3}$	2.9	$1.7 \cdot 10^{-3}$
1.0	0	0	$-4.3 \cdot 10^{-16}$	$3.7 \cdot 10^{-16}$	$-1.9 \cdot 10^8$	$-9.4 \cdot 10^8$	$-2.8 \cdot 10^7$	$-2.7 \cdot 10^9$
z^6	1.0	0	1.0	0	1.0	0	1.0	0
z^5	0	0	$1.8 \cdot 10^{-15}$	$-9.9 \cdot 10^{-17}$	$-2.3 \cdot 10^9$	$-1.2 \cdot 10^{10}$	$-3.5 \cdot 10^8$	$-3.3 \cdot 10^{10}$
z^4	11.0	0	12.0	$-1.1 \cdot 10^{-14}$	11.0	$-1.9 \cdot 10^{-3}$	11.0	$2.1 \cdot 10^{-3}$
z^3	0	0	$-4.6 \cdot 10^{-15}$	$1.6 \cdot 10^{-14}$	$-5.8 \cdot 10^9$	$-2.9 \cdot 10^{10}$	$-8.7 \cdot 10^8$	$-8.3 \cdot 10^{10}$
z^2	8.4	0	10.0	$-1.8 \cdot 10^{-14}$	8.5	$-2.8 \cdot 10^{-3}$	8.4	$3.6 \cdot 10^{-3}$
z	0	0	$7.3 \cdot 10^{-19}$	$3.8 \cdot 10^{-15}$	$-1.3 \cdot 10^9$	$-6.5 \cdot 10^9$	$-2.0 \cdot 10^8$	$-1.9 \cdot 10^{10}$
1.0	0.42	0	0.56	$1.4 \cdot 10^{-16}$	0.42	$-2.4 \cdot 10^{-4}$	0.42	$2.7 \cdot 10^{-4}$

Table 39: Case 1a (cpv PCWIN64 MR2025b), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-4.4 \cdot 10^8$	$-2.2 \cdot 10^9$	$-6.7 \cdot 10^7$	$-6.4 \cdot 10^9$
z^5	5.2	0	5.5	$-4.0 \cdot 10^{-15}$	5.2	$-4.6 \cdot 10^{-4}$	5.2	$5.3 \cdot 10^{-4}$
z^4	0	0	$-2.4 \cdot 10^{-16}$	$1.1 \cdot 10^{-14}$	$-5.0 \cdot 10^9$	$-2.5 \cdot 10^{10}$	$-7.5 \cdot 10^8$	$-7.2 \cdot 10^{10}$
z^3	13.0	0	15.0	$-1.9 \cdot 10^{-14}$	13.0	$-3.2 \cdot 10^{-3}$	13.0	$3.8 \cdot 10^{-3}$
z^2	0	0	$-1.8 \cdot 10^{-15}$	$8.2 \cdot 10^{-15}$	$-3.7 \cdot 10^9$	$-1.9 \cdot 10^{10}$	$-5.6 \cdot 10^8$	$-5.4 \cdot 10^{10}$
z	2.9	0	3.7	$-5.4 \cdot 10^{-15}$	2.9	$-1.3 \cdot 10^{-3}$	2.9	$1.7 \cdot 10^{-3}$
1.0	0	0	$-4.3 \cdot 10^{-16}$	$3.7 \cdot 10^{-16}$	$-1.9 \cdot 10^8$	$-9.4 \cdot 10^8$	$-2.8 \cdot 10^7$	$-2.7 \cdot 10^9$
z^6	1.0	0	1.0	0	1.0	0	1.0	0
z^5	0	0	$1.8 \cdot 10^{-15}$	$-9.9 \cdot 10^{-17}$	$-2.3 \cdot 10^9$	$-1.2 \cdot 10^{10}$	$-3.5 \cdot 10^8$	$-3.3 \cdot 10^{10}$
z^4	11.0	0	12.0	$-1.1 \cdot 10^{-14}$	11.0	$-1.9 \cdot 10^{-3}$	11.0	$2.1 \cdot 10^{-3}$
z^3	0	0	$-4.6 \cdot 10^{-15}$	$1.6 \cdot 10^{-14}$	$-5.8 \cdot 10^9$	$-2.9 \cdot 10^{10}$	$-8.7 \cdot 10^8$	$-8.3 \cdot 10^{10}$
z^2	8.4	0	10.0	$-1.8 \cdot 10^{-14}$	8.5	$-2.8 \cdot 10^{-3}$	8.4	$3.6 \cdot 10^{-3}$
z	0	0	$7.3 \cdot 10^{-19}$	$3.8 \cdot 10^{-15}$	$-1.3 \cdot 10^9$	$-6.5 \cdot 10^9$	$-2.0 \cdot 10^8$	$-1.9 \cdot 10^{10}$
1.0	0.42	0	0.56	$1.4 \cdot 10^{-16}$	0.42	$-2.4 \cdot 10^{-4}$	0.42	$2.7 \cdot 10^{-4}$

Table 40: Case 1a (AlexReis PCWIN64 MR2025b), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

5.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$5.55 \cdot 10^{-17} + 3.23i$	$8.54 \cdot 10^{-16} - 0.243i$	$1.9 \cdot 10^{-11} - 0.232i$	$-4.84 \cdot 10^{-13} - 0.232i$
2.0	$5.55 \cdot 10^{-17} - 3.23i$	$1.13 \cdot 10^{-15} + 0.243i$	$1.96 \cdot 10^{-11} + 0.232i$	$-7.24 \cdot 10^{-14} + 0.232i$
3.0	$-1.39 \cdot 10^{-16} + 0.866i$	$1.4 \cdot 10^{-15} - 0.908i$	$3.5 \cdot 10^{-11} - 0.866i$	$-1.3 \cdot 10^{-12} - 0.866i$
4.0	$-1.39 \cdot 10^{-16} - 0.866i$	$1.03 \cdot 10^{-16} + 0.908i$	$3.7 \cdot 10^{-11} + 0.866i$	$2.56 \cdot 10^{-13} + 0.866i$
5.0	$1.21 \cdot 10^{-17} + 0.232i$	$-4.06 \cdot 10^{-16} - 3.4i$	$2.65 \cdot 10^{-10} - 3.23i$	$-7.0 \cdot 10^{-12} - 3.23i$
6.0	$1.21 \cdot 10^{-17} - 0.232i$	$3.94 \cdot 10^{-15} + 3.4i$	$2.72 \cdot 10^{-10} + 3.23i$	$-1.24 \cdot 10^{-12} + 3.23i$

Table 41: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4: poles.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$1.39 \cdot 10^{-16} + 3.23i$	$7.01 \cdot 10^{-16} - 0.243i$	$1.93 \cdot 10^{-11} - 0.232i$	$2.13 \cdot 10^{-13} + 0.232i$
2.0	$1.39 \cdot 10^{-16} - 3.23i$	$3.33 \cdot 10^{-16} + 0.243i$	$1.89 \cdot 10^{-11} + 0.232i$	$5.06 \cdot 10^{-13} - 0.232i$
3.0	$-2.78 \cdot 10^{-17} + 0.866i$	$1.39 \cdot 10^{-16} - 0.908i$	$3.64 \cdot 10^{-11} - 0.866i$	$-3.95 \cdot 10^{-13} + 0.866i$
4.0	$-2.78 \cdot 10^{-17} - 0.866i$	$1.34 \cdot 10^{-15} + 0.908i$	$3.49 \cdot 10^{-11} + 0.866i$	$1.43 \cdot 10^{-12} - 0.866i$
5.0	$1.73 \cdot 10^{-18} + 0.232i$	$1.83 \cdot 10^{-15} - 3.4i$	$2.69 \cdot 10^{-10} - 3.23i$	$4.22 \cdot 10^{-13} + 3.23i$
6.0	$1.73 \cdot 10^{-18} - 0.232i$	$3.22 \cdot 10^{-16} + 3.4i$	$2.63 \cdot 10^{-10} + 3.23i$	$7.46 \cdot 10^{-12} - 3.23i$

Table 42: Case 1a (cpv MACI64 MR2022b), $r = 6$, Z4: poles.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$5.55 \cdot 10^{-17} + 3.23i$	$-2.65 \cdot 10^{-16} + 0.243i$	$1.25 \cdot 10^{-11} - 6.27 \cdot 10^{-11}i$	$2.52 \cdot 10^{-13} - 2.27 \cdot 10^{-11}i$
2.0	$5.55 \cdot 10^{-17} - 3.23i$	$2.35 \cdot 10^{-16} - 0.243i$	$1.66 \cdot 10^{-11} + 0.5i$	$3.42 \cdot 10^{-13} + 0.5i$
3.0	$-1.39 \cdot 10^{-16} + 0.866i$	$7.55 \cdot 10^{-16} + 0.908i$	$1.68 \cdot 10^{-11} - 0.5i$	$3.23 \cdot 10^{-13} - 0.5i$
4.0	$-1.39 \cdot 10^{-16} - 0.866i$	$-1.43 \cdot 10^{-16} - 0.908i$	$4.96 \cdot 10^{-11} + 1.5i$	$9.76 \cdot 10^{-13} + 1.5i$
5.0	$1.21 \cdot 10^{-17} + 0.232i$	$3.11 \cdot 10^{-17} + 3.4i$	$5.05 \cdot 10^{-11} - 1.5i$	$9.48 \cdot 10^{-13} - 1.5i$
6.0	$1.21 \cdot 10^{-17} - 0.232i$	$-2.45 \cdot 10^{-15} - 3.4i$	$2.31 \cdot 10^9 + 1.15 \cdot 10^{10}i$	$3.47 \cdot 10^8 + 3.3 \cdot 10^{10}i$

Table 43: Case 1a (cpv PCWIN64 MR2025b), $r = 6$, Z4: poles.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$5.55 \cdot 10^{-17} + 3.23_z$	$-2.65 \cdot 10^{-16} + 0.243_z$	$1.25 \cdot 10^{-11} - 6.27 \cdot 10^{-11}_z$	$2.52 \cdot 10^{-13} - 2.27 \cdot 10^{-11}_z$
2.0	$5.55 \cdot 10^{-17} - 3.23_z$	$2.35 \cdot 10^{-16} - 0.243_z$	$1.66 \cdot 10^{-11} + 0.5_z$	$3.42 \cdot 10^{-13} + 0.5_z$
3.0	$-1.39 \cdot 10^{-16} + 0.866_z$	$7.55 \cdot 10^{-16} + 0.908_z$	$1.68 \cdot 10^{-11} - 0.5_z$	$3.23 \cdot 10^{-13} - 0.5_z$
4.0	$-1.39 \cdot 10^{-16} - 0.866_z$	$-1.43 \cdot 10^{-16} - 0.908_z$	$4.96 \cdot 10^{-11} + 1.5_z$	$9.76 \cdot 10^{-13} + 1.5_z$
5.0	$1.21 \cdot 10^{-17} + 0.232_z$	$3.11 \cdot 10^{-17} + 3.4_z$	$5.05 \cdot 10^{-11} - 1.5_z$	$9.48 \cdot 10^{-13} - 1.5_z$
6.0	$1.21 \cdot 10^{-17} - 0.232_z$	$-2.45 \cdot 10^{-15} - 3.4_z$	$2.31 \cdot 10^9 + 1.15 \cdot 10^{10}_z$	$3.47 \cdot 10^8 + 3.3 \cdot 10^{10}_z$

Table 44: Case 1a (AlexReis PCWIN64 MR2025b), $r = 6$, Z4: poles.

5.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.8 \cdot 10^{-11} + 2.97 \cdot 10^{-11}_z$	$-2.66 \cdot 10^{-13} + 1.23 \cdot 10^{-11}_z$
2.0	0	$1.1 \cdot 10^{-15} + 7.9 \cdot 10^{-16}_z$	$2.34 \cdot 10^{-11} - 0.5_z$	$-7.85 \cdot 10^{-13} - 0.5_z$
3.0	1.5_z	$1.13 \cdot 10^{-15} - 0.523_z$	$2.46 \cdot 10^{-11} + 0.5_z$	$1.08 \cdot 10^{-13} + 0.5_z$
4.0	-1.5_z	$6.11 \cdot 10^{-16} + 0.523_z$	$7.02 \cdot 10^{-11} - 1.5_z$	$-2.43 \cdot 10^{-12} - 1.5_z$
5.0	0.5_z	$8.46 \cdot 10^{-16} - 1.57_z$	$7.38 \cdot 10^{-11} + 1.5_z$	$2.62 \cdot 10^{-13} + 1.5_z$
6.0	-0.5_z	$5.09 \cdot 10^{-16} + 1.57_z$	$1.18 \cdot 10^{10} - 1.88 \cdot 10^{10}_z$	$-1.58 \cdot 10^9 - 6.51 \cdot 10^{10}_z$

Table 45: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4: zeros.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.78 \cdot 10^{-11} + 8.65 \cdot 10^{-12}_z$	$2.47 \cdot 10^{-13} - 5.44 \cdot 10^{-11}_z$
2.0	0	$2.1 \cdot 10^{-16} + 1.53 \cdot 10^{-16}_z$	$2.42 \cdot 10^{-11} - 0.5_z$	$-1.87 \cdot 10^{-13} + 0.5_z$
3.0	1.5_z	$5.51 \cdot 10^{-16} - 0.523_z$	$2.33 \cdot 10^{-11} + 0.5_z$	$8.55 \cdot 10^{-13} - 0.5_z$
4.0	-1.5_z	$1.27 \cdot 10^{-15} + 0.523_z$	$7.25 \cdot 10^{-11} - 1.5_z$	$-5.55 \cdot 10^{-13} + 1.5_z$
5.0	0.5_z	$9.91 \cdot 10^{-16} - 1.57_z$	$7.0 \cdot 10^{-11} + 1.5_z$	$2.66 \cdot 10^{-12} - 1.5_z$
6.0	-0.5_z	$8.9 \cdot 10^{-16} + 1.57_z$	$3.3 \cdot 10^{10} - 1.73 \cdot 10^{10}_z$	$6.91 \cdot 10^7 + 1.4 \cdot 10^{10}_z$

Table 46: Case 1a (cpv MACI64 MR2022b), $r = 6$, Z4: zeros.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.34 \cdot 10^{-11} + 0.232_z$	$2.75 \cdot 10^{-13} + 0.232_z$
2.0	0	$1.18 \cdot 10^{-16} - 5.87 \cdot 10^{-17}_z$	$1.35 \cdot 10^{-11} - 0.232_z$	$2.66 \cdot 10^{-13} - 0.232_z$
3.0	1.5_z	$3.18 \cdot 10^{-17} + 0.523_z$	$2.48 \cdot 10^{-11} + 0.866_z$	$5.0 \cdot 10^{-13} + 0.866_z$
4.0	-1.5_z	$-1.45 \cdot 10^{-17} - 0.523_z$	$2.53 \cdot 10^{-11} - 0.866_z$	$4.74 \cdot 10^{-13} - 0.866_z$
5.0	0.5_z	$4.28 \cdot 10^{-16} + 1.57_z$	$1.86 \cdot 10^{-10} + 3.23_z$	$3.6 \cdot 10^{-12} + 3.23_z$
6.0	-0.5_z	$-5.15 \cdot 10^{-16} - 1.57_z$	$1.88 \cdot 10^{-10} - 3.23_z$	$3.54 \cdot 10^{-12} - 3.23_z$

Table 47: Case 1a (cpv PCWIN64 MR2025b), $r = 6$, Z4: zeros.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.34 \cdot 10^{-11} + 0.232_z$	$2.75 \cdot 10^{-13} + 0.232_z$
2.0	0	$1.18 \cdot 10^{-16} - 5.87 \cdot 10^{-17}_z$	$1.35 \cdot 10^{-11} - 0.232_z$	$2.66 \cdot 10^{-13} - 0.232_z$
3.0	1.5_z	$3.18 \cdot 10^{-17} + 0.523_z$	$2.48 \cdot 10^{-11} + 0.866_z$	$5.0 \cdot 10^{-13} + 0.866_z$
4.0	-1.5_z	$-1.45 \cdot 10^{-17} - 0.523_z$	$2.53 \cdot 10^{-11} - 0.866_z$	$4.74 \cdot 10^{-13} - 0.866_z$
5.0	0.5_z	$4.28 \cdot 10^{-16} + 1.57_z$	$1.86 \cdot 10^{-10} + 3.23_z$	$3.6 \cdot 10^{-12} + 3.23_z$
6.0	-0.5_z	$-5.15 \cdot 10^{-16} - 1.57_z$	$1.88 \cdot 10^{-10} - 3.23_z$	$3.54 \cdot 10^{-12} - 3.23_z$

Table 48: Case 1a (AlexReis PCWIN64 MR2025b), $r = 6$, Z4: zeros.

5.4 Poles and zeros visualisation

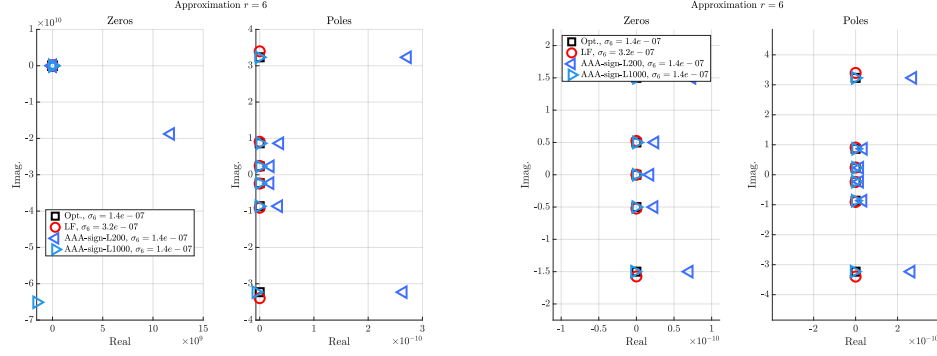


Figure 7: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

5.5 Approximation visualization

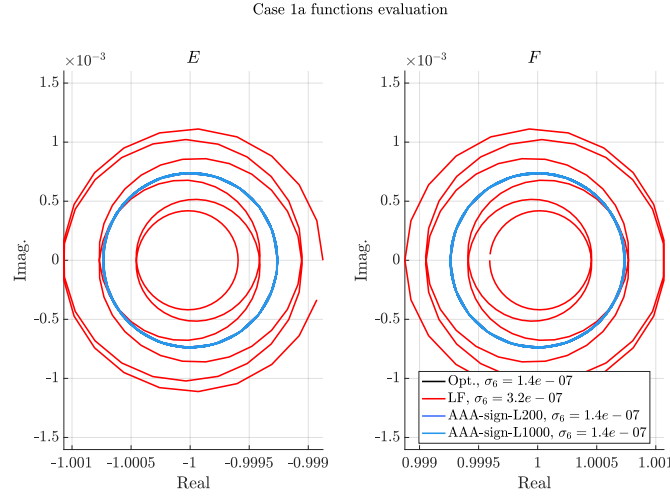


Figure 8: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4. E and F evaluations.

6 Approximation objective $r = 7$

6.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$		AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	$-2.5 \cdot 10^6$	$-1.8 \cdot 10^8$	$-5.6 \cdot 10^5$	$-2.6 \cdot 10^8$
z^6	6.1	0	7.2	$-1.9 \cdot 10^{-14}$	6.0	$1.1 \cdot 10^{-3}$	6.0	$7.4 \cdot 10^{-4}$
z^5	0	0	$-1.5 \cdot 10^{-13}$	$-5.8 \cdot 10^{-15}$	$-4.0 \cdot 10^7$	$-2.8 \cdot 10^9$	$-8.8 \cdot 10^6$	$-4.0 \cdot 10^9$
z^4	23.0	0	36.0	$-7.5 \cdot 10^{-14}$	22.0	0.012	22.0	$8.5 \cdot 10^{-3}$
z^3	0	0	$-3.2 \cdot 10^{-13}$	$-4.4 \cdot 10^{-14}$	$-5.0 \cdot 10^7$	$-3.5 \cdot 10^9$	$-1.1 \cdot 10^7$	$-5.0 \cdot 10^9$
z^2	10.0	0	20.0	$-1.2 \cdot 10^{-13}$	10.0	$8.7 \cdot 10^{-3}$	10.0	$6.6 \cdot 10^{-3}$
z	0	0	$-5.3 \cdot 10^{-14}$	$-3.8 \cdot 10^{-14}$	$-7.5 \cdot 10^6$	$-5.2 \cdot 10^8$	$-1.6 \cdot 10^6$	$-7.6 \cdot 10^8$
1.0	0.37	0	0.85	$-1.2 \cdot 10^{-14}$	0.35	$4.3 \cdot 10^{-4}$	0.35	$3.4 \cdot 10^{-4}$
z^7	1.0	0	1.0	0	1.0	$1.7 \cdot 10^{-17}$	1.0	0
z^6	0	0	$-2.9 \cdot 10^{-14}$	$-1.6 \cdot 10^{-14}$	$-1.5 \cdot 10^7$	$-1.1 \cdot 10^9$	$-3.4 \cdot 10^6$	$-1.6 \cdot 10^9$
z^5	16.0	0	22.0	$-5.2 \cdot 10^{-14}$	16.0	$5.5 \cdot 10^{-3}$	16.0	$3.9 \cdot 10^{-3}$
z^4	0	0	$-3.1 \cdot 10^{-13}$	$1.8 \cdot 10^{-16}$	$-5.8 \cdot 10^7$	$-4.0 \cdot 10^9$	$-1.3 \cdot 10^7$	$-5.8 \cdot 10^9$
z^3	20.0	0	35.0	$-1.1 \cdot 10^{-13}$	19.0	0.014	19.0	0.01
z^2	0	0	$-1.8 \cdot 10^{-13}$	$-7.8 \cdot 10^{-14}$	$-2.6 \cdot 10^7$	$-1.8 \cdot 10^9$	$-5.7 \cdot 10^6$	$-2.6 \cdot 10^9$
z	3.0	0	6.3	$-6.4 \cdot 10^{-14}$	2.9	$3.0 \cdot 10^{-3}$	2.9	$2.3 \cdot 10^{-3}$
1.0	0	0	$-6.8 \cdot 10^{-15}$	$-3.7 \cdot 10^{-16}$	$-9.3 \cdot 10^5$	$-6.5 \cdot 10^7$	$-2.0 \cdot 10^5$	$-9.4 \cdot 10^7$

Table 49: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 1.1 \cdot 10^{-08}$		AAA-sign-L1000, $\sigma_7 = 9.9 \cdot 10^{-09}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	-93.0	0.99	$1.5 \cdot 10^{-4}$	$-7.1 \cdot 10^{-5}$
z^6	6.1	0	7.2	$1.3 \cdot 10^{-14}$	4.5	$8.6 \cdot 10^{-3}$	6.1	$-1.5 \cdot 10^{-5}$
z^5	0	0	$-2.3 \cdot 10^{-15}$	$5.2 \cdot 10^{-14}$	$-1.5 \cdot 10^3$	16.0	$-3.9 \cdot 10^{-4}$	$-1.0 \cdot 10^{-3}$
z^4	23.0	0	36.0	$9.9 \cdot 10^{-14}$	5.9	0.097	23.0	$-1.1 \cdot 10^{-4}$
z^3	0	0	$6.8 \cdot 10^{-15}$	$1.2 \cdot 10^{-13}$	$-1.8 \cdot 10^3$	19.0	$-3.7 \cdot 10^{-3}$	$-1.3 \cdot 10^{-3}$
z^2	10.0	0	20.0	$1.1 \cdot 10^{-13}$	-2.2	0.074	10.0	$-4.2 \cdot 10^{-5}$
z	0	0	$1.1 \cdot 10^{-15}$	$3.6 \cdot 10^{-14}$	-270.0	2.9	$-9.8 \cdot 10^{-4}$	$-1.9 \cdot 10^{-4}$
1.0	0.37	0	0.85	$3.8 \cdot 10^{-15}$	-0.25	$3.8 \cdot 10^{-3}$	0.37	$5.3 \cdot 10^{-7}$
z^7	1.0	0	1.0	$5.2 \cdot 10^{-17}$	1.0	0	1.0	0
z^6	0	0	$-4.1 \cdot 10^{-15}$	$9.4 \cdot 10^{-15}$	-560.0	6.0	$3.7 \cdot 10^{-4}$	$-4.1 \cdot 10^{-4}$
z^5	16.0	0	22.0	$5.5 \cdot 10^{-14}$	7.9	0.044	16.0	$-6.1 \cdot 10^{-5}$
z^4	0	0	$1.6 \cdot 10^{-14}$	$1.3 \cdot 10^{-13}$	$-2.1 \cdot 10^3$	22.0	$-2.4 \cdot 10^{-3}$	$-1.5 \cdot 10^{-3}$
z^3	20.0	0	35.0	$1.3 \cdot 10^{-13}$	0.34	0.11	20.0	$-9.6 \cdot 10^{-5}$
z^2	0	0	$1.6 \cdot 10^{-15}$	$7.5 \cdot 10^{-14}$	-940.0	10.0	$-2.7 \cdot 10^{-3}$	$-6.5 \cdot 10^{-4}$
z	3.0	0	6.3	$4.2 \cdot 10^{-14}$	-1.3	0.026	3.0	$-5.5 \cdot 10^{-6}$
1.0	0	0	$-4.0 \cdot 10^{-16}$	$4.8 \cdot 10^{-15}$	-33.0	0.36	$-1.4 \cdot 10^{-4}$	$-2.5 \cdot 10^{-5}$

Table 50: Case 1a (cpv MACI64 MR2022b), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$		AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	$-2.0 \cdot 10^6$	$-1.5 \cdot 10^8$	$6.3 \cdot 10^4$	$-1.7 \cdot 10^8$
z^6	6.1	0	7.2	$-1.5 \cdot 10^{-14}$	6.1	$-3.2 \cdot 10^{-5}$	6.1	$4.4 \cdot 10^{-4}$
z^5	0	0	$-4.4 \cdot 10^{-14}$	$1.0 \cdot 10^{-13}$	$-3.1 \cdot 10^7$	$-2.3 \cdot 10^9$	$9.9 \cdot 10^5$	$-2.7 \cdot 10^9$
z^4	23.0	0	36.0	$-1.1 \cdot 10^{-13}$	23.0	$-3.5 \cdot 10^{-4}$	23.0	$4.4 \cdot 10^{-3}$
z^3	0	0	$-4.8 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-3.9 \cdot 10^7$	$-2.9 \cdot 10^9$	$1.2 \cdot 10^6$	$-3.4 \cdot 10^9$
z^2	10.0	0	20.0	$-6.7 \cdot 10^{-14}$	10.0	$-3.1 \cdot 10^{-4}$	10.0	$3.0 \cdot 10^{-3}$
z	0	0	$-2.2 \cdot 10^{-15}$	$4.1 \cdot 10^{-14}$	$-5.8 \cdot 10^6$	$-4.4 \cdot 10^8$	$1.9 \cdot 10^5$	$-5.1 \cdot 10^8$
1.0	0.37	0	0.85	$-9.7 \cdot 10^{-15}$	0.37	$-1.6 \cdot 10^{-5}$	0.37	$1.3 \cdot 10^{-4}$
z^7	1.0	0	1.0	$-5.2 \cdot 10^{-17}$	1.0	0	1.0	0
z^6	0	0	$-7.7 \cdot 10^{-15}$	$1.5 \cdot 10^{-14}$	$-1.2 \cdot 10^7$	$-9.0 \cdot 10^8$	$3.8 \cdot 10^5$	$-1.0 \cdot 10^9$
z^5	16.0	0	22.0	$-6.0 \cdot 10^{-14}$	16.0	$-1.7 \cdot 10^{-4}$	16.0	$2.1 \cdot 10^{-3}$
z^4	0	0	$-7.1 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-4.5 \cdot 10^7$	$-3.4 \cdot 10^9$	$1.4 \cdot 10^6$	$-3.9 \cdot 10^9$
z^3	20.0	0	35.0	$-1.1 \cdot 10^{-13}$	20.0	$-4.2 \cdot 10^{-4}$	20.0	$4.9 \cdot 10^{-3}$
z^2	0	0	$-1.6 \cdot 10^{-14}$	$1.2 \cdot 10^{-13}$	$-2.0 \cdot 10^7$	$-1.5 \cdot 10^9$	$6.4 \cdot 10^5$	$-1.7 \cdot 10^9$
z	3.0	0	6.3	$-3.5 \cdot 10^{-14}$	3.0	$-1.1 \cdot 10^{-4}$	3.0	$9.7 \cdot 10^{-4}$
1.0	0	0	$1.0 \cdot 10^{-15}$	$7.7 \cdot 10^{-15}$	$-7.2 \cdot 10^5$	$-5.4 \cdot 10^7$	$2.3 \cdot 10^4$	$-6.2 \cdot 10^7$

Table 51: Case 1a (cpv PCWIN64 MR2025b), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$		AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	$-2.0 \cdot 10^6$	$-1.5 \cdot 10^8$	$6.3 \cdot 10^4$	$-1.7 \cdot 10^8$
z^6	6.1	0	7.2	$-1.5 \cdot 10^{-14}$	6.1	$-3.2 \cdot 10^{-5}$	6.1	$4.4 \cdot 10^{-4}$
z^5	0	0	$-4.4 \cdot 10^{-14}$	$1.0 \cdot 10^{-13}$	$-3.1 \cdot 10^7$	$-2.3 \cdot 10^9$	$9.9 \cdot 10^5$	$-2.7 \cdot 10^9$
z^4	23.0	0	36.0	$-1.1 \cdot 10^{-13}$	23.0	$-3.5 \cdot 10^{-4}$	23.0	$4.4 \cdot 10^{-3}$
z^3	0	0	$-4.8 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-3.9 \cdot 10^7$	$-2.9 \cdot 10^9$	$1.2 \cdot 10^6$	$-3.4 \cdot 10^9$
z^2	10.0	0	20.0	$-6.7 \cdot 10^{-14}$	10.0	$-3.1 \cdot 10^{-4}$	10.0	$3.0 \cdot 10^{-3}$
z	0	0	$-2.2 \cdot 10^{-15}$	$4.1 \cdot 10^{-14}$	$-5.8 \cdot 10^6$	$-4.4 \cdot 10^8$	$1.9 \cdot 10^5$	$-5.1 \cdot 10^8$
1.0	0.37	0	0.85	$-9.7 \cdot 10^{-15}$	0.37	$-1.6 \cdot 10^{-5}$	0.37	$1.3 \cdot 10^{-4}$
z^7	1.0	0	1.0	$-5.2 \cdot 10^{-17}$	1.0	0	1.0	0
z^6	0	0	$-7.7 \cdot 10^{-15}$	$1.5 \cdot 10^{-14}$	$-1.2 \cdot 10^7$	$-9.0 \cdot 10^8$	$3.8 \cdot 10^5$	$-1.0 \cdot 10^9$
z^5	16.0	0	22.0	$-6.0 \cdot 10^{-14}$	16.0	$-1.7 \cdot 10^{-4}$	16.0	$2.1 \cdot 10^{-3}$
z^4	0	0	$-7.1 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-4.5 \cdot 10^7$	$-3.4 \cdot 10^9$	$1.4 \cdot 10^6$	$-3.9 \cdot 10^9$
z^3	20.0	0	35.0	$-1.1 \cdot 10^{-13}$	20.0	$-4.2 \cdot 10^{-4}$	20.0	$4.9 \cdot 10^{-3}$
z^2	0	0	$-1.6 \cdot 10^{-14}$	$1.2 \cdot 10^{-13}$	$-2.0 \cdot 10^7$	$-1.5 \cdot 10^9$	$6.4 \cdot 10^5$	$-1.7 \cdot 10^9$
z	3.0	0	6.3	$-3.5 \cdot 10^{-14}$	3.0	$-1.1 \cdot 10^{-4}$	3.0	$9.7 \cdot 10^{-4}$
1.0	0	0	$1.0 \cdot 10^{-15}$	$7.7 \cdot 10^{-15}$	$-7.2 \cdot 10^5$	$-5.4 \cdot 10^7$	$2.3 \cdot 10^4$	$-6.2 \cdot 10^7$

Table 52: Case 1a (AlexReis PCWIN64 MR2025b), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

6.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	0	$1.08 \cdot 10^{-15} + 5.86 \cdot 10^{-17}i$	$1.63 \cdot 10^{-11} + 0.198i$	$5.75 \cdot 10^{-12} + 0.198i$
2.0	$-7.63 \cdot 10^{-17} + 3.79i$	$-1.46 \cdot 10^{-16} - 0.458i$	$5.67 \cdot 10^{-12} - 0.198i$	$-2.73 \cdot 10^{-12} - 0.198i$
3.0	$-7.63 \cdot 10^{-17} - 3.79i$	$3.4 \cdot 10^{-15} + 0.458i$	$3.53 \cdot 10^{-11} + 0.691i$	$1.68 \cdot 10^{-11} + 0.691i$
4.0	$2.78 \cdot 10^{-17} + 1.09i$	$3.32 \cdot 10^{-15} + 1.23i$	$-1.59 \cdot 10^{-12} - 0.691i$	$-1.26 \cdot 10^{-11} - 0.691i$
5.0	$2.78 \cdot 10^{-17} - 1.09i$	$5.36 \cdot 10^{-15} - 1.23i$	$1.01 \cdot 10^{-10} + 1.8i$	$4.36 \cdot 10^{-11} + 1.8i$
6.0	$7.63 \cdot 10^{-17} + 0.417i$	$1.82 \cdot 10^{-15} - 4.47i$	$6.14 \cdot 10^{-12} - 1.8i$	$-3.18 \cdot 10^{-11} - 1.8i$
7.0	$7.63 \cdot 10^{-17} - 0.417i$	$1.53 \cdot 10^{-14} + 4.47i$	$1.54 \cdot 10^7 + 1.07 \cdot 10^9i$	$3.38 \cdot 10^6 + 1.55 \cdot 10^9i$

Table 53: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4: poles.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 1.1 \cdot 10^{-08}$	AAA-sign-L1000, $\sigma_7 = 9.9 \cdot 10^{-09}$
1.0	0	$6.4 \cdot 10^{-17} - 7.54 \cdot 10^{-16}z$	$-8.57 \cdot 10^{-4} - 0.197z$	$4.89 \cdot 10^{-5} + 8.52 \cdot 10^{-6}z$
2.0	$-1.46 \cdot 10^{-16} + 3.79z$	$-7.57 \cdot 10^{-16} + 0.458z$	$-8.53 \cdot 10^{-4} + 0.197z$	$4.97 \cdot 10^{-5} - 0.417z$
3.0	$-1.46 \cdot 10^{-16} - 3.79z$	$6.74 \cdot 10^{-16} - 0.458z$	$-1.38 \cdot 10^{-4} - 0.689z$	$4.82 \cdot 10^{-5} + 0.417z$
4.0	$2.78 \cdot 10^{-17} + 1.09z$	$-7.92 \cdot 10^{-16} + 1.23z$	$-1.19 \cdot 10^{-4} + 0.689z$	$3.1 \cdot 10^{-5} - 1.09z$
5.0	$2.78 \cdot 10^{-17} - 1.09z$	$-2.55 \cdot 10^{-16} - 1.23z$	$0.00463 - 1.79z$	$3.26 \cdot 10^{-5} + 1.09z$
6.0	$-3.47 \cdot 10^{-17} + 0.417z$	$-4.93 \cdot 10^{-15} + 4.47z$	$0.0047 + 1.79z$	$-2.98 \cdot 10^{-4} - 3.79z$
7.0	$-3.47 \cdot 10^{-17} - 0.417z$	$1.15 \cdot 10^{-14} - 4.47z$	$565.0 - 6.0z$	$-2.83 \cdot 10^{-4} + 3.8z$

Table 54: Case 1a (cpv MACI64 MR2022b), $r = 7$, Z4: poles.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	0	$-1.61 \cdot 10^{-16} - 1.23 \cdot 10^{-15}z$	$1.16 \cdot 10^{-11} + 0.198z$	$-2.97 \cdot 10^{-12} + 0.198z$
2.0	$-7.63 \cdot 10^{-17} + 3.79z$	$6.42 \cdot 10^{-16} + 0.458z$	$1.16 \cdot 10^{-11} - 0.198z$	$2.94 \cdot 10^{-12} - 0.198z$
3.0	$-7.63 \cdot 10^{-17} - 3.79z$	$-4.86 \cdot 10^{-16} - 0.458z$	$1.81 \cdot 10^{-11} + 0.691z$	$-1.0 \cdot 10^{-11} + 0.691z$
4.0	$2.78 \cdot 10^{-17} + 1.09z$	$1.24 \cdot 10^{-15} + 1.23z$	$1.81 \cdot 10^{-11} - 0.691z$	$9.84 \cdot 10^{-12} - 0.691z$
5.0	$2.78 \cdot 10^{-17} - 1.09z$	$1.68 \cdot 10^{-15} - 1.23z$	$5.9 \cdot 10^{-11} + 1.8z$	$-2.56 \cdot 10^{-11} + 1.8z$
6.0	$7.63 \cdot 10^{-17} + 0.417z$	$8.32 \cdot 10^{-15} + 4.47z$	$5.86 \cdot 10^{-11} - 1.8z$	$2.36 \cdot 10^{-11} - 1.8z$
7.0	$7.63 \cdot 10^{-17} - 0.417z$	$-9.17 \cdot 10^{-16} - 4.47z$	$1.2 \cdot 10^7 + 9.01 \cdot 10^8z$	$-3.8 \cdot 10^5 + 1.04 \cdot 10^9z$

Table 55: Case 1a (cpv PCWIN64 MR2025b), $r = 7$, Z4: poles.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	0	$-1.61 \cdot 10^{-16} - 1.23 \cdot 10^{-15}z$	$1.16 \cdot 10^{-11} + 0.198z$	$-2.97 \cdot 10^{-12} + 0.198z$
2.0	$-7.63 \cdot 10^{-17} + 3.79z$	$6.42 \cdot 10^{-16} + 0.458z$	$1.16 \cdot 10^{-11} - 0.198z$	$2.94 \cdot 10^{-12} - 0.198z$
3.0	$-7.63 \cdot 10^{-17} - 3.79z$	$-4.86 \cdot 10^{-16} - 0.458z$	$1.81 \cdot 10^{-11} + 0.691z$	$-1.0 \cdot 10^{-11} + 0.691z$
4.0	$2.78 \cdot 10^{-17} + 1.09z$	$1.24 \cdot 10^{-15} + 1.23z$	$1.81 \cdot 10^{-11} - 0.691z$	$9.84 \cdot 10^{-12} - 0.691z$
5.0	$2.78 \cdot 10^{-17} - 1.09z$	$1.68 \cdot 10^{-15} - 1.23z$	$5.9 \cdot 10^{-11} + 1.8z$	$-2.56 \cdot 10^{-11} + 1.8z$
6.0	$7.63 \cdot 10^{-17} + 0.417z$	$8.32 \cdot 10^{-15} + 4.47z$	$5.86 \cdot 10^{-11} - 1.8z$	$2.36 \cdot 10^{-11} - 1.8z$
7.0	$7.63 \cdot 10^{-17} - 0.417z$	$-9.17 \cdot 10^{-16} - 4.47z$	$1.2 \cdot 10^7 + 9.01 \cdot 10^8z$	$-3.8 \cdot 10^5 + 1.04 \cdot 10^9z$

Table 56: Case 1a (AlexReis PCWIN64 MR2025b), $r = 7$, Z4: poles.

6.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	NaN	NaN	$1.05 \cdot 10^{-11} - 6.74 \cdot 10^{-10}z$	$1.46 \cdot 10^{-12} - 4.66 \cdot 10^{-10}z$
2.0	$1.73 \cdot 10^{-17} + 1.8z$	$2.11 \cdot 10^{-16} - 0.216z$	$2.4 \cdot 10^{-11} + 0.417z$	$1.06 \cdot 10^{-11} + 0.417z$
3.0	$1.73 \cdot 10^{-17} - 1.8z$	$2.13 \cdot 10^{-15} + 0.216z$	$1.61 \cdot 10^{-12} - 0.417z$	$-7.23 \cdot 10^{-12} - 0.417z$
4.0	$5.55 \cdot 10^{-17} + 0.691z$	$1.4 \cdot 10^{-15} - 0.768z$	$5.5 \cdot 10^{-11} + 1.09z$	$2.6 \cdot 10^{-11} + 1.09z$
5.0	$5.55 \cdot 10^{-17} - 0.691z$	$4.64 \cdot 10^{-15} + 0.768z$	$-2.6 \cdot 10^{-12} - 1.09z$	$-1.99 \cdot 10^{-11} - 1.09z$
6.0	$1.21 \cdot 10^{-17} + 0.198z$	$5.1 \cdot 10^{-15} + 2.07z$	$3.02 \cdot 10^{-10} + 3.79z$	$1.01 \cdot 10^{-10} + 3.79z$
7.0	$1.21 \cdot 10^{-17} - 0.198z$	$7.22 \cdot 10^{-15} - 2.07z$	$1.02 \cdot 10^{-10} - 3.79z$	$-5.76 \cdot 10^{-11} - 3.79z$

Table 57: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4: zeros.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 1.1 \cdot 10^{-08}$	AAA-sign-L1000, $\sigma_7 = 9.9 \cdot 10^{-09}$
1.0	NaN	NaN	$-9.17 \cdot 10^{-4} + 4.3 \cdot 10^{-6}z$	$4.98 \cdot 10^{-5} - 0.198z$
2.0	$-3.12 \cdot 10^{-17} + 1.8z$	$2.07 \cdot 10^{-16} + 0.216z$	$-6.4 \cdot 10^{-4} - 0.416z$	$4.86 \cdot 10^{-5} + 0.198z$
3.0	$-3.12 \cdot 10^{-17} - 1.8z$	$-1.59 \cdot 10^{-16} - 0.216z$	$-6.31 \cdot 10^{-4} + 0.416z$	$4.58 \cdot 10^{-5} - 0.691z$
4.0	$4.16 \cdot 10^{-17} + 0.691z$	$-1.16 \cdot 10^{-15} + 0.768z$	$0.00106 - 1.08z$	$4.52 \cdot 10^{-5} + 0.691z$
5.0	$4.16 \cdot 10^{-17} - 0.691z$	$1.19 \cdot 10^{-15} - 0.768z$	$0.0011 + 1.08z$	$-1.89 \cdot 10^{-5} - 1.8z$
6.0	$5.2 \cdot 10^{-18} + 0.198z$	$-1.59 \cdot 10^{-16} + 2.07z$	$0.0242 - 3.78z$	$-1.35 \cdot 10^{-5} + 1.8z$
7.0	$5.2 \cdot 10^{-18} - 0.198z$	$8.43 \cdot 10^{-16} - 2.07z$	$0.0244 + 3.78z$	$-3.31 \cdot 10^4 - 1.58 \cdot 10^4z$

Table 58: Case 1a (cpv MACI64 MR2022b), $r = 7$, Z4: zeros.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	NaN	NaN	$1.11 \cdot 10^{-11} - 8.34 \cdot 10^{-10}i$	$-2.09 \cdot 10^{-14} - 7.35 \cdot 10^{-10}i$
2.0	$1.73 \cdot 10^{-17} + 1.8i$	$8.87 \cdot 10^{-16} + 0.216i$	$1.36 \cdot 10^{-11} + 0.417i$	$-6.16 \cdot 10^{-12} + 0.417i$
3.0	$1.73 \cdot 10^{-17} - 1.8i$	$-8.68 \cdot 10^{-16} - 0.216i$	$1.36 \cdot 10^{-11} - 0.417i$	$6.12 \cdot 10^{-12} - 0.417i$
4.0	$5.55 \cdot 10^{-17} + 0.691i$	$2.39 \cdot 10^{-16} + 0.768i$	$2.86 \cdot 10^{-11} + 1.09i$	$-1.55 \cdot 10^{-11} + 1.09i$
5.0	$5.55 \cdot 10^{-17} - 0.691i$	$4.76 \cdot 10^{-16} - 0.768i$	$2.84 \cdot 10^{-11} - 1.09i$	$1.49 \cdot 10^{-11} - 1.09i$
6.0	$1.21 \cdot 10^{-17} + 0.198i$	$3.76 \cdot 10^{-15} + 2.07i$	$2.24 \cdot 10^{-10} + 3.79i$	$-5.59 \cdot 10^{-11} + 3.79i$
7.0	$1.21 \cdot 10^{-17} - 0.198i$	$1.55 \cdot 10^{-15} - 2.07i$	$2.23 \cdot 10^{-10} - 3.79i$	$4.61 \cdot 10^{-11} - 3.79i$

Table 59: Case 1a (cpv PCWIN64 MR2025b), $r = 7$, Z4: zeros.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	NaN	NaN	$1.11 \cdot 10^{-11} - 8.34 \cdot 10^{-10}i$	$-2.09 \cdot 10^{-14} - 7.35 \cdot 10^{-10}i$
2.0	$1.73 \cdot 10^{-17} + 1.8i$	$8.87 \cdot 10^{-16} + 0.216i$	$1.36 \cdot 10^{-11} + 0.417i$	$-6.16 \cdot 10^{-12} + 0.417i$
3.0	$1.73 \cdot 10^{-17} - 1.8i$	$-8.68 \cdot 10^{-16} - 0.216i$	$1.36 \cdot 10^{-11} - 0.417i$	$6.12 \cdot 10^{-12} - 0.417i$
4.0	$5.55 \cdot 10^{-17} + 0.691i$	$2.39 \cdot 10^{-16} + 0.768i$	$2.86 \cdot 10^{-11} + 1.09i$	$-1.55 \cdot 10^{-11} + 1.09i$
5.0	$5.55 \cdot 10^{-17} - 0.691i$	$4.76 \cdot 10^{-16} - 0.768i$	$2.84 \cdot 10^{-11} - 1.09i$	$1.49 \cdot 10^{-11} - 1.09i$
6.0	$1.21 \cdot 10^{-17} + 0.198i$	$3.76 \cdot 10^{-15} + 2.07i$	$2.24 \cdot 10^{-10} + 3.79i$	$-5.59 \cdot 10^{-11} + 3.79i$
7.0	$1.21 \cdot 10^{-17} - 0.198i$	$1.55 \cdot 10^{-15} - 2.07i$	$2.23 \cdot 10^{-10} - 3.79i$	$4.61 \cdot 10^{-11} - 3.79i$

Table 60: Case 1a (AlexReis PCWIN64 MR2025b), $r = 7$, Z4: zeros.

6.4 Poles and zeros visualisation

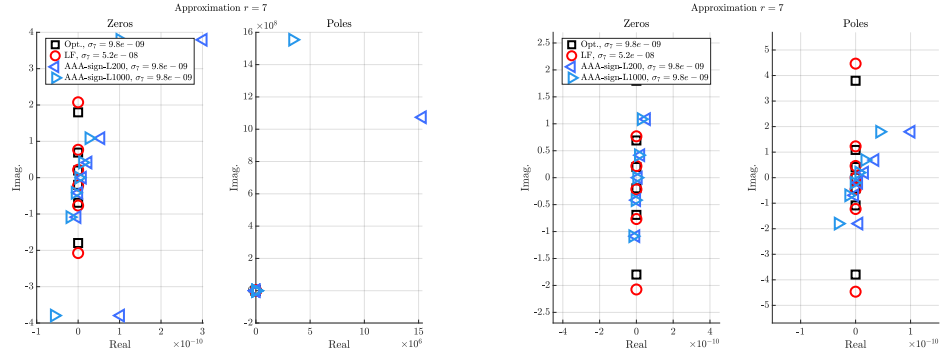


Figure 9: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

6.5 Approximation visualization

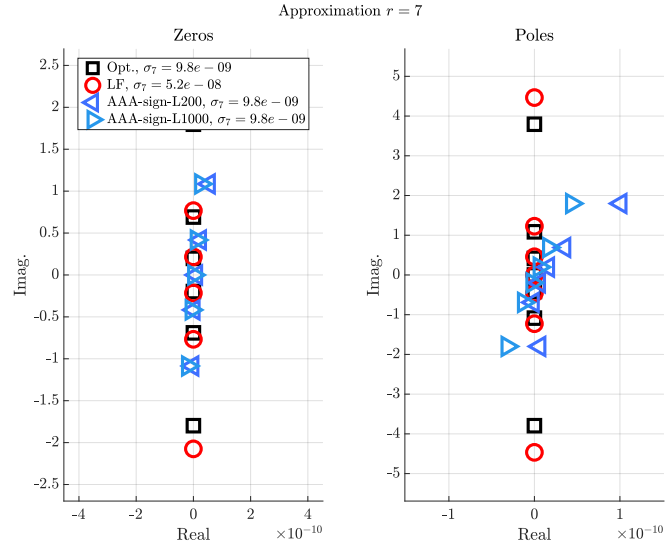


Figure 10: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4. E and F evaluations.

7 Approximation objective $r = 8$

7.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-8.7 \cdot 10^5$	$1.7 \cdot 10^8$	$-5.1 \cdot 10^6$	$-7.9 \cdot 10^8$
z^7	6.9	0	7.2	$-1.9 \cdot 10^{-14}$	7.0	$-4.9 \cdot 10^{-4}$	7.0	$-3.4 \cdot 10^{-3}$
z^6	0	0	$-5.3 \cdot 10^{-14}$	$-1.4 \cdot 10^{-13}$	$-1.8 \cdot 10^7$	$3.7 \cdot 10^9$	$-1.1 \cdot 10^8$	$-1.7 \cdot 10^{10}$
z^5	36.0	0	41.0	$-7.4 \cdot 10^{-14}$	37.0	$-9.8 \cdot 10^{-3}$	37.0	-0.046
z^4	0	0	$7.7 \cdot 10^{-14}$	$-3.2 \cdot 10^{-13}$	$-3.4 \cdot 10^7$	$6.9 \cdot 10^9$	$-2.0 \cdot 10^8$	$-3.1 \cdot 10^{10}$
z^3	27.0	0	33.0	$-2.2 \cdot 10^{-14}$	28.0	-0.014	28.0	-0.051
z^2	0	0	$1.6 \cdot 10^{-13}$	$-7.9 \cdot 10^{-14}$	$-1.0 \cdot 10^7$	$2.1 \cdot 10^9$	$-6.0 \cdot 10^7$	$-9.3 \cdot 10^9$
z	2.9	0	3.7	$3.3 \cdot 10^{-14}$	3.0	$-2.4 \cdot 10^{-3}$	3.1	$-7.0 \cdot 10^{-3}$
1.0	0	0	$1.0 \cdot 10^{-14}$	$1.0 \cdot 10^{-14}$	$-2.8 \cdot 10^5$	$5.5 \cdot 10^7$	$-1.6 \cdot 10^6$	$-2.5 \cdot 10^8$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-1.4 \cdot 10^{-14}$	$-4.1 \cdot 10^{-14}$	$-6.0 \cdot 10^6$	$1.2 \cdot 10^9$	$-3.5 \cdot 10^7$	$-5.4 \cdot 10^9$
z^6	21.0	0	23.0	$-5.5 \cdot 10^{-14}$	21.0	$-3.4 \cdot 10^{-3}$	21.0	-0.019
z^5	0	0	$-5.6 \cdot 10^{-14}$	$-2.7 \cdot 10^{-13}$	$-3.2 \cdot 10^7$	$6.3 \cdot 10^9$	$-1.9 \cdot 10^8$	$-2.9 \cdot 10^{10}$
z^4	39.0	0	46.0	$-6.5 \cdot 10^{-14}$	40.0	-0.015	40.0	-0.062
z^3	0	0	$1.9 \cdot 10^{-13}$	$-2.4 \cdot 10^{-13}$	$-2.4 \cdot 10^7$	$4.7 \cdot 10^9$	$-1.4 \cdot 10^8$	$-2.1 \cdot 10^{10}$
z^2	12.0	0	15.0	$3.1 \cdot 10^{-14}$	12.0	$-8.0 \cdot 10^{-3}$	12.0	-0.025
z	0	0	$6.2 \cdot 10^{-14}$	$1.9 \cdot 10^{-14}$	$-2.5 \cdot 10^6$	$5.1 \cdot 10^8$	$-1.5 \cdot 10^7$	$-2.3 \cdot 10^9$
1.0	0.32	0	0.42	$8.3 \cdot 10^{-15}$	0.33	$-3.2 \cdot 10^{-4}$	0.33	$-8.5 \cdot 10^{-4}$

Table 61: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.4 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.4 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$4.6 \cdot 10^{-3}$	$1.0 \cdot 10^{-4}$	$4.6 \cdot 10^{-3}$	$1.3 \cdot 10^{-4}$
z^7	6.9	0	7.2	$1.3 \cdot 10^{-14}$	6.9	$-8.5 \cdot 10^{-5}$	6.9	$-8.5 \cdot 10^{-5}$
z^6	0	0	$6.3 \cdot 10^{-14}$	$-1.2 \cdot 10^{-13}$	0.12	$1.5 \cdot 10^{-3}$	0.12	$2.0 \cdot 10^{-3}$
z^5	36.0	0	41.0	$1.9 \cdot 10^{-13}$	36.0	$-1.3 \cdot 10^{-3}$	36.0	$-1.3 \cdot 10^{-3}$
z^4	0	0	$3.4 \cdot 10^{-13}$	$-3.0 \cdot 10^{-13}$	0.25	$1.8 \cdot 10^{-3}$	0.25	$2.8 \cdot 10^{-3}$
z^3	27.0	0	33.0	$3.6 \cdot 10^{-13}$	27.0	$-1.5 \cdot 10^{-3}$	27.0	$-1.6 \cdot 10^{-3}$
z^2	0	0	$1.6 \cdot 10^{-13}$	$-1.4 \cdot 10^{-13}$	0.086	$2.6 \cdot 10^{-4}$	0.086	$5.5 \cdot 10^{-4}$
z	2.9	0	3.7	$3.8 \cdot 10^{-14}$	2.9	$-2.2 \cdot 10^{-4}$	2.9	$-2.3 \cdot 10^{-4}$
1.0	0	0	$6.1 \cdot 10^{-15}$	$-6.4 \cdot 10^{-15}$	$2.6 \cdot 10^{-3}$	$-3.5 \cdot 10^{-7}$	$2.6 \cdot 10^{-3}$	$7.3 \cdot 10^{-6}$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$3.0 \cdot 10^{-15}$	$-2.1 \cdot 10^{-14}$	0.035	$6.0 \cdot 10^{-4}$	0.035	$7.7 \cdot 10^{-4}$
z^6	21.0	0	23.0	$7.1 \cdot 10^{-14}$	21.0	$-5.0 \cdot 10^{-4}$	21.0	$-5.1 \cdot 10^{-4}$
z^5	0	0	$2.2 \cdot 10^{-13}$	$-2.3 \cdot 10^{-13}$	0.22	$2.2 \cdot 10^{-3}$	0.22	$3.0 \cdot 10^{-3}$
z^4	39.0	0	46.0	$3.4 \cdot 10^{-13}$	39.0	$-1.8 \cdot 10^{-3}$	39.0	$-1.8 \cdot 10^{-3}$
z^3	0	0	$3.1 \cdot 10^{-13}$	$-2.5 \cdot 10^{-13}$	0.19	$9.3 \cdot 10^{-4}$	0.19	$1.6 \cdot 10^{-3}$
z^2	12.0	0	15.0	$1.9 \cdot 10^{-13}$	12.0	$-7.9 \cdot 10^{-4}$	12.0	$-7.9 \cdot 10^{-4}$
z	0	0	$4.5 \cdot 10^{-14}$	$-5.3 \cdot 10^{-14}$	0.023	$3.0 \cdot 10^{-5}$	0.023	$1.0 \cdot 10^{-4}$
1.0	0.32	0	0.42	$1.8 \cdot 10^{-15}$	0.32	$-2.7 \cdot 10^{-5}$	0.32	$-2.7 \cdot 10^{-5}$

Table 62: Case 1a (cpv MACI64 MR2022b), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-5.8 \cdot 10^{-11}$	$-1.2 \cdot 10^{-8}$	$2.5 \cdot 10^{-12}$	$1.4 \cdot 10^{-9}$
z^7	6.9	0	7.2	$-1.5 \cdot 10^{-14}$	6.9	$-2.1 \cdot 10^{-10}$	6.9	$-5.3 \cdot 10^{-11}$
z^6	0	0	$-5.0 \cdot 10^{-14}$	$3.3 \cdot 10^{-14}$	$-1.2 \cdot 10^{-9}$	$-2.6 \cdot 10^{-7}$	$6.3 \cdot 10^{-11}$	$2.9 \cdot 10^{-8}$
z^5	36.0	0	41.0	$-2.6 \cdot 10^{-14}$	36.0	$-3.3 \cdot 10^{-9}$	36.0	$-8.3 \cdot 10^{-10}$
z^4	0	0	$-1.7 \cdot 10^{-13}$	$1.5 \cdot 10^{-13}$	$-2.1 \cdot 10^{-9}$	$-4.8 \cdot 10^{-7}$	$1.3 \cdot 10^{-10}$	$5.5 \cdot 10^{-8}$
z^3	27.0	0	33.0	$1.7 \cdot 10^{-13}$	27.0	$-4.1 \cdot 10^{-9}$	27.0	$-1.0 \cdot 10^{-9}$
z^2	0	0	$-8.5 \cdot 10^{-14}$	$9.5 \cdot 10^{-14}$	$-6.3 \cdot 10^{-10}$	$-1.5 \cdot 10^{-7}$	$4.3 \cdot 10^{-11}$	$1.6 \cdot 10^{-8}$
z	2.9	0	3.7	$4.3 \cdot 10^{-14}$	2.9	$-6.2 \cdot 10^{-10}$	2.9	$-1.6 \cdot 10^{-10}$
1.0	0	0	$-2.2 \cdot 10^{-15}$	$4.0 \cdot 10^{-15}$	$-1.7 \cdot 10^{-11}$	$-3.9 \cdot 10^{-9}$	$1.3 \cdot 10^{-12}$	$4.3 \cdot 10^{-10}$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-7.1 \cdot 10^{-15}$	$-2.0 \cdot 10^{-15}$	$-3.9 \cdot 10^{-10}$	$-8.4 \cdot 10^{-8}$	$1.9 \cdot 10^{-11}$	$9.7 \cdot 10^{-9}$
z^6	21.0	0	23.0	$-4.0 \cdot 10^{-14}$	21.0	$-1.3 \cdot 10^{-9}$	21.0	$-3.2 \cdot 10^{-10}$
z^5	0	0	$-1.2 \cdot 10^{-13}$	$9.2 \cdot 10^{-14}$	$-2.0 \cdot 10^{-9}$	$-4.5 \cdot 10^{-7}$	$1.2 \cdot 10^{-10}$	$5.1 \cdot 10^{-8}$
z^4	39.0	0	46.0	$9.0 \cdot 10^{-14}$	39.0	$-4.8 \cdot 10^{-9}$	39.0	$-1.2 \cdot 10^{-9}$
z^3	0	0	$-1.6 \cdot 10^{-13}$	$1.6 \cdot 10^{-13}$	$-1.5 \cdot 10^{-9}$	$-3.4 \cdot 10^{-7}$	$9.7 \cdot 10^{-11}$	$3.8 \cdot 10^{-8}$
z^2	12.0	0	15.0	$1.2 \cdot 10^{-13}$	12.0	$-2.1 \cdot 10^{-9}$	12.0	$-5.4 \cdot 10^{-10}$
z	0	0	$-2.1 \cdot 10^{-14}$	$3.3 \cdot 10^{-14}$	$-1.5 \cdot 10^{-10}$	$-3.6 \cdot 10^{-8}$	$1.1 \cdot 10^{-11}$	$4.0 \cdot 10^{-9}$
1.0	0.32	0	0.42	$6.7 \cdot 10^{-15}$	0.32	$-7.6 \cdot 10^{-11}$	0.32	$-1.9 \cdot 10^{-11}$

Table 63: Case 1a (cpv PCWIN64 MR2025b), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-5.8 \cdot 10^{-11}$	$-1.2 \cdot 10^{-8}$	$2.5 \cdot 10^{-12}$	$1.4 \cdot 10^{-9}$
z^7	6.9	0	7.2	$-1.5 \cdot 10^{-14}$	6.9	$-2.1 \cdot 10^{-10}$	6.9	$-5.3 \cdot 10^{-11}$
z^6	0	0	$-5.0 \cdot 10^{-14}$	$3.3 \cdot 10^{-14}$	$-1.2 \cdot 10^{-9}$	$-2.6 \cdot 10^{-7}$	$6.3 \cdot 10^{-11}$	$2.9 \cdot 10^{-8}$
z^5	36.0	0	41.0	$-2.6 \cdot 10^{-14}$	36.0	$-3.3 \cdot 10^{-9}$	36.0	$-8.3 \cdot 10^{-10}$
z^4	0	0	$-1.7 \cdot 10^{-13}$	$1.5 \cdot 10^{-13}$	$-2.1 \cdot 10^{-9}$	$-4.8 \cdot 10^{-7}$	$1.3 \cdot 10^{-10}$	$5.5 \cdot 10^{-8}$
z^3	27.0	0	33.0	$1.7 \cdot 10^{-13}$	27.0	$-4.1 \cdot 10^{-9}$	27.0	$-1.0 \cdot 10^{-9}$
z^2	0	0	$-8.5 \cdot 10^{-14}$	$9.5 \cdot 10^{-14}$	$-6.3 \cdot 10^{-10}$	$-1.5 \cdot 10^{-7}$	$4.3 \cdot 10^{-11}$	$1.6 \cdot 10^{-8}$
z	2.9	0	3.7	$4.3 \cdot 10^{-14}$	2.9	$-6.2 \cdot 10^{-10}$	2.9	$-1.6 \cdot 10^{-10}$
1.0	0	0	$-2.2 \cdot 10^{-15}$	$4.0 \cdot 10^{-15}$	$-1.7 \cdot 10^{-11}$	$-3.9 \cdot 10^{-9}$	$1.3 \cdot 10^{-12}$	$4.3 \cdot 10^{-10}$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-7.1 \cdot 10^{-15}$	$-2.0 \cdot 10^{-15}$	$-3.9 \cdot 10^{-10}$	$-8.4 \cdot 10^{-8}$	$1.9 \cdot 10^{-11}$	$9.7 \cdot 10^{-9}$
z^6	21.0	0	23.0	$-4.0 \cdot 10^{-14}$	21.0	$-1.3 \cdot 10^{-9}$	21.0	$-3.2 \cdot 10^{-10}$
z^5	0	0	$-1.2 \cdot 10^{-13}$	$9.2 \cdot 10^{-14}$	$-2.0 \cdot 10^{-9}$	$-4.5 \cdot 10^{-7}$	$1.2 \cdot 10^{-10}$	$5.1 \cdot 10^{-8}$
z^4	39.0	0	46.0	$9.0 \cdot 10^{-14}$	39.0	$-4.8 \cdot 10^{-9}$	39.0	$-1.2 \cdot 10^{-9}$
z^3	0	0	$-1.6 \cdot 10^{-13}$	$1.6 \cdot 10^{-13}$	$-1.5 \cdot 10^{-9}$	$-3.4 \cdot 10^{-7}$	$9.7 \cdot 10^{-11}$	$3.8 \cdot 10^{-8}$
z^2	12.0	0	15.0	$1.2 \cdot 10^{-13}$	12.0	$-2.1 \cdot 10^{-9}$	12.0	$-5.4 \cdot 10^{-10}$
z	0	0	$-2.1 \cdot 10^{-14}$	$3.3 \cdot 10^{-14}$	$-1.5 \cdot 10^{-10}$	$-3.6 \cdot 10^{-8}$	$1.1 \cdot 10^{-11}$	$4.0 \cdot 10^{-9}$
1.0	0.32	0	0.42	$6.7 \cdot 10^{-15}$	0.32	$-7.6 \cdot 10^{-11}$	0.32	$-1.9 \cdot 10^{-11}$

Table 64: Case 1a (AlexReis PCWIN64 MR2025b), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

7.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	$6.66 \cdot 10^{-16} + 4.35z$	$-4.06 \cdot 10^{-15} + 0.178z$	$3.84 \cdot 10^{-12} + 6.42 \cdot 10^{-10}z$	$5.64 \cdot 10^{-13} - 1.45 \cdot 10^{-10}z$
2.0	$6.66 \cdot 10^{-16} - 4.35z$	$-6.53 \cdot 10^{-16} - 0.178z$	$-4.23 \cdot 10^{-12} - 0.359z$	$-2.48 \cdot 10^{-12} + 0.359z$
3.0	$2.22 \cdot 10^{-16} + 1.3z$	$-7.33 \cdot 10^{-17} - 0.599z$	$1.28 \cdot 10^{-11} + 0.359z$	$3.91 \cdot 10^{-12} - 0.359z$
4.0	$2.22 \cdot 10^{-16} - 1.3z$	$-1.58 \cdot 10^{-15} + 0.599z$	$-1.39 \cdot 10^{-11} - 0.866z$	$-6.57 \cdot 10^{-12} + 0.866z$
5.0	$1.39 \cdot 10^{-17} + 0.579z$	$4.61 \cdot 10^{-15} - 1.34z$	$2.73 \cdot 10^{-11} + 0.866z$	$9.26 \cdot 10^{-12} - 0.866z$
6.0	$1.39 \cdot 10^{-17} - 0.579z$	$3.38 \cdot 10^{-15} + 1.34z$	$-2.86 \cdot 10^{-11} - 2.09z$	$-1.37 \cdot 10^{-11} + 2.09z$
7.0	$1.47 \cdot 10^{-17} + 0.172z$	$-4.63 \cdot 10^{-16} - 4.52z$	$7.16 \cdot 10^{-11} + 2.09z$	$2.47 \cdot 10^{-11} - 2.09z$
8.0	$1.47 \cdot 10^{-17} - 0.172z$	$1.25 \cdot 10^{-14} + 4.52z$	$6.04 \cdot 10^6 - 1.21 \cdot 10^9z$	$3.54 \cdot 10^7 + 5.44 \cdot 10^9z$

Table 65: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4: poles.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.4 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.4 \cdot 10^{-10}$
1.0	$8.88 \cdot 10^{-16} + 4.35_z$	$-2.44 \cdot 10^{-15} - 0.178_z$	$-8.97 \cdot 10^{-4} + 0.172_z$	$-8.97 \cdot 10^{-4} + 0.172_z$
2.0	$8.88 \cdot 10^{-16} - 4.35_z$	$-5.6 \cdot 10^{-16} + 0.178_z$	$-9.0 \cdot 10^{-4} - 0.172_z$	$-9.0 \cdot 10^{-4} - 0.172_z$
3.0	$8.33 \cdot 10^{-17} + 1.3_z$	$-5.87 \cdot 10^{-17} - 0.599_z$	$-0.0011 + 0.579_z$	$-0.0011 + 0.579_z$
4.0	$8.33 \cdot 10^{-17} - 1.3_z$	$-3.93 \cdot 10^{-15} + 0.599_z$	$-0.00112 - 0.579_z$	$-0.00112 - 0.579_z$
5.0	$4.16 \cdot 10^{-17} + 0.579_z$	$1.3 \cdot 10^{-15} - 1.34_z$	$-0.002 + 1.3_z$	$-0.002 + 1.3_z$
6.0	$4.16 \cdot 10^{-17} - 0.579_z$	$-5.03 \cdot 10^{-15} + 1.34_z$	$-0.00203 - 1.3_z$	$-0.00203 - 1.3_z$
7.0	$-2.6 \cdot 10^{-18} + 0.172_z$	$1.06 \cdot 10^{-14} - 4.52_z$	$-0.0135 + 4.36_z$	$-0.0135 + 4.36_z$
8.0	$-2.6 \cdot 10^{-18} - 0.172_z$	$-2.57 \cdot 10^{-15} + 4.52_z$	$-0.0136 - 4.36_z$	$-0.0136 - 4.36_z$

Table 66: Case 1a (cpv MACI64 MR2022b), $r = 8$, Z4: poles.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	$7.22 \cdot 10^{-16} + 4.35_z$	$1.37 \cdot 10^{-17} + 0.178_z$	$7.54 \cdot 10^{-13} - 0.172_z$	$8.64 \cdot 10^{-13} + 0.172_z$
2.0	$7.22 \cdot 10^{-16} - 4.35_z$	$1.31 \cdot 10^{-15} - 0.178_z$	$1.1 \cdot 10^{-11} + 0.172_z$	$-1.73 \cdot 10^{-12} - 0.172_z$
3.0	$1.94 \cdot 10^{-16} + 1.3_z$	$-1.07 \cdot 10^{-15} + 0.599_z$	$-9.06 \cdot 10^{-12} - 0.579_z$	$3.82 \cdot 10^{-12} + 0.579_z$
4.0	$1.94 \cdot 10^{-16} - 1.3_z$	$3.39 \cdot 10^{-15} - 0.599_z$	$2.57 \cdot 10^{-11} + 0.579_z$	$-4.98 \cdot 10^{-12} - 0.579_z$
5.0	$-4.16 \cdot 10^{-17} + 0.579_z$	$-1.82 \cdot 10^{-15} + 1.34_z$	$-2.0 \cdot 10^{-11} - 1.3_z$	$8.73 \cdot 10^{-12} + 1.3_z$
6.0	$-4.16 \cdot 10^{-17} - 0.579_z$	$3.47 \cdot 10^{-15} - 1.34_z$	$5.85 \cdot 10^{-11} + 1.3_z$	$-1.1 \cdot 10^{-11} - 1.3_z$
7.0	$1.73 \cdot 10^{-18} + 0.172_z$	$-5.1 \cdot 10^{-15} - 4.52_z$	$2.98 \cdot 10^{-11} - 4.35_z$	$2.57 \cdot 10^{-11} + 4.35_z$
8.0	$1.73 \cdot 10^{-18} - 0.172_z$	$6.85 \cdot 10^{-15} + 4.52_z$	$2.95 \cdot 10^{-10} + 4.35_z$	$-4.05 \cdot 10^{-11} - 4.35_z$

Table 67: Case 1a (cpv PCWIN64 MR2025b), $r = 8$, Z4: poles.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	$7.22 \cdot 10^{-16} + 4.35_z$	$1.37 \cdot 10^{-17} + 0.178_z$	$7.54 \cdot 10^{-13} - 0.172_z$	$8.64 \cdot 10^{-13} + 0.172_z$
2.0	$7.22 \cdot 10^{-16} - 4.35_z$	$1.31 \cdot 10^{-15} - 0.178_z$	$1.1 \cdot 10^{-11} + 0.172_z$	$-1.73 \cdot 10^{-12} - 0.172_z$
3.0	$1.94 \cdot 10^{-16} + 1.3_z$	$-1.07 \cdot 10^{-15} + 0.599_z$	$-9.06 \cdot 10^{-12} - 0.579_z$	$3.82 \cdot 10^{-12} + 0.579_z$
4.0	$1.94 \cdot 10^{-16} - 1.3_z$	$3.39 \cdot 10^{-15} - 0.599_z$	$2.57 \cdot 10^{-11} + 0.579_z$	$-4.98 \cdot 10^{-12} - 0.579_z$
5.0	$-4.16 \cdot 10^{-17} + 0.579_z$	$-1.82 \cdot 10^{-15} + 1.34_z$	$-2.0 \cdot 10^{-11} - 1.3_z$	$8.73 \cdot 10^{-12} + 1.3_z$
6.0	$-4.16 \cdot 10^{-17} - 0.579_z$	$3.47 \cdot 10^{-15} - 1.34_z$	$5.85 \cdot 10^{-11} + 1.3_z$	$-1.1 \cdot 10^{-11} - 1.3_z$
7.0	$1.73 \cdot 10^{-18} + 0.172_z$	$-5.1 \cdot 10^{-15} - 4.52_z$	$2.98 \cdot 10^{-11} - 4.35_z$	$2.57 \cdot 10^{-11} + 4.35_z$
8.0	$1.73 \cdot 10^{-18} - 0.172_z$	$6.85 \cdot 10^{-15} + 4.52_z$	$2.95 \cdot 10^{-10} + 4.35_z$	$-4.05 \cdot 10^{-11} - 4.35_z$

Table 68: Case 1a (AlexReis PCWIN64 MR2025b), $r = 8$, Z4: poles.

7.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	NaN	NaN	$-1.46 \cdot 10^{-13} - 0.172_z$	$-9.08 \cdot 10^{-13} + 0.172_z$
2.0	0	$-2.76 \cdot 10^{-15} - 2.73 \cdot 10^{-15}_z$	$8.04 \cdot 10^{-12} + 0.172_z$	$2.11 \cdot 10^{-12} - 0.172_z$
3.0	$1.39 \cdot 10^{-16} + 2.09_z$	$4.64 \cdot 10^{-16} - 0.371_z$	$-8.67 \cdot 10^{-12} - 0.579_z$	$-4.32 \cdot 10^{-12} + 0.579_z$
4.0	$1.39 \cdot 10^{-16} - 2.09_z$	$-3.67 \cdot 10^{-15} + 0.371_z$	$1.88 \cdot 10^{-11} + 0.579_z$	$6.15 \cdot 10^{-12} - 0.579_z$
5.0	$-5.55 \cdot 10^{-17} + 0.866_z$	$8.79 \cdot 10^{-16} - 0.897_z$	$-2.05 \cdot 10^{-11} - 1.3_z$	$-9.53 \cdot 10^{-12} + 1.3_z$
6.0	$-5.55 \cdot 10^{-17} - 0.866_z$	$1.64 \cdot 10^{-15} + 0.897_z$	$4.14 \cdot 10^{-11} + 1.3_z$	$1.43 \cdot 10^{-11} - 1.3_z$
7.0	$-6.94 \cdot 10^{-18} + 0.359_z$	$6.16 \cdot 10^{-15} - 2.17_z$	$-2.26 \cdot 10^{-11} - 4.35_z$	$-1.71 \cdot 10^{-11} + 4.35_z$
8.0	$-6.94 \cdot 10^{-18} - 0.359_z$	$4.41 \cdot 10^{-15} + 2.17_z$	$1.86 \cdot 10^{-10} + 4.35_z$	$6.27 \cdot 10^{-11} - 4.35_z$

Table 69: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4: zeros.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.4 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.4 \cdot 10^{-10}$
1.0	NaN	NaN	$-8.78 \cdot 10^{-4} + 5.09 \cdot 10^{-8}i$	$-8.78 \cdot 10^{-4} - 2.54 \cdot 10^{-6}i$
2.0	0	$-1.63 \cdot 10^{-15} + 1.75 \cdot 10^{-15}i$	$-9.64 \cdot 10^{-4} + 0.359i$	$-9.64 \cdot 10^{-4} + 0.359i$
3.0	$-3.33 \cdot 10^{-16} + 2.09i$	$-2.19 \cdot 10^{-15} - 0.371i$	$-9.71 \cdot 10^{-4} - 0.359i$	$-9.71 \cdot 10^{-4} - 0.359i$
4.0	$-3.33 \cdot 10^{-16} - 2.09i$	$-1.01 \cdot 10^{-15} + 0.371i$	$-0.00138 + 0.866i$	$-0.00138 + 0.866i$
5.0	$-5.55 \cdot 10^{-17} + 0.866i$	$1.73 \cdot 10^{-15} - 0.897i$	$-0.0014 - 0.866i$	$-0.0014 - 0.866i$
6.0	$-5.55 \cdot 10^{-17} - 0.866i$	$-5.71 \cdot 10^{-15} + 0.897i$	$-0.0038 + 2.09i$	$-0.0038 + 2.09i$
7.0	$-6.25 \cdot 10^{-17} + 0.359i$	$2.2 \cdot 10^{-15} - 2.17i$	$-0.00385 - 2.09i$	$-0.00385 - 2.09i$
8.0	$-6.25 \cdot 10^{-17} - 0.359i$	$-2.01 \cdot 10^{-15} + 2.17i$	$-1500.0 + 32.5i$	$-1500.0 + 40.5i$

Table 70: Case 1a (cpv MACI64 MR2022b), $r = 8$, Z4: zeros.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	NaN	NaN	$5.66 \cdot 10^{-12} + 1.34 \cdot 10^{-9}i$	$-4.28 \cdot 10^{-13} - 1.47 \cdot 10^{-10}i$
2.0	0	$5.49 \cdot 10^{-16} - 1.11 \cdot 10^{-15}i$	$-4.07 \cdot 10^{-12} - 0.359i$	$2.24 \cdot 10^{-12} + 0.359i$
3.0	$1.39 \cdot 10^{-16} + 2.09i$	$-4.33 \cdot 10^{-17} + 0.371i$	$1.74 \cdot 10^{-11} + 0.359i$	$-3.19 \cdot 10^{-12} - 0.359i$
4.0	$1.39 \cdot 10^{-16} - 2.09i$	$2.25 \cdot 10^{-15} - 0.371i$	$-1.44 \cdot 10^{-11} - 0.866i$	$5.83 \cdot 10^{-12} + 0.866i$
5.0	$-5.55 \cdot 10^{-17} + 0.866i$	$-2.49 \cdot 10^{-15} + 0.897i$	$3.78 \cdot 10^{-11} + 0.866i$	$-7.39 \cdot 10^{-12} - 0.866i$
6.0	$-5.55 \cdot 10^{-17} - 0.866i$	$3.87 \cdot 10^{-15} - 0.897i$	$-2.21 \cdot 10^{-11} - 2.09i$	$1.38 \cdot 10^{-11} + 2.09i$
7.0	$-6.94 \cdot 10^{-18} + 0.359i$	$8.84 \cdot 10^{-16} + 2.17i$	$1.05 \cdot 10^{-10} + 2.09i$	$-1.81 \cdot 10^{-11} - 2.09i$
8.0	$-6.94 \cdot 10^{-18} - 0.359i$	$2.13 \cdot 10^{-15} - 2.17i$	$2.7 \cdot 10^5 - 5.7 \cdot 10^8i$	$-8.64 \cdot 10^6 + 4.91 \cdot 10^9i$

Table 71: Case 1a (cpv PCWIN64 MR2025b), $r = 8$, Z4: zeros.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	NaN	NaN	$5.66 \cdot 10^{-12} + 1.34 \cdot 10^{-9}i$	$-4.28 \cdot 10^{-13} - 1.47 \cdot 10^{-10}i$
2.0	0	$5.49 \cdot 10^{-16} - 1.11 \cdot 10^{-15}i$	$-4.07 \cdot 10^{-12} - 0.359i$	$2.24 \cdot 10^{-12} + 0.359i$
3.0	$1.39 \cdot 10^{-16} + 2.09i$	$-4.33 \cdot 10^{-17} + 0.371i$	$1.74 \cdot 10^{-11} + 0.359i$	$-3.19 \cdot 10^{-12} - 0.359i$
4.0	$1.39 \cdot 10^{-16} - 2.09i$	$2.25 \cdot 10^{-15} - 0.371i$	$-1.44 \cdot 10^{-11} - 0.866i$	$5.83 \cdot 10^{-12} + 0.866i$
5.0	$-5.55 \cdot 10^{-17} + 0.866i$	$-2.49 \cdot 10^{-15} + 0.897i$	$3.78 \cdot 10^{-11} + 0.866i$	$-7.39 \cdot 10^{-12} - 0.866i$
6.0	$-5.55 \cdot 10^{-17} - 0.866i$	$3.87 \cdot 10^{-15} - 0.897i$	$-2.21 \cdot 10^{-11} - 2.09i$	$1.38 \cdot 10^{-11} + 2.09i$
7.0	$-6.94 \cdot 10^{-18} + 0.359i$	$8.84 \cdot 10^{-16} + 2.17i$	$1.05 \cdot 10^{-10} + 2.09i$	$-1.81 \cdot 10^{-11} - 2.09i$
8.0	$-6.94 \cdot 10^{-18} - 0.359i$	$2.13 \cdot 10^{-15} - 2.17i$	$2.7 \cdot 10^5 - 5.7 \cdot 10^8i$	$-8.64 \cdot 10^6 + 4.91 \cdot 10^9i$

Table 72: Case 1a (AlexReis PCWIN64 MR2025b), $r = 8$, Z4: zeros.

7.4 Poles and zeros visualisation

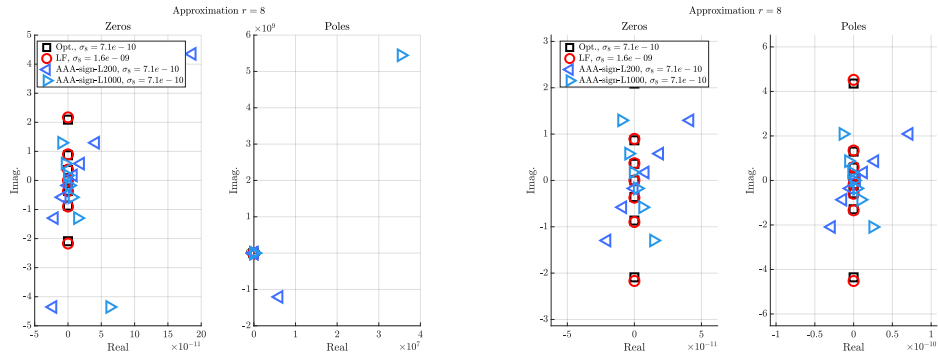


Figure 11: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

7.5 Approximation visualization

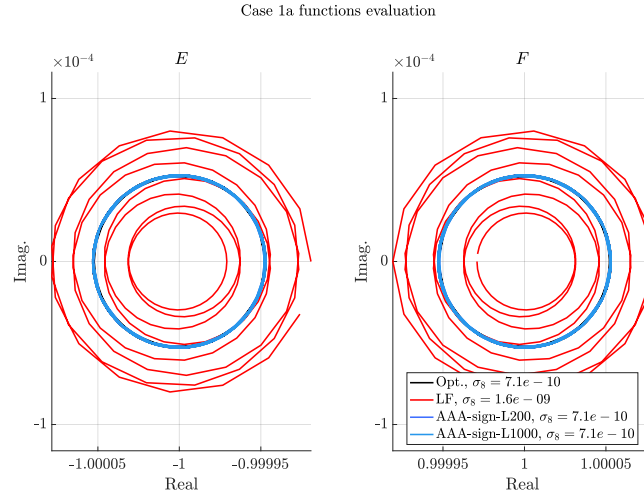


Figure 12: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4. E and F evaluations.

8 Approximation objective $r = 9$

8.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	$-2.1 \cdot 10^3$	$1.4 \cdot 10^6$	-940.0	$-5.1 \cdot 10^5$
z^8	7.8	0	8.9	$5.1 \cdot 10^{-14}$	7.8	$-8.0 \cdot 10^{-4}$	7.8	$3.7 \cdot 10^{-4}$
z^7	0	0	$9.2 \cdot 10^{-13}$	$1.1 \cdot 10^{-12}$	$-5.7 \cdot 10^4$	$3.7 \cdot 10^7$	$-2.5 \cdot 10^4$	$-1.4 \cdot 10^7$
z^6	55.0	0	78.0	$1.6 \cdot 10^{-13}$	54.0	-0.016	55.0	$7.1 \cdot 10^{-3}$
z^5	0	0	$4.0 \cdot 10^{-12}$	$6.7 \cdot 10^{-12}$	$-1.5 \cdot 10^5$	$9.8 \cdot 10^7$	$-6.6 \cdot 10^4$	$-3.6 \cdot 10^7$
z^4	61.0	0	110.0	$3.2 \cdot 10^{-13}$	60.0	-0.027	62.0	0.012
z^3	0	0	$2.2 \cdot 10^{-12}$	$4.8 \cdot 10^{-12}$	$-7.4 \cdot 10^4$	$4.9 \cdot 10^7$	$-3.3 \cdot 10^4$	$-1.8 \cdot 10^7$
z^2	13.0	0	27.0	$3.1 \cdot 10^{-13}$	13.0	$-7.6 \cdot 10^{-3}$	13.0	$3.1 \cdot 10^{-3}$
z	0	0	$6.9 \cdot 10^{-14}$	$5.3 \cdot 10^{-13}$	$-6.0 \cdot 10^3$	$3.9 \cdot 10^6$	$-2.7 \cdot 10^3$	$-1.5 \cdot 10^6$
1.0	0.27	0	0.64	$-4.9 \cdot 10^{-17}$	0.27	$-1.9 \cdot 10^{-4}$	0.28	$7.3 \cdot 10^{-5}$
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$1.5 \cdot 10^{-13}$	$1.3 \cdot 10^{-13}$	$-1.6 \cdot 10^4$	$1.1 \cdot 10^7$	$-7.3 \cdot 10^3$	$-4.0 \cdot 10^6$
z^7	27.0	0	35.0	$1.7 \cdot 10^{-13}$	27.0	$-5.3 \cdot 10^{-3}$	27.0	$2.4 \cdot 10^{-3}$
z^6	0	0	$2.5 \cdot 10^{-12}$	$3.7 \cdot 10^{-12}$	$-1.1 \cdot 10^5$	$7.5 \cdot 10^7$	$-5.1 \cdot 10^4$	$-2.8 \cdot 10^7$
z^5	71.0	0	110.0	$1.0 \cdot 10^{-13}$	70.0	-0.026	72.0	0.012
z^4	0	0	$3.8 \cdot 10^{-12}$	$7.1 \cdot 10^{-12}$	$-1.3 \cdot 10^5$	$8.5 \cdot 10^7$	$-5.8 \cdot 10^4$	$-3.2 \cdot 10^7$
z^3	35.0	0	67.0	$5.1 \cdot 10^{-13}$	35.0	-0.018	36.0	$7.7 \cdot 10^{-3}$
z^2	0	0	$6.6 \cdot 10^{-13}$	$2.1 \cdot 10^{-12}$	$-2.8 \cdot 10^4$	$1.8 \cdot 10^7$	$-1.2 \cdot 10^4$	$-6.8 \cdot 10^6$
z	2.8	0	6.2	$6.5 \cdot 10^{-14}$	2.8	$-1.8 \cdot 10^{-3}$	2.9	$7.3 \cdot 10^{-4}$
1.0	0	0	$-6.9 \cdot 10^{-15}$	$6.8 \cdot 10^{-14}$	-570.0	$3.8 \cdot 10^5$	-260.0	$-1.4 \cdot 10^5$

Table 73: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.2 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.2 \cdot 10^{-11}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	190.0	32.0	190.0	32.0
z^8	7.8	0	8.9	$-1.1 \cdot 10^{-13}$	8.3	0.29	8.3	0.29
z^7	0	0	$-1.7 \cdot 10^{-13}$	$3.8 \cdot 10^{-13}$	$5.2 \cdot 10^3$	850.0	$5.2 \cdot 10^3$	850.0
z^6	55.0	0	78.0	$-3.4 \cdot 10^{-12}$	65.0	6.0	65.0	6.0
z^5	0	0	$-3.2 \cdot 10^{-13}$	$3.3 \cdot 10^{-12}$	$1.4 \cdot 10^4$	$2.2 \cdot 10^3$	$1.4 \cdot 10^4$	$2.2 \cdot 10^3$
z^4	61.0	0	110.0	$-6.6 \cdot 10^{-12}$	81.0	11.0	81.0	11.0
z^3	0	0	$-7.6 \cdot 10^{-14}$	$2.8 \cdot 10^{-12}$	$6.8 \cdot 10^3$	$1.1 \cdot 10^3$	$6.8 \cdot 10^3$	$1.1 \cdot 10^3$
z^2	13.0	0	27.0	$-2.2 \cdot 10^{-12}$	19.0	3.2	19.0	3.2
z	0	0	$3.2 \cdot 10^{-14}$	$3.4 \cdot 10^{-13}$	550.0	89.0	550.0	89.0
1.0	0.27	0	0.64	$-4.4 \cdot 10^{-14}$	0.43	0.085	0.43	0.085
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$-4.9 \cdot 10^{-14}$	$2.4 \cdot 10^{-14}$	$1.5 \cdot 10^3$	250.0	$1.5 \cdot 10^3$	250.0
z^7	27.0	0	35.0	$-1.0 \cdot 10^{-12}$	30.0	2.0	30.0	2.0
z^6	0	0	$-2.7 \cdot 10^{-13}$	$1.6 \cdot 10^{-12}$	$1.0 \cdot 10^4$	$1.7 \cdot 10^3$	$1.0 \cdot 10^4$	$1.7 \cdot 10^3$
z^5	71.0	0	110.0	$-6.0 \cdot 10^{-12}$	89.0	10.0	89.0	10.0
z^4	0	0	$-2.2 \cdot 10^{-13}$	$3.9 \cdot 10^{-12}$	$1.2 \cdot 10^4$	$1.9 \cdot 10^3$	$1.2 \cdot 10^4$	$1.9 \cdot 10^3$
z^3	35.0	0	67.0	$-4.8 \cdot 10^{-12}$	49.0	7.6	49.0	7.6
z^2	0	0	$2.4 \cdot 10^{-14}$	$1.3 \cdot 10^{-12}$	$2.5 \cdot 10^3$	410.0	$2.5 \cdot 10^3$	410.0
z	2.8	0	6.2	$-5.1 \cdot 10^{-13}$	4.3	0.79	4.3	0.79
1.0	0	0	$6.7 \cdot 10^{-15}$	$3.6 \cdot 10^{-14}$	52.0	8.6	52.0	8.6

Table 74: Case 1a (cpv MACI64 MR2022b), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	$2.5 \cdot 10^{-9}$	$-2.9 \cdot 10^{-6}$	$-3.5 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$
z^8	7.8	0	8.9	$-2.6 \cdot 10^{-13}$	7.8	$3.9 \cdot 10^{-8}$	7.8	$1.9 \cdot 10^{-9}$
z^7	0	0	$3.5 \cdot 10^{-13}$	$-1.3 \cdot 10^{-13}$	$7.7 \cdot 10^{-8}$	$-7.9 \cdot 10^{-5}$	$-8.9 \cdot 10^{-9}$	$-2.1 \cdot 10^{-5}$
z^6	55.0	0	78.0	$-4.4 \cdot 10^{-12}$	55.0	$8.3 \cdot 10^{-7}$	55.0	$4.1 \cdot 10^{-8}$
z^5	0	0	$2.0 \cdot 10^{-12}$	$3.8 \cdot 10^{-13}$	$2.2 \cdot 10^{-7}$	$-2.1 \cdot 10^{-4}$	$-2.2 \cdot 10^{-8}$	$-5.6 \cdot 10^{-5}$
z^4	61.0	0	110.0	$-7.9 \cdot 10^{-12}$	61.0	$1.6 \cdot 10^{-6}$	61.0	$7.7 \cdot 10^{-8}$
z^3	0	0	$1.7 \cdot 10^{-12}$	$6.9 \cdot 10^{-13}$	$1.2 \cdot 10^{-7}$	$-1.0 \cdot 10^{-4}$	$-1.1 \cdot 10^{-8}$	$-2.8 \cdot 10^{-5}$
z^2	13.0	0	27.0	$-2.4 \cdot 10^{-12}$	13.0	$4.7 \cdot 10^{-7}$	13.0	$2.3 \cdot 10^{-8}$
z	0	0	$2.1 \cdot 10^{-13}$	$1.2 \cdot 10^{-13}$	$9.6 \cdot 10^{-9}$	$-8.2 \cdot 10^{-6}$	$-8.4 \cdot 10^{-10}$	$-2.2 \cdot 10^{-6}$
1.0	0.27	0	0.64	$-6.1 \cdot 10^{-14}$	0.27	$1.3 \cdot 10^{-8}$	0.27	$6.3 \cdot 10^{-10}$
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$5.0 \cdot 10^{-14}$	$-5.1 \cdot 10^{-14}$	$2.1 \cdot 10^{-8}$	$-2.3 \cdot 10^{-5}$	$-2.6 \cdot 10^{-9}$	$-6.1 \cdot 10^{-6}$
z^7	27.0	0	35.0	$-1.6 \cdot 10^{-12}$	27.0	$2.7 \cdot 10^{-7}$	27.0	$1.3 \cdot 10^{-8}$
z^6	0	0	$1.1 \cdot 10^{-12}$	$-4.6 \cdot 10^{-14}$	$1.6 \cdot 10^{-7}$	$-1.6 \cdot 10^{-4}$	$-1.8 \cdot 10^{-8}$	$-4.3 \cdot 10^{-5}$
z^5	71.0	0	110.0	$-7.4 \cdot 10^{-12}$	71.0	$1.4 \cdot 10^{-6}$	71.0	$7.1 \cdot 10^{-8}$
z^4	0	0	$2.3 \cdot 10^{-12}$	$7.6 \cdot 10^{-13}$	$2.0 \cdot 10^{-7}$	$-1.8 \cdot 10^{-4}$	$-1.9 \cdot 10^{-8}$	$-4.8 \cdot 10^{-5}$
z^3	35.0	0	67.0	$-5.6 \cdot 10^{-12}$	35.0	$1.1 \cdot 10^{-6}$	35.0	$5.4 \cdot 10^{-8}$
z^2	0	0	$8.0 \cdot 10^{-13}$	$3.8 \cdot 10^{-13}$	$4.4 \cdot 10^{-8}$	$-3.8 \cdot 10^{-5}$	$-3.9 \cdot 10^{-9}$	$-1.0 \cdot 10^{-5}$
z	2.8	0	6.2	$-5.9 \cdot 10^{-13}$	2.8	$1.2 \cdot 10^{-7}$	2.8	$5.8 \cdot 10^{-9}$
1.0	0	0	$2.3 \cdot 10^{-14}$	$1.9 \cdot 10^{-14}$	$9.3 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$	$-8.0 \cdot 10^{-11}$	$-2.1 \cdot 10^{-7}$

Table 75: Case 1a (cpv PCWIN64 MR2025b), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	$2.5 \cdot 10^{-9}$	$-2.9 \cdot 10^{-6}$	$-3.5 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$
z^8	7.8	0	8.9	$-2.6 \cdot 10^{-13}$	7.8	$3.9 \cdot 10^{-8}$	7.8	$1.9 \cdot 10^{-9}$
z^7	0	0	$3.5 \cdot 10^{-13}$	$-1.3 \cdot 10^{-13}$	$7.7 \cdot 10^{-8}$	$-7.9 \cdot 10^{-5}$	$-8.9 \cdot 10^{-9}$	$-2.1 \cdot 10^{-5}$
z^6	55.0	0	78.0	$-4.4 \cdot 10^{-12}$	55.0	$8.3 \cdot 10^{-7}$	55.0	$4.1 \cdot 10^{-8}$
z^5	0	0	$2.0 \cdot 10^{-12}$	$3.8 \cdot 10^{-13}$	$2.2 \cdot 10^{-7}$	$-2.1 \cdot 10^{-4}$	$-2.2 \cdot 10^{-8}$	$-5.6 \cdot 10^{-5}$
z^4	61.0	0	110.0	$-7.9 \cdot 10^{-12}$	61.0	$1.6 \cdot 10^{-6}$	61.0	$7.7 \cdot 10^{-8}$
z^3	0	0	$1.7 \cdot 10^{-12}$	$6.9 \cdot 10^{-13}$	$1.2 \cdot 10^{-7}$	$-1.0 \cdot 10^{-4}$	$-1.1 \cdot 10^{-8}$	$-2.8 \cdot 10^{-5}$
z^2	13.0	0	27.0	$-2.4 \cdot 10^{-12}$	13.0	$4.7 \cdot 10^{-7}$	13.0	$2.3 \cdot 10^{-8}$
z	0	0	$2.1 \cdot 10^{-13}$	$1.2 \cdot 10^{-13}$	$9.6 \cdot 10^{-9}$	$-8.2 \cdot 10^{-6}$	$-8.4 \cdot 10^{-10}$	$-2.2 \cdot 10^{-6}$
1.0	0.27	0	0.64	$-6.1 \cdot 10^{-14}$	0.27	$1.3 \cdot 10^{-8}$	0.27	$6.3 \cdot 10^{-10}$
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$5.0 \cdot 10^{-14}$	$-5.1 \cdot 10^{-14}$	$2.1 \cdot 10^{-8}$	$-2.3 \cdot 10^{-5}$	$-2.6 \cdot 10^{-9}$	$-6.1 \cdot 10^{-6}$
z^7	27.0	0	35.0	$-1.6 \cdot 10^{-12}$	27.0	$2.7 \cdot 10^{-7}$	27.0	$1.3 \cdot 10^{-8}$
z^6	0	0	$1.1 \cdot 10^{-12}$	$-4.6 \cdot 10^{-14}$	$1.6 \cdot 10^{-7}$	$-1.6 \cdot 10^{-4}$	$-1.8 \cdot 10^{-8}$	$-4.3 \cdot 10^{-5}$
z^5	71.0	0	110.0	$-7.4 \cdot 10^{-12}$	71.0	$1.4 \cdot 10^{-6}$	71.0	$7.1 \cdot 10^{-8}$
z^4	0	0	$2.3 \cdot 10^{-12}$	$7.6 \cdot 10^{-13}$	$2.0 \cdot 10^{-7}$	$-1.8 \cdot 10^{-4}$	$-1.9 \cdot 10^{-8}$	$-4.8 \cdot 10^{-5}$
z^3	35.0	0	67.0	$-5.6 \cdot 10^{-12}$	35.0	$1.1 \cdot 10^{-6}$	35.0	$5.4 \cdot 10^{-8}$
z^2	0	0	$8.0 \cdot 10^{-13}$	$3.8 \cdot 10^{-13}$	$4.4 \cdot 10^{-8}$	$-3.8 \cdot 10^{-5}$	$-3.9 \cdot 10^{-9}$	$-1.0 \cdot 10^{-5}$
z	2.8	0	6.2	$-5.9 \cdot 10^{-13}$	2.8	$1.2 \cdot 10^{-7}$	2.8	$5.8 \cdot 10^{-9}$
1.0	0	0	$2.3 \cdot 10^{-14}$	$1.9 \cdot 10^{-14}$	$9.3 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$	$-8.0 \cdot 10^{-11}$	$-2.1 \cdot 10^{-7}$

Table 76: Case 1a (AlexReis PCWIN64 MR2025b), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

8.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	0	$1.11 \cdot 10^{-15} - 1.09 \cdot 10^{-14}i$	$5.97 \cdot 10^{-10} - 0.153i$	$-1.86 \cdot 10^{-10} + 0.153i$
2.0	$-1.11 \cdot 10^{-15} + 4.91i$	$-4.13 \cdot 10^{-15} + 0.338i$	$-2.87 \cdot 10^{-10} + 0.153i$	$1.01 \cdot 10^{-9} - 0.153i$
3.0	$-1.11 \cdot 10^{-15} - 4.91i$	$-3.5 \cdot 10^{-15} - 0.338i$	$1.65 \cdot 10^{-9} - 0.5i$	$-1.45 \cdot 10^{-9} + 0.5i$
4.0	$-5.55 \cdot 10^{-17} + 1.5i$	$-1.64 \cdot 10^{-14} + 0.789i$	$-1.26 \cdot 10^{-9} + 0.5i$	$2.51 \cdot 10^{-9} - 0.5i$
5.0	$-5.55 \cdot 10^{-17} - 1.5i$	$-6.96 \cdot 10^{-15} - 0.789i$	$3.35 \cdot 10^{-9} - 1.03i$	$-3.19 \cdot 10^{-9} + 1.03i$
6.0	$-1.39 \cdot 10^{-17} + 0.727i$	$-1.08 \cdot 10^{-14} + 1.67i$	$-2.7 \cdot 10^{-9} + 1.03i$	$5.04 \cdot 10^{-9} - 1.03i$
7.0	$-1.39 \cdot 10^{-17} - 0.727i$	$-2.77 \cdot 10^{-14} - 1.67i$	$7.99 \cdot 10^{-9} - 2.38i$	$-6.5 \cdot 10^{-9} + 2.38i$
8.0	$1.73 \cdot 10^{-18} + 0.315i$	$-5.92 \cdot 10^{-14} + 5.6i$	$-6.02 \cdot 10^{-9} + 2.38i$	$1.26 \cdot 10^{-8} - 2.38i$
9.0	$1.73 \cdot 10^{-18} - 0.315i$	$-2.58 \cdot 10^{-14} - 5.6i$	$1.63 \cdot 10^4 - 1.07 \cdot 10^7i$	$7310.0 + 4.0 \cdot 10^6i$

Table 77: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4: poles.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.2 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.2 \cdot 10^{-11}$
1.0	0	$-1.08 \cdot 10^{-15} - 5.74 \cdot 10^{-15}i$	$-7.88 \cdot 10^{-4} + 0.153i$	$-7.88 \cdot 10^{-4} + 0.153i$
2.0	$5.55 \cdot 10^{-16} + 4.91i$	$1.48 \cdot 10^{-15} + 0.338i$	$-8.16 \cdot 10^{-4} - 0.153i$	$-8.16 \cdot 10^{-4} - 0.153i$
3.0	$5.55 \cdot 10^{-16} - 4.91i$	$-1.07 \cdot 10^{-15} - 0.338i$	$-9.03 \cdot 10^{-4} + 0.5i$	$-9.03 \cdot 10^{-4} + 0.5i$
4.0	$1.11 \cdot 10^{-16} + 1.5i$	$8.32 \cdot 10^{-15} + 0.789i$	$-0.001 - 0.5i$	$-0.001 - 0.5i$
5.0	$1.11 \cdot 10^{-16} - 1.5i$	$-6.52 \cdot 10^{-15} - 0.789i$	$-0.0016 - 1.03i$	$-0.0016 - 1.03i$
6.0	$2.78 \cdot 10^{-17} + 0.727i$	$2.26 \cdot 10^{-14} + 1.67i$	$-0.00138 + 1.03i$	$-0.00138 + 1.03i$
7.0	$2.78 \cdot 10^{-17} - 0.727i$	$-1.99 \cdot 10^{-14} - 1.67i$	$-0.00476 - 2.38i$	$-0.00476 - 2.38i$
8.0	$1.73 \cdot 10^{-18} + 0.315i$	$-6.04 \cdot 10^{-14} - 5.6i$	$-0.0042 + 2.38i$	$-0.0042 + 2.38i$
9.0	$1.73 \cdot 10^{-18} - 0.315i$	$1.06 \cdot 10^{-13} + 5.6i$	$-1500.0 - 247.0i$	$-1500.0 - 246.0i$

Table 78: Case 1a (cpv MACI64 MR2022b), $r = 9$, Z4: poles.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	0	$-3.67 \cdot 10^{-15} - 3.11 \cdot 10^{-15}i$	$-3.27 \cdot 10^{-10} + 2.77 \cdot 10^{-7}i$	$2.81 \cdot 10^{-11} + 7.54 \cdot 10^{-8}i$
2.0	$-1.11 \cdot 10^{-15} + 4.91i$	$-2.8 \cdot 10^{-15} + 0.338i$	$1.28 \cdot 10^{-9} - 0.315i$	$1.14 \cdot 10^{-10} - 0.315i$
3.0	$-1.11 \cdot 10^{-15} - 4.91i$	$-5.57 \cdot 10^{-15} - 0.338i$	$-2.03 \cdot 10^{-9} + 0.315i$	$-4.97 \cdot 10^{-11} + 0.315i$
4.0	$-5.55 \cdot 10^{-17} + 1.5i$	$3.63 \cdot 10^{-15} + 0.789i$	$3.22 \cdot 10^{-9} - 0.727i$	$2.35 \cdot 10^{-10} - 0.727i$
5.0	$-5.55 \cdot 10^{-17} - 1.5i$	$-1.15 \cdot 10^{-14} - 0.789i$	$-4.32 \cdot 10^{-9} + 0.727i$	$-1.36 \cdot 10^{-10} + 0.727i$
6.0	$-1.39 \cdot 10^{-17} + 0.727i$	$-2.13 \cdot 10^{-14} - 1.67i$	$6.52 \cdot 10^{-9} - 1.5i$	$5.02 \cdot 10^{-10} - 1.5i$
7.0	$-1.39 \cdot 10^{-17} - 0.727i$	$9.08 \cdot 10^{-15} + 1.67i$	$-8.83 \cdot 10^{-9} + 1.5i$	$-2.53 \cdot 10^{-10} + 1.5i$
8.0	$1.73 \cdot 10^{-18} + 0.315i$	$-1.42 \cdot 10^{-13} - 5.6i$	$1.66 \cdot 10^{-8} - 4.91i$	$2.31 \cdot 10^{-9} - 4.91i$
9.0	$1.73 \cdot 10^{-18} - 0.315i$	$1.23 \cdot 10^{-13} + 5.6i$	$-3.32 \cdot 10^{-8} + 4.91i$	$-1.3 \cdot 10^{-10} + 4.91i$

Table 79: Case 1a (cpv PCWIN64 MR2025b), $r = 9$, Z4: poles.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	0	$-3.67 \cdot 10^{-15} - 3.11 \cdot 10^{-15}i$	$-3.27 \cdot 10^{-10} + 2.77 \cdot 10^{-7}i$	$2.81 \cdot 10^{-11} + 7.54 \cdot 10^{-8}i$
2.0	$-1.11 \cdot 10^{-15} + 4.91i$	$-2.8 \cdot 10^{-15} + 0.338i$	$1.28 \cdot 10^{-9} - 0.315i$	$1.14 \cdot 10^{-10} - 0.315i$
3.0	$-1.11 \cdot 10^{-15} - 4.91i$	$-5.57 \cdot 10^{-15} - 0.338i$	$-2.03 \cdot 10^{-9} + 0.315i$	$-4.97 \cdot 10^{-11} + 0.315i$
4.0	$-5.55 \cdot 10^{-17} + 1.5i$	$3.63 \cdot 10^{-15} + 0.789i$	$3.22 \cdot 10^{-9} - 0.727i$	$2.35 \cdot 10^{-10} - 0.727i$
5.0	$-5.55 \cdot 10^{-17} - 1.5i$	$-1.15 \cdot 10^{-14} - 0.789i$	$-4.32 \cdot 10^{-9} + 0.727i$	$-1.36 \cdot 10^{-10} + 0.727i$
6.0	$-1.39 \cdot 10^{-17} + 0.727i$	$-2.13 \cdot 10^{-14} - 1.67i$	$6.52 \cdot 10^{-9} - 1.5i$	$5.02 \cdot 10^{-10} - 1.5i$
7.0	$-1.39 \cdot 10^{-17} - 0.727i$	$9.08 \cdot 10^{-15} + 1.67i$	$-8.83 \cdot 10^{-9} + 1.5i$	$-2.53 \cdot 10^{-10} + 1.5i$
8.0	$1.73 \cdot 10^{-18} + 0.315i$	$-1.42 \cdot 10^{-13} - 5.6i$	$1.66 \cdot 10^{-8} - 4.91i$	$2.31 \cdot 10^{-9} - 4.91i$
9.0	$1.73 \cdot 10^{-18} - 0.315i$	$1.23 \cdot 10^{-13} + 5.6i$	$-3.32 \cdot 10^{-8} + 4.91i$	$-1.3 \cdot 10^{-10} + 4.91i$

Table 80: Case 1a (AlexReis PCWIN64 MR2025b), $r = 9$, Z4: poles.

8.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	NaN	NaN	$1.51 \cdot 10^{-10} + 6.79 \cdot 10^{-8}i$	$4.01 \cdot 10^{-10} - 1.92 \cdot 10^{-7}i$
2.0	$-8.33 \cdot 10^{-17} + 2.38i$	$8.36 \cdot 10^{-16} + 0.163i$	$1.08 \cdot 10^{-9} - 0.315i$	$-7.87 \cdot 10^{-10} + 0.315i$
3.0	$-8.33 \cdot 10^{-17} - 2.38i$	$-1.5 \cdot 10^{-15} - 0.163i$	$-7.46 \cdot 10^{-10} + 0.315i$	$1.69 \cdot 10^{-9} - 0.315i$
4.0	$2.78 \cdot 10^{-17} + 1.03i$	$-1.15 \cdot 10^{-14} + 0.539i$	$2.36 \cdot 10^{-9} - 0.727i$	$-2.22 \cdot 10^{-9} + 0.727i$
5.0	$2.78 \cdot 10^{-17} - 1.03i$	$-3.68 \cdot 10^{-15} - 0.539i$	$-1.88 \cdot 10^{-9} + 0.727i$	$3.55 \cdot 10^{-9} - 0.727i$
6.0	$-6.07 \cdot 10^{-18} + 0.5i$	$-1.44 \cdot 10^{-14} + 1.13i$	$4.9 \cdot 10^{-9} - 1.5i$	$-4.53 \cdot 10^{-9} + 1.5i$
7.0	$-6.07 \cdot 10^{-18} - 0.5i$	$-1.54 \cdot 10^{-14} - 1.13i$	$-3.91 \cdot 10^{-9} + 1.5i$	$7.48 \cdot 10^{-9} - 1.5i$
8.0	$1.99 \cdot 10^{-17} + 0.153i$	$-2.2 \cdot 10^{-14} + 2.68i$	$1.81 \cdot 10^{-8} - 4.91i$	$-8.23 \cdot 10^{-9} + 4.91i$
9.0	$1.99 \cdot 10^{-17} - 0.153i$	$-3.49 \cdot 10^{-14} - 2.68i$	$-1.09 \cdot 10^{-8} + 4.91i$	$3.12 \cdot 10^{-8} - 4.91i$

Table 81: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4: zeros.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.2 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.2 \cdot 10^{-11}$
1.0	NaN	NaN	$-7.86 \cdot 10^{-4} - 2.65 \cdot 10^{-5}i$	$-7.86 \cdot 10^{-4} - 2.68 \cdot 10^{-5}i$
2.0	$-2.78 \cdot 10^{-16} + 2.38i$	$-2.12 \cdot 10^{-15} + 0.163i$	$-8.23 \cdot 10^{-4} + 0.315i$	$-8.23 \cdot 10^{-4} + 0.315i$
3.0	$-2.78 \cdot 10^{-16} - 2.38i$	$4.75 \cdot 10^{-16} - 0.163i$	$-8.82 \cdot 10^{-4} - 0.315i$	$-8.82 \cdot 10^{-4} - 0.315i$
4.0	$-4.16 \cdot 10^{-17} + 1.03i$	$6.99 \cdot 10^{-15} + 0.539i$	$-0.00121 - 0.726i$	$-0.00121 - 0.726i$
5.0	$-4.16 \cdot 10^{-17} - 1.03i$	$-5.34 \cdot 10^{-15} - 0.539i$	$-0.00106 + 0.726i$	$-0.00106 + 0.726i$
6.0	$1.73 \cdot 10^{-18} + 0.5i$	$8.08 \cdot 10^{-15} + 1.13i$	$-0.00243 - 1.5i$	$-0.00243 - 1.5i$
7.0	$1.73 \cdot 10^{-18} - 0.5i$	$-8.11 \cdot 10^{-15} - 1.13i$	$-0.00209 + 1.5i$	$-0.00209 + 1.5i$
8.0	$-8.67 \cdot 10^{-19} + 0.153i$	$-3.84 \cdot 10^{-14} - 2.68i$	$-0.017 - 4.91i$	$-0.017 - 4.91i$
9.0	$-8.67 \cdot 10^{-19} - 0.153i$	$5.6 \cdot 10^{-14} + 2.68i$	$-0.0159 + 4.91i$	$-0.0159 + 4.91i$

Table 82: Case 1a (cpv MACI64 MR2022b), $r = 9$, Z4: zeros.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	NaN	NaN	$4.67 \cdot 10^{-10} - 0.153i$	$6.88 \cdot 10^{-11} - 0.153i$
2.0	$1.11 \cdot 10^{-16} + 2.38i$	$-3.53 \cdot 10^{-15} + 0.163i$	$-1.14 \cdot 10^{-9} + 0.153i$	$-1.08 \cdot 10^{-11} + 0.153i$
3.0	$1.11 \cdot 10^{-16} - 2.38i$	$-4.0 \cdot 10^{-15} - 0.163i$	$2.17 \cdot 10^{-9} - 0.5i$	$1.67 \cdot 10^{-10} - 0.5i$
4.0	$6.94 \cdot 10^{-17} + 1.03i$	$2.91 \cdot 10^{-16} + 0.539i$	$-3.05 \cdot 10^{-9} + 0.5i$	$-9.07 \cdot 10^{-11} + 0.5i$
5.0	$6.94 \cdot 10^{-17} - 1.03i$	$-8.48 \cdot 10^{-15} - 0.539i$	$4.57 \cdot 10^{-9} - 1.03i$	$3.34 \cdot 10^{-10} - 1.03i$
6.0	$2.08 \cdot 10^{-17} + 0.5i$	$4.72 \cdot 10^{-15} + 1.13i$	$-6.06 \cdot 10^{-9} + 1.03i$	$-1.89 \cdot 10^{-10} + 1.03i$
7.0	$2.08 \cdot 10^{-17} - 0.5i$	$-1.45 \cdot 10^{-14} - 1.13i$	$9.83 \cdot 10^{-9} - 2.38i$	$8.7 \cdot 10^{-10} - 2.38i$
8.0	$5.2 \cdot 10^{-18} + 0.153i$	$-4.7 \cdot 10^{-14} - 2.68i$	$-1.44 \cdot 10^{-8} + 2.38i$	$-3.2 \cdot 10^{-10} + 2.38i$
9.0	$5.2 \cdot 10^{-18} - 0.153i$	$3.31 \cdot 10^{-14} + 2.68i$	$-2310.0 - 2.66 \cdot 10^6i$	$4370.0 - 9.93 \cdot 10^6i$

Table 83: Case 1a (cpv PCWIN64 MR2025b), $r = 9$, Z4: zeros.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	NaN	NaN	$4.67 \cdot 10^{-10} - 0.153i$	$6.88 \cdot 10^{-11} - 0.153i$
2.0	$1.11 \cdot 10^{-16} + 2.38i$	$-3.53 \cdot 10^{-15} + 0.163i$	$-1.14 \cdot 10^{-9} + 0.153i$	$-1.08 \cdot 10^{-11} + 0.153i$
3.0	$1.11 \cdot 10^{-16} - 2.38i$	$-4.0 \cdot 10^{-15} - 0.163i$	$2.17 \cdot 10^{-9} - 0.5i$	$1.67 \cdot 10^{-10} - 0.5i$
4.0	$6.94 \cdot 10^{-17} + 1.03i$	$2.91 \cdot 10^{-16} + 0.539i$	$-3.05 \cdot 10^{-9} + 0.5i$	$-9.07 \cdot 10^{-11} + 0.5i$
5.0	$6.94 \cdot 10^{-17} - 1.03i$	$-8.48 \cdot 10^{-15} - 0.539i$	$4.57 \cdot 10^{-9} - 1.03i$	$3.34 \cdot 10^{-10} - 1.03i$
6.0	$2.08 \cdot 10^{-17} + 0.5i$	$4.72 \cdot 10^{-15} + 1.13i$	$-6.06 \cdot 10^{-9} + 1.03i$	$-1.89 \cdot 10^{-10} + 1.03i$
7.0	$2.08 \cdot 10^{-17} - 0.5i$	$-1.45 \cdot 10^{-14} - 1.13i$	$9.83 \cdot 10^{-9} - 2.38i$	$8.7 \cdot 10^{-10} - 2.38i$
8.0	$5.2 \cdot 10^{-18} + 0.153i$	$-4.7 \cdot 10^{-14} - 2.68i$	$-1.44 \cdot 10^{-8} + 2.38i$	$-3.2 \cdot 10^{-10} + 2.38i$
9.0	$5.2 \cdot 10^{-18} - 0.153i$	$3.31 \cdot 10^{-14} + 2.68i$	$-2310.0 - 2.66 \cdot 10^6i$	$4370.0 - 9.93 \cdot 10^6i$

Table 84: Case 1a (AlexReis PCWIN64 MR2025b), $r = 9$, Z4: zeros.

8.4 Poles and zeros visualisation

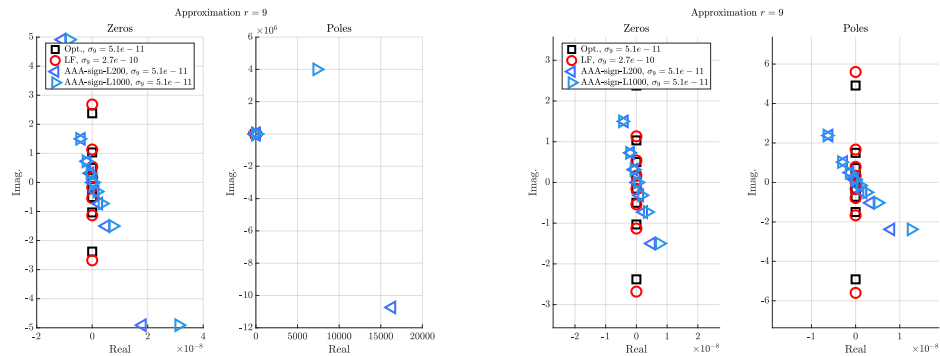


Figure 13: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

8.5 Approximation visualization

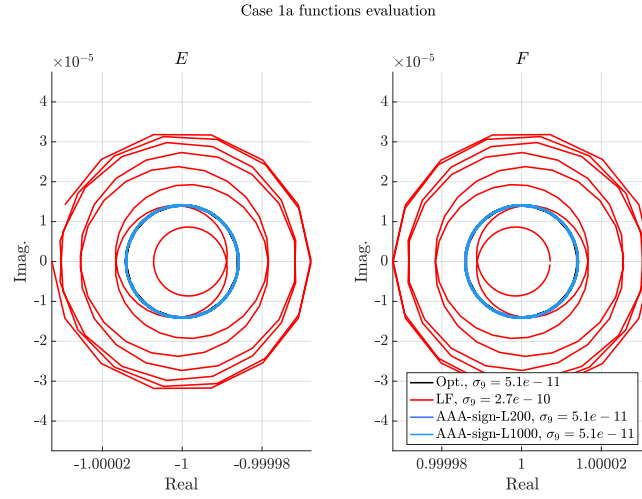


Figure 14: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4. E and F evaluations.

9 Approximation objective $r = 10$

9.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$7.7 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$	$7.7 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$
z^9	8.7	0	NaN	NaN	8.7	$4.6 \cdot 10^{-3}$	8.7	$4.6 \cdot 10^{-3}$
z^8	0	0	0	0	0.14	0.07	0.14	0.069
z^7	78.0	0	NaN	NaN	78.0	0.13	78.0	0.13
z^6	0	0	0	0	0.036	0.12	0.035	0.12
z^5	120.0	0	NaN	NaN	120.0	0.34	120.0	0.34
z^4	0	0	0	0	-0.31	$-3.8 \cdot 10^{-3}$	-0.31	$-7.4 \cdot 10^{-3}$
z^3	44.0	0	NaN	NaN	44.0	0.17	44.0	0.17
z^2	0	0	0	0	-0.1	-0.016	-0.11	-0.017
z	2.7	0	NaN	NaN	2.8	0.014	2.8	0.014
1.0	0	0	0	0	$-2.7 \cdot 10^{-3}$	$-5.5 \cdot 10^{-4}$	$-2.7 \cdot 10^{-3}$	$-5.6 \cdot 10^{-4}$
z^{10}	1.0	0	NaN	NaN	1.0	$-3.4 \cdot 10^{-17}$	1.0	0
z^9	0	0	0	0	0.051	0.023	0.051	0.022
z^8	34.0	0	NaN	NaN	34.0	0.036	34.0	0.036
z^7	0	0	0	0	0.17	0.12	0.17	0.12
z^6	120.0	0	NaN	NaN	120.0	0.26	120.0	0.26
z^5	0	0	0	0	-0.2	0.062	-0.2	0.057
z^4	89.0	0	NaN	NaN	90.0	0.3	90.0	0.3
z^3	0	0	0	0	-0.24	-0.026	-0.24	-0.028
z^2	14.0	0	NaN	NaN	14.0	0.064	14.0	0.064
z	0	0	0	0	-0.026	$-4.7 \cdot 10^{-3}$	-0.026	$-4.9 \cdot 10^{-3}$
1.0	0.24	0	NaN	NaN	0.24	$1.4 \cdot 10^{-3}$	0.24	$1.4 \cdot 10^{-3}$

Table 85: Case 1a (cpv MACA64 MR2023b), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$4.2 \cdot 10^{-8}$	$-3.6 \cdot 10^{-5}$	$7.5 \cdot 10^{-8}$	$4.1 \cdot 10^{-6}$
z^9	8.7	0	8.9	$-1.1 \cdot 10^{-13}$	8.7	$-6.5 \cdot 10^{-7}$	8.7	$1.3 \cdot 10^{-7}$
z^8	0	0	$-1.6 \cdot 10^{-14}$	$-3.8 \cdot 10^{-12}$	$1.3 \cdot 10^{-6}$	$-1.2 \cdot 10^{-3}$	$2.6 \cdot 10^{-6}$	$1.4 \cdot 10^{-4}$
z^7	78.0	0	85.0	$-3.7 \cdot 10^{-12}$	78.0	$-1.8 \cdot 10^{-5}$	78.0	$3.5 \cdot 10^{-6}$
z^6	0	0	$1.9 \cdot 10^{-12}$	$-2.6 \cdot 10^{-11}$	$4.0 \cdot 10^{-6}$	$-4.3 \cdot 10^{-3}$	$9.1 \cdot 10^{-6}$	$4.9 \cdot 10^{-4}$
z^5	120.0	0	140.0	$-7.9 \cdot 10^{-12}$	120.0	$-4.8 \cdot 10^{-5}$	120.0	$9.3 \cdot 10^{-6}$
z^4	0	0	$4.0 \cdot 10^{-12}$	$-3.0 \cdot 10^{-11}$	$2.6 \cdot 10^{-6}$	$-3.2 \cdot 10^{-3}$	$6.9 \cdot 10^{-6}$	$3.7 \cdot 10^{-4}$
z^3	44.0	0	54.0	$-2.9 \cdot 10^{-12}$	44.0	$-2.4 \cdot 10^{-5}$	44.0	$4.7 \cdot 10^{-6}$
z^2	0	0	$1.2 \cdot 10^{-12}$	$-6.9 \cdot 10^{-12}$	$3.7 \cdot 10^{-7}$	$-5.2 \cdot 10^{-4}$	$1.1 \cdot 10^{-6}$	$6.0 \cdot 10^{-5}$
z	2.7	0	3.5	$-1.3 \cdot 10^{-13}$	2.7	$-2.0 \cdot 10^{-6}$	2.7	$3.9 \cdot 10^{-7}$
1.0	0	0	$3.2 \cdot 10^{-14}$	$-1.5 \cdot 10^{-13}$	$5.1 \cdot 10^{-9}$	$-8.7 \cdot 10^{-6}$	$1.9 \cdot 10^{-8}$	$1.0 \cdot 10^{-6}$
z^{10}	1.0	0	1.0	0	1.0	0	1.0	0
z^9	0	0	$-4.1 \cdot 10^{-14}$	$-5.2 \cdot 10^{-13}$	$3.5 \cdot 10^{-7}$	$-3.1 \cdot 10^{-4}$	$6.6 \cdot 10^{-7}$	$3.5 \cdot 10^{-5}$
z^8	34.0	0	36.0	$-1.1 \cdot 10^{-12}$	34.0	$-5.1 \cdot 10^{-6}$	34.0	$1.0 \cdot 10^{-6}$
z^7	0	0	$5.3 \cdot 10^{-13}$	$-1.3 \cdot 10^{-11}$	$2.8 \cdot 10^{-6}$	$-2.8 \cdot 10^{-3}$	$6.0 \cdot 10^{-6}$	$3.2 \cdot 10^{-4}$
z^6	120.0	0	130.0	$-6.8 \cdot 10^{-12}$	120.0	$-3.7 \cdot 10^{-5}$	120.0	$7.1 \cdot 10^{-6}$
z^5	0	0	$3.6 \cdot 10^{-12}$	$-3.4 \cdot 10^{-11}$	$3.9 \cdot 10^{-6}$	$-4.5 \cdot 10^{-3}$	$9.5 \cdot 10^{-6}$	$5.1 \cdot 10^{-4}$
z^4	89.0	0	110.0	$-6.0 \cdot 10^{-12}$	89.0	$-4.2 \cdot 10^{-5}$	89.0	$8.1 \cdot 10^{-6}$
z^3	0	0	$2.8 \cdot 10^{-12}$	$-1.8 \cdot 10^{-11}$	$1.2 \cdot 10^{-6}$	$-1.6 \cdot 10^{-3}$	$3.4 \cdot 10^{-6}$	$1.8 \cdot 10^{-4}$
z^2	14.0	0	18.0	$-8.5 \cdot 10^{-13}$	14.0	$-9.1 \cdot 10^{-6}$	14.0	$1.8 \cdot 10^{-6}$
z	0	0	$2.9 \cdot 10^{-13}$	$-1.5 \cdot 10^{-12}$	$6.5 \cdot 10^{-8}$	$-1.0 \cdot 10^{-4}$	$2.2 \cdot 10^{-7}$	$1.2 \cdot 10^{-5}$
1.0	0.24	0	0.31	$-8.1 \cdot 10^{-15}$	0.24	$-1.9 \cdot 10^{-7}$	0.24	$3.7 \cdot 10^{-8}$

Table 86: Case 1a (cpv MACI64 MR2022b), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$2.4 \cdot 10^{-9}$	$4.3 \cdot 10^{-6}$	$2.2 \cdot 10^{-9}$	$-3.9 \cdot 10^{-8}$
z^9	8.7	0	NaN	NaN	8.7	$-1.0 \cdot 10^{-7}$	8.7	$-1.3 \cdot 10^{-8}$
z^8	0	0	0	0	$5.9 \cdot 10^{-8}$	$1.5 \cdot 10^{-4}$	$7.2 \cdot 10^{-8}$	$-1.4 \cdot 10^{-6}$
z^7	78.0	0	NaN	NaN	78.0	$-2.8 \cdot 10^{-6}$	78.0	$-3.7 \cdot 10^{-7}$
z^6	0	0	0	0	$1.4 \cdot 10^{-7}$	$5.0 \cdot 10^{-4}$	$2.4 \cdot 10^{-7}$	$-5.2 \cdot 10^{-6}$
z^5	120.0	0	NaN	NaN	120.0	$-7.5 \cdot 10^{-6}$	120.0	$-9.7 \cdot 10^{-7}$
z^4	0	0	0	0	$7.2 \cdot 10^{-8}$	$3.8 \cdot 10^{-4}$	$1.8 \cdot 10^{-7}$	$-4.2 \cdot 10^{-6}$
z^3	44.0	0	NaN	NaN	44.0	$-3.7 \cdot 10^{-6}$	44.0	$-4.9 \cdot 10^{-7}$
z^2	0	0	0	0	$7.8 \cdot 10^{-9}$	$6.0 \cdot 10^{-5}$	$2.8 \cdot 10^{-8}$	$-7.1 \cdot 10^{-7}$
z	2.7	0	NaN	NaN	2.7	$-3.0 \cdot 10^{-7}$	2.7	$-3.9 \cdot 10^{-8}$
1.0	0	0	0	0	$9.9 \cdot 10^{-11}$	$1.0 \cdot 10^{-6}$	$4.6 \cdot 10^{-10}$	$-1.3 \cdot 10^{-8}$
z^{10}	1.0	0	NaN	NaN	1.0	0	1.0	0
z^9	0	0	0	0	$1.8 \cdot 10^{-8}$	$3.7 \cdot 10^{-5}$	$1.9 \cdot 10^{-8}$	$-3.5 \cdot 10^{-7}$
z^8	34.0	0	NaN	NaN	34.0	$-8.1 \cdot 10^{-7}$	34.0	$-1.1 \cdot 10^{-7}$
z^7	0	0	0	0	$1.1 \cdot 10^{-7}$	$3.3 \cdot 10^{-4}$	$1.6 \cdot 10^{-7}$	$-3.3 \cdot 10^{-6}$
z^6	120.0	0	NaN	NaN	120.0	$-5.7 \cdot 10^{-6}$	120.0	$-7.4 \cdot 10^{-7}$
z^5	0	0	0	0	$1.2 \cdot 10^{-7}$	$5.2 \cdot 10^{-4}$	$2.5 \cdot 10^{-7}$	$-5.6 \cdot 10^{-6}$
z^4	89.0	0	NaN	NaN	89.0	$-6.5 \cdot 10^{-6}$	89.0	$-8.4 \cdot 10^{-7}$
z^3	0	0	0	0	$2.9 \cdot 10^{-8}$	$1.9 \cdot 10^{-4}$	$8.7 \cdot 10^{-8}$	$-2.1 \cdot 10^{-6}$
z^2	14.0	0	NaN	NaN	14.0	$-1.4 \cdot 10^{-6}$	14.0	$-1.8 \cdot 10^{-7}$
z	0	0	0	0	$1.3 \cdot 10^{-9}$	$1.2 \cdot 10^{-5}$	$5.4 \cdot 10^{-9}$	$-1.4 \cdot 10^{-7}$
1.0	0.24	0	NaN	NaN	0.24	$-2.9 \cdot 10^{-8}$	0.24	$-3.8 \cdot 10^{-9}$

Table 87: Case 1a (cpv PCWIN64 MR2025b), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$2.4 \cdot 10^{-9}$	$4.3 \cdot 10^{-6}$	$2.2 \cdot 10^{-9}$	$-3.9 \cdot 10^{-8}$
z^9	8.7	0	NaN	NaN	8.7	$-1.0 \cdot 10^{-7}$	8.7	$-1.3 \cdot 10^{-8}$
z^8	0	0	0	0	$5.9 \cdot 10^{-8}$	$1.5 \cdot 10^{-4}$	$7.2 \cdot 10^{-8}$	$-1.4 \cdot 10^{-6}$
z^7	78.0	0	NaN	NaN	78.0	$-2.8 \cdot 10^{-6}$	78.0	$-3.7 \cdot 10^{-7}$
z^6	0	0	0	0	$1.4 \cdot 10^{-7}$	$5.0 \cdot 10^{-4}$	$2.4 \cdot 10^{-7}$	$-5.2 \cdot 10^{-6}$
z^5	120.0	0	NaN	NaN	120.0	$-7.5 \cdot 10^{-6}$	120.0	$-9.7 \cdot 10^{-7}$
z^4	0	0	0	0	$7.2 \cdot 10^{-8}$	$3.8 \cdot 10^{-4}$	$1.8 \cdot 10^{-7}$	$-4.2 \cdot 10^{-6}$
z^3	44.0	0	NaN	NaN	44.0	$-3.7 \cdot 10^{-6}$	44.0	$-4.9 \cdot 10^{-7}$
z^2	0	0	0	0	$7.8 \cdot 10^{-9}$	$6.0 \cdot 10^{-5}$	$2.8 \cdot 10^{-8}$	$-7.1 \cdot 10^{-7}$
z	2.7	0	NaN	NaN	2.7	$-3.0 \cdot 10^{-7}$	2.7	$-3.9 \cdot 10^{-8}$
1.0	0	0	0	0	$9.9 \cdot 10^{-11}$	$1.0 \cdot 10^{-6}$	$4.6 \cdot 10^{-10}$	$-1.3 \cdot 10^{-8}$
z^{10}	1.0	0	NaN	NaN	1.0	0	1.0	0
z^9	0	0	0	0	$1.8 \cdot 10^{-8}$	$3.7 \cdot 10^{-5}$	$1.9 \cdot 10^{-8}$	$-3.5 \cdot 10^{-7}$
z^8	34.0	0	NaN	NaN	34.0	$-8.1 \cdot 10^{-7}$	34.0	$-1.1 \cdot 10^{-7}$
z^7	0	0	0	0	$1.1 \cdot 10^{-7}$	$3.3 \cdot 10^{-4}$	$1.6 \cdot 10^{-7}$	$-3.3 \cdot 10^{-6}$
z^6	120.0	0	NaN	NaN	120.0	$-5.7 \cdot 10^{-6}$	120.0	$-7.4 \cdot 10^{-7}$
z^5	0	0	0	0	$1.2 \cdot 10^{-7}$	$5.2 \cdot 10^{-4}$	$2.5 \cdot 10^{-7}$	$-5.6 \cdot 10^{-6}$
z^4	89.0	0	NaN	NaN	89.0	$-6.5 \cdot 10^{-6}$	89.0	$-8.4 \cdot 10^{-7}$
z^3	0	0	0	0	$2.9 \cdot 10^{-8}$	$1.9 \cdot 10^{-4}$	$8.7 \cdot 10^{-8}$	$-2.1 \cdot 10^{-6}$
z^2	14.0	0	NaN	NaN	14.0	$-1.4 \cdot 10^{-6}$	14.0	$-1.8 \cdot 10^{-7}$
z	0	0	0	0	$1.3 \cdot 10^{-9}$	$1.2 \cdot 10^{-5}$	$5.4 \cdot 10^{-9}$	$-1.4 \cdot 10^{-7}$
1.0	0.24	0	NaN	NaN	0.24	$-2.9 \cdot 10^{-8}$	0.24	$-3.8 \cdot 10^{-9}$

Table 88: Case 1a (AlexReis PCWIN64 MR2025b), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

9.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$
1.0	$2.22 \cdot 10^{-16} + 5.47i$	$1.27 \cdot 10^{-14} + 0.141i$	$0.00103 - 0.137i$	$0.00103 - 0.137i$
2.0	$2.22 \cdot 10^{-16} - 5.47i$	$1.93 \cdot 10^{-14} - 0.141i$	$8.67 \cdot 10^{-4} + 0.138i$	$8.68 \cdot 10^{-4} + 0.138i$
3.0	$2.78 \cdot 10^{-16} + 1.7i$	$1.97 \cdot 10^{-14} + 0.453i$	$0.00103 - 0.442i$	$0.00103 - 0.442i$
4.0	$2.78 \cdot 10^{-16} - 1.7i$	$3.14 \cdot 10^{-14} - 0.453i$	$5.19 \cdot 10^{-4} + 0.442i$	$5.2 \cdot 10^{-4} + 0.442i$
5.0	$-1.25 \cdot 10^{-16} + 0.866i$	$9.09 \cdot 10^{-16} + 0.891i$	$-2.36 \cdot 10^{-4} + 0.867i$	$-2.35 \cdot 10^{-4} + 0.867i$
6.0	$-1.25 \cdot 10^{-16} - 0.866i$	$2.9 \cdot 10^{-14} - 0.891i$	$7.39 \cdot 10^{-4} - 0.868i$	$7.4 \cdot 10^{-4} - 0.868i$
7.0	$-6.94 \cdot 10^{-18} + 0.441i$	$-7.64 \cdot 10^{-15} + 1.75i$	$-0.00261 + 1.7i$	$-0.00261 + 1.7i$
8.0	$-6.94 \cdot 10^{-18} - 0.441i$	$-1.98 \cdot 10^{-14} - 1.75i$	$-7.39 \cdot 10^{-4} - 1.7i$	$-7.38 \cdot 10^{-4} - 1.7i$
9.0	$-1.34 \cdot 10^{-17} + 0.137i$	$-2.66 \cdot 10^{-14} - 5.64i$	$-0.0286 + 5.47i$	$-0.0285 + 5.47i$
10.0	$-1.34 \cdot 10^{-17} - 0.137i$	$-5.92 \cdot 10^{-14} + 5.64i$	$-0.0228 - 5.49i$	$-0.0228 - 5.49i$

Table 89: Case 1a (cpv MAC64 MR2023b), $r = 10$, Z4: poles.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	$6.66 \cdot 10^{-16} + 5.47i$	$-6.77 \cdot 10^{-15} - 0.141i$	$-1.38 \cdot 10^{-8} - 0.137i$	$-9.33 \cdot 10^{-9} + 0.137i$
2.0	$6.66 \cdot 10^{-16} - 5.47i$	$-1.0 \cdot 10^{-14} + 0.141i$	$9.86 \cdot 10^{-9} + 0.137i$	$-4.69 \cdot 10^{-9} - 0.137i$
3.0	$2.22 \cdot 10^{-16} + 1.7i$	$-3.47 \cdot 10^{-15} - 0.453i$	$-3.99 \cdot 10^{-8} - 0.441i$	$-1.58 \cdot 10^{-8} + 0.441i$
4.0	$2.22 \cdot 10^{-16} - 1.7i$	$-1.04 \cdot 10^{-14} + 0.453i$	$3.43 \cdot 10^{-8} + 0.441i$	$-1.28 \cdot 10^{-9} - 0.441i$
5.0	$-5.2 \cdot 10^{-17} + 0.866i$	$-3.18 \cdot 10^{-15} - 0.891i$	$-7.56 \cdot 10^{-8} - 0.866i$	$-2.7 \cdot 10^{-8} + 0.866i$
6.0	$-5.2 \cdot 10^{-17} - 0.866i$	$1.9 \cdot 10^{-15} + 0.891i$	$6.45 \cdot 10^{-8} + 0.866i$	$2.74 \cdot 10^{-10} - 0.866i$
7.0	$3.47 \cdot 10^{-18} + 0.441i$	$-1.86 \cdot 10^{-14} - 1.75i$	$-1.48 \cdot 10^{-7} - 1.7i$	$-5.77 \cdot 10^{-8} + 1.7i$
8.0	$3.47 \cdot 10^{-18} - 0.441i$	$3.05 \cdot 10^{-14} + 1.75i$	$1.16 \cdot 10^{-7} + 1.7i$	$-6.33 \cdot 10^{-9} - 1.7i$
9.0	$-1.52 \cdot 10^{-18} + 0.137i$	$-5.53 \cdot 10^{-14} - 5.64i$	$-5.63 \cdot 10^{-7} - 5.47i$	$-3.48 \cdot 10^{-7} + 5.47i$
10.0	$-1.52 \cdot 10^{-18} - 0.137i$	$1.18 \cdot 10^{-13} + 5.64i$	$2.67 \cdot 10^{-7} + 5.47i$	$-1.87 \cdot 10^{-7} - 5.47i$

Table 90: Case 1a (cpv MACI64 MR2022b), $r = 10$, Z4: poles.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	$2.22 \cdot 10^{-16} + 5.47i$	$-1.13 \cdot 10^{-14} + 0.141i$	$1.68 \cdot 10^{-9} + 0.137i$	$-3.96 \cdot 10^{-10} - 0.137i$
2.0	$2.22 \cdot 10^{-16} - 5.47i$	$-3.06 \cdot 10^{-14} - 0.141i$	$-1.75 \cdot 10^{-9} - 0.137i$	$4.93 \cdot 10^{-11} + 0.137i$
3.0	$2.78 \cdot 10^{-16} + 1.7i$	$6.86 \cdot 10^{-15} + 0.453i$	$5.46 \cdot 10^{-9} + 0.441i$	$-9.24 \cdot 10^{-10} - 0.441i$
4.0	$2.78 \cdot 10^{-16} - 1.7i$	$-3.6 \cdot 10^{-14} - 0.453i$	$-5.53 \cdot 10^{-9} - 0.441i$	$5.0 \cdot 10^{-10} + 0.441i$
5.0	$-1.25 \cdot 10^{-16} + 0.866i$	$1.56 \cdot 10^{-14} + 0.891i$	$1.05 \cdot 10^{-8} + 0.866i$	$-1.72 \cdot 10^{-9} - 0.866i$
6.0	$-1.25 \cdot 10^{-16} - 0.866i$	$-4.39 \cdot 10^{-14} - 0.891i$	$-1.08 \cdot 10^{-8} - 0.866i$	$1.04 \cdot 10^{-9} + 0.866i$
7.0	$-6.94 \cdot 10^{-18} + 0.441i$	$8.07 \cdot 10^{-15} + 1.75i$	$2.0 \cdot 10^{-8} + 1.7i$	$-3.56 \cdot 10^{-9} - 1.7i$
8.0	$-6.94 \cdot 10^{-18} - 0.441i$	$-5.94 \cdot 10^{-14} - 1.75i$	$-2.13 \cdot 10^{-8} - 1.7i$	$1.79 \cdot 10^{-9} + 1.7i$
9.0	$-1.34 \cdot 10^{-17} + 0.137i$	$-1.74 \cdot 10^{-13} - 5.64i$	$5.76 \cdot 10^{-8} + 5.47i$	$-1.63 \cdot 10^{-8} - 5.47i$
10.0	$-1.34 \cdot 10^{-17} - 0.137i$	$1.08 \cdot 10^{-13} + 5.64i$	$-7.4 \cdot 10^{-8} - 5.47i$	$7.27 \cdot 10^{-10} + 5.47i$

Table 91: Case 1a (cpv PCWIN64 MR2025b), $r = 10$, Z4: poles.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	$2.22 \cdot 10^{-16} + 5.47i$	$-1.13 \cdot 10^{-14} + 0.141i$	$1.68 \cdot 10^{-9} + 0.137i$	$-3.96 \cdot 10^{-10} - 0.137i$
2.0	$2.22 \cdot 10^{-16} - 5.47i$	$-3.06 \cdot 10^{-14} - 0.141i$	$-1.75 \cdot 10^{-9} - 0.137i$	$4.93 \cdot 10^{-11} + 0.137i$
3.0	$2.78 \cdot 10^{-16} + 1.7i$	$6.86 \cdot 10^{-15} + 0.453i$	$5.46 \cdot 10^{-9} + 0.441i$	$-9.24 \cdot 10^{-10} - 0.441i$
4.0	$2.78 \cdot 10^{-16} - 1.7i$	$-3.6 \cdot 10^{-14} - 0.453i$	$-5.53 \cdot 10^{-9} - 0.441i$	$5.0 \cdot 10^{-10} + 0.441i$
5.0	$-1.25 \cdot 10^{-16} + 0.866i$	$1.56 \cdot 10^{-14} + 0.891i$	$1.05 \cdot 10^{-8} + 0.866i$	$-1.72 \cdot 10^{-9} - 0.866i$
6.0	$-1.25 \cdot 10^{-16} - 0.866i$	$-4.39 \cdot 10^{-14} - 0.891i$	$-1.08 \cdot 10^{-8} - 0.866i$	$1.04 \cdot 10^{-9} + 0.866i$
7.0	$-6.94 \cdot 10^{-18} + 0.441i$	$8.07 \cdot 10^{-15} + 1.75i$	$2.0 \cdot 10^{-8} + 1.7i$	$-3.56 \cdot 10^{-9} - 1.7i$
8.0	$-6.94 \cdot 10^{-18} - 0.441i$	$-5.94 \cdot 10^{-14} - 1.75i$	$-2.13 \cdot 10^{-8} - 1.7i$	$1.79 \cdot 10^{-9} + 1.7i$
9.0	$-1.34 \cdot 10^{-17} + 0.137i$	$-1.74 \cdot 10^{-13} - 5.64i$	$5.76 \cdot 10^{-8} + 5.47i$	$-1.63 \cdot 10^{-8} - 5.47i$
10.0	$-1.34 \cdot 10^{-17} - 0.137i$	$1.08 \cdot 10^{-13} + 5.64i$	$-7.4 \cdot 10^{-8} - 5.47i$	$7.27 \cdot 10^{-10} + 5.47i$

Table 92: Case 1a (AlexReis PCWIN64 MR2025b), $r = 10$, Z4: poles.

9.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$
1.0	NaN	NaN	$9.66 \cdot 10^{-4} + 1.92 \cdot 10^{-4}_z$	$9.67 \cdot 10^{-4} + 1.96 \cdot 10^{-4}_z$
2.0	0	$1.35 \cdot 10^{-14} - 8.67 \cdot 10^{-15}_z$	$0.00105 - 0.282_z$	$0.00105 - 0.282_z$
3.0	$3.33 \cdot 10^{-16} + 2.67_z$	$1.74 \cdot 10^{-14} + 0.289_z$	$7.24 \cdot 10^{-4} + 0.282_z$	$7.24 \cdot 10^{-4} + 0.282_z$
4.0	$3.33 \cdot 10^{-16} - 2.67_z$	$2.58 \cdot 10^{-14} - 0.289_z$	$9.39 \cdot 10^{-4} - 0.63_z$	$9.39 \cdot 10^{-4} - 0.63_z$
5.0	$2.43 \cdot 10^{-16} + 1.19_z$	$1.34 \cdot 10^{-14} + 0.647_z$	$2.23 \cdot 10^{-4} + 0.63_z$	$2.23 \cdot 10^{-4} + 0.63_z$
6.0	$2.43 \cdot 10^{-16} - 1.19_z$	$3.58 \cdot 10^{-14} - 0.647_z$	$-0.00102 + 1.19_z$	$-0.00102 + 1.19_z$
7.0	$4.16 \cdot 10^{-17} + 0.629_z$	$-5.1 \cdot 10^{-15} + 1.23_z$	$3.05 \cdot 10^{-4} - 1.19_z$	$3.06 \cdot 10^{-4} - 1.19_z$
8.0	$4.16 \cdot 10^{-17} - 0.629_z$	$6.08 \cdot 10^{-15} - 1.23_z$	$-0.00688 + 2.67_z$	$-0.00687 + 2.67_z$
9.0	$-3.47 \cdot 10^{-18} + 0.281_z$	$-2.24 \cdot 10^{-14} + 2.75_z$	$-0.00398 - 2.67_z$	$-0.00398 - 2.67_z$
10.0	$-3.47 \cdot 10^{-18} - 0.281_z$	$-3.62 \cdot 10^{-14} - 2.75_z$	$-971.0 + 392.0_z$	$-975.0 + 389.0_z$

Table 93: Case 1a (cpv MACA64 MR2023b), $r = 10$, Z4: zeros.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	NaN	NaN	$-1.87 \cdot 10^{-9} + 3.18 \cdot 10^{-6}_z$	$-6.85 \cdot 10^{-9} - 3.69 \cdot 10^{-7}_z$
2.0	0	$-9.07 \cdot 10^{-15} + 4.27 \cdot 10^{-14}_z$	$-2.63 \cdot 10^{-8} - 0.281_z$	$-1.22 \cdot 10^{-8} + 0.281_z$
3.0	$3.33 \cdot 10^{-16} + 2.67_z$	$-4.41 \cdot 10^{-15} - 0.289_z$	$2.18 \cdot 10^{-8} + 0.281_z$	$-2.83 \cdot 10^{-9} - 0.281_z$
4.0	$3.33 \cdot 10^{-16} - 2.67_z$	$-1.09 \cdot 10^{-14} + 0.289_z$	$-5.58 \cdot 10^{-8} - 0.629_z$	$-2.05 \cdot 10^{-8} + 0.629_z$
5.0	$-2.22 \cdot 10^{-16} + 1.19_z$	$-4.14 \cdot 10^{-15} - 0.647_z$	$4.82 \cdot 10^{-8} + 0.629_z$	$-1.51 \cdot 10^{-10} - 0.629_z$
6.0	$-2.22 \cdot 10^{-16} - 1.19_z$	$-5.14 \cdot 10^{-15} + 0.647_z$	$-1.03 \cdot 10^{-7} - 1.19_z$	$-3.76 \cdot 10^{-8} + 1.19_z$
7.0	$-4.16 \cdot 10^{-17} + 0.629_z$	$-4.28 \cdot 10^{-15} - 1.23_z$	$8.56 \cdot 10^{-8} + 1.19_z$	$-8.47 \cdot 10^{-10} - 1.19_z$
8.0	$-4.16 \cdot 10^{-17} - 0.629_z$	$9.49 \cdot 10^{-15} + 1.23_z$	$-2.41 \cdot 10^{-7} - 2.67_z$	$-1.08 \cdot 10^{-7} + 2.67_z$
9.0	$1.73 \cdot 10^{-18} + 0.281_z$	$-3.66 \cdot 10^{-14} - 2.75_z$	$1.68 \cdot 10^{-7} + 2.67_z$	$-2.91 \cdot 10^{-8} - 2.67_z$
10.0	$1.73 \cdot 10^{-18} - 0.281_z$	$6.7 \cdot 10^{-14} + 2.75_z$	$-284.0 - 2.41 \cdot 10^5_z$	$-3.94 \cdot 10^4 + 2.13 \cdot 10^6_z$

Table 94: Case 1a (cpv MACI64 MR2022b), $r = 10$, Z4: zeros.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	NaN	NaN	$-3.62 \cdot 10^{-11} - 3.64 \cdot 10^{-7}_z$	$-1.7 \cdot 10^{-10} + 4.59 \cdot 10^{-9}_z$
2.0	0	$-2.24 \cdot 10^{-14} - 5.81 \cdot 10^{-15}_z$	$3.48 \cdot 10^{-9} + 0.281_z$	$-6.42 \cdot 10^{-10} - 0.281_z$
3.0	$3.33 \cdot 10^{-16} + 2.67_z$	$-4.04 \cdot 10^{-16} + 0.289_z$	$-3.55 \cdot 10^{-9} - 0.281_z$	$2.69 \cdot 10^{-10} + 0.281_z$
4.0	$3.33 \cdot 10^{-16} - 2.67_z$	$-3.45 \cdot 10^{-14} - 0.289_z$	$7.74 \cdot 10^{-9} + 0.629_z$	$-1.27 \cdot 10^{-9} - 0.629_z$
5.0	$2.43 \cdot 10^{-16} + 1.19_z$	$-3.9 \cdot 10^{-14} - 0.647_z$	$-7.85 \cdot 10^{-9} - 0.629_z$	$7.51 \cdot 10^{-10} + 0.629_z$
6.0	$2.43 \cdot 10^{-16} - 1.19_z$	$1.22 \cdot 10^{-14} + 0.647_z$	$1.43 \cdot 10^{-8} + 1.19_z$	$-2.4 \cdot 10^{-9} - 1.19_z$
7.0	$4.16 \cdot 10^{-17} + 0.629_z$	$1.25 \cdot 10^{-14} + 1.23_z$	$-1.48 \cdot 10^{-8} - 1.19_z$	$1.38 \cdot 10^{-9} + 1.19_z$
8.0	$4.16 \cdot 10^{-17} - 0.629_z$	$-5.01 \cdot 10^{-14} - 1.23_z$	$3.04 \cdot 10^{-8} + 2.67_z$	$-6.13 \cdot 10^{-9} - 2.67_z$
9.0	$-3.47 \cdot 10^{-18} + 0.281_z$	$-8.32 \cdot 10^{-14} - 2.75_z$	$-3.4 \cdot 10^{-8} - 2.67_z$	$2.21 \cdot 10^{-9} + 2.67_z$
10.0	$-3.47 \cdot 10^{-18} - 0.281_z$	$2.31 \cdot 10^{-14} + 2.75_z$	$-1130.0 + 2.0 \cdot 10^6_z$	$-1.28 \cdot 10^7 - 2.24 \cdot 10^8_z$

Table 95: Case 1a (cpv PCWIN64 MR2025b), $r = 10$, Z4: zeros.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	NaN	NaN	$-3.62 \cdot 10^{-11} - 3.64 \cdot 10^{-7}_z$	$-1.7 \cdot 10^{-10} + 4.59 \cdot 10^{-9}_z$
2.0	0	$-2.24 \cdot 10^{-14} - 5.81 \cdot 10^{-15}_z$	$3.48 \cdot 10^{-9} + 0.281_z$	$-6.42 \cdot 10^{-10} - 0.281_z$
3.0	$3.33 \cdot 10^{-16} + 2.67_z$	$-4.04 \cdot 10^{-16} + 0.289_z$	$-3.55 \cdot 10^{-9} - 0.281_z$	$2.69 \cdot 10^{-10} + 0.281_z$
4.0	$3.33 \cdot 10^{-16} - 2.67_z$	$-3.45 \cdot 10^{-14} - 0.289_z$	$7.74 \cdot 10^{-9} + 0.629_z$	$-1.27 \cdot 10^{-9} - 0.629_z$
5.0	$2.43 \cdot 10^{-16} + 1.19_z$	$-3.9 \cdot 10^{-14} - 0.647_z$	$-7.85 \cdot 10^{-9} - 0.629_z$	$7.51 \cdot 10^{-10} + 0.629_z$
6.0	$2.43 \cdot 10^{-16} - 1.19_z$	$1.22 \cdot 10^{-14} + 0.647_z$	$1.43 \cdot 10^{-8} + 1.19_z$	$-2.4 \cdot 10^{-9} - 1.19_z$
7.0	$4.16 \cdot 10^{-17} + 0.629_z$	$1.25 \cdot 10^{-14} + 1.23_z$	$-1.48 \cdot 10^{-8} - 1.19_z$	$1.38 \cdot 10^{-9} + 1.19_z$
8.0	$4.16 \cdot 10^{-17} - 0.629_z$	$-5.01 \cdot 10^{-14} - 1.23_z$	$3.04 \cdot 10^{-8} + 2.67_z$	$-6.13 \cdot 10^{-9} - 2.67_z$
9.0	$-3.47 \cdot 10^{-18} + 0.281_z$	$-8.32 \cdot 10^{-14} - 2.75_z$	$-3.4 \cdot 10^{-8} - 2.67_z$	$2.21 \cdot 10^{-9} + 2.67_z$
10.0	$-3.47 \cdot 10^{-18} - 0.281_z$	$2.31 \cdot 10^{-14} + 2.75_z$	$-1130.0 + 2.0 \cdot 10^6_z$	$-1.28 \cdot 10^7 - 2.24 \cdot 10^8_z$

Table 96: Case 1a (AlexReis PCWIN64 MR2025b), $r = 10$, Z4: zeros.

9.4 Poles and zeros visualisation

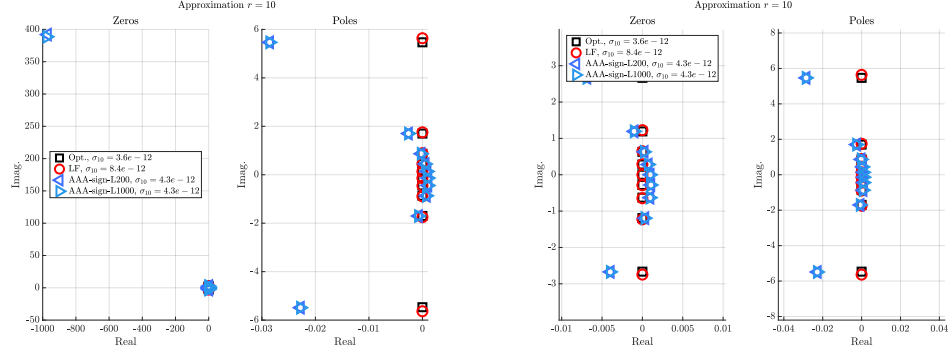


Figure 15: Case 1a (cpv MACA64 MR2023b), $r = 10$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

9.5 Approximation visualization

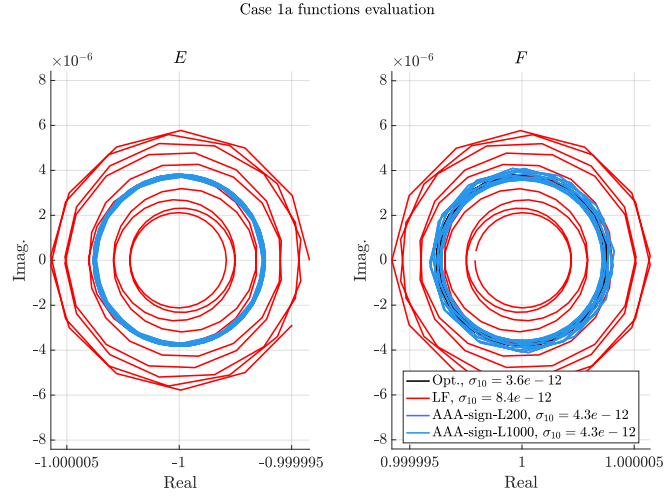


Figure 16: Case 1a (cpv MACA64 MR2023b), $r = 10$, Z4. E and F evaluations.